

The impact of arbitrage on market liquidity[☆]

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Abstract

I study how arbitrage affects liquidity by analyzing several billion trades in the American Depositary Receipt (ADR) market from 2001 to 2016. Price deviations persist, on average, for 12 minutes, and mainly arise because of price pressure. Impulse response functions estimated at 1-minute intervals indicate that a positive shock to arbitrage—simultaneous trades of the ADR and the home-market share in the opposite direction—decreases deviations and bid-ask spreads. I confirm these findings by exploiting institutional details that create exogenous variation in the impediments to arbitrage across days. Overall, these results suggest that arbitrage decreases price pressure and provides liquidity.

Keywords: Arbitrage, liquidity, ADR, efficiency, fragmentation, market integration

JEL: G14, G15

[☆]I thank the editor (Bill Schwert) and an anonymous referee for many valuable suggestions which greatly improved the paper. I also thank Lamont Black, Ekkehart Boehmer, Dion Bongaerts, Jonathan Brogaard, Howard Chan, Tarun Chordia, Ruben Cox, Thierry Foucault, Louis Gagnon, Nicolae Gârleanu, Michael Goldstein, Amit Goyal, Allaudeen Hameed, Terry Hendershott, Shing-yang Hu, Jonathan Kalodimos, Andrew Karolyi, Albert Kyle, Su Li, Albert Menkveld, Pamela Moulton, Maureen O'Hara, Louis Piccotti, Gideon Saar, Piet Sercu, René Stulz, Avanidhar Subrahmanyam, Raman Uppal, Dimitrios Vagias, Mathijs van Dijk, Manuel Vasconcelos, Kumar Venkataraman, Axel Vischer, Avi Wohl, and participants at the 2016 Research in Behavioral Finance Conference (Amsterdam), 2014 FMA (Nashville), the 2014 NFA (Ottawa), the 2014 Asian FA (Bali), 2014 Eastern FA (Pittsburgh), the 2013 Erasmus Liquidity Conference (Rotterdam), the 2013 Conference on the Theories and Practices of Securities and Financial Markets (Kaohsiung), the 2013 World Finance and Banking Symposium (Beijing), and at seminars at Babson College, Cass Business School, City University of Hong Kong, Cornell University, Erasmus University, Frankfurt School Of Management, and the University at Buffalo for valuable comments. I am grateful for the hospitality of the Department of Finance at the Johnson Graduate School of Management (Cornell University), where some of the work on this paper was carried out, especially from my hosts, Andrew Karolyi and Pamela Moulton. This work was carried out on the National e-infrastructure with the support of SURF Foundation. This work is supported in part by NSF ACI-1541215. I thank OneMarket Data for the use of their OneTick software.

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1. Introduction

Arbitrage enforces the law of one price and thereby improves the informational efficiency of financial markets. But how arbitrage affects other measures of market quality, in particular market liquidity, is less well understood. If arbitrage opportunities arise due to information, liquidity providers face higher adverse selection risk. In this case arbitrage can be “toxic.” Foucault, Kozhan, and Tham (2017) provide theoretical and empirical evidence that liquidity is lower if toxic arbitrage is likely. But if arbitrage opportunities arise as a result of price pressure—such as when intermediaries post bid and ask prices away from fundamentals to attract inventory-offsetting trades (Hendershott and Menkveld, 2014)—then arbitrageurs trade against intermediaries’ inventory, decreasing inventory holding costs. This observation is made succinctly in a survey of the theoretical limits-of-arbitrage literature conducted by Gromb and Vayanos (2010), who state that “arbitrageurs provide liquidity” (p. 258).

Inspired by this observation, I investigate why price deviations arise and how arbitrage affects market liquidity.¹ To date the empirical literature finds evidence mainly of illiquidity as an impediment to arbitrage; meanwhile, the evidence of how arbitrage affects liquidity is scarce. This is not surprising: potential reverse causality and simultaneity between arbitrage activity and liquidity—higher liquidity decreases impediments to arbitrage, encouraging arbitrage, which affects liquidity—is challenging to address.

I address the challenge twofold. First, I use high-frequency data. For example, I estimate panel vector autoregressions (VARs) and impulse response functions (IRFs) at the 1-minute interval. Using data at such high frequency preserves the order in which things unfold, addressing reverse causality and simultaneity concerns. Second, I exploit institutional details of the American Depositary Receipt (ADR) market that create exogenous variation in the impediments to arbitrage across days. For many ADRs impediments to arbitrage vary exogenously across days because corporate actions, such as dividend payments or stock splits,

¹ Empirically, arbitrage opportunities should manifest as price deviations; however, as has been extensively argued by the limits-to-arbitrage literature, not all price deviations are necessarily arbitrage opportunities.

on ADRs and their home market share often do not occur on the same day. During days between corporate actions—i.e., when one stock is ex- and the other is cum-dividend—prices are not directly comparable and institutions that normally facilitate arbitrage shut down.² For this and for other reasons reviewed in Section 2, on these days impediments to arbitrage are higher.

I examine tick-by-tick prices for 194 ADRs and their associated home market shares from February 2001—after decimalization in the U.S.—to December 2016. With these data I construct my proxy for arbitrage trades as simultaneous trades—defined as trades on both the host- and home-market within 0.1 or 1 seconds—for which one is buyer- and the other is seller-initiated (signing trades as in Lee and Ready (1991)). I denote these trades as *pair* trades and all non-simultaneous trades as *single* trades.

I first investigate the extent to which price deviations are toxic. Following Foucault et al. (2017), a price deviation is toxic for one market if the share price of this market moves to eliminate a price deviation created by a price movement in the other market. In the ADR market less than 10% of all price deviations are toxic (compared to around 50% in the foreign exchange market studied by Foucault et al. (2017)).

Alternatively, in partially segmented markets price deviations might reflect price pressure. Trading patterns during price deviations are consistent with this hypothesis. When prices deviate, order imbalance (the difference in the number of buyer- and seller-initiated trades) from *single* and from *pair* trades is, respectively, in the same and opposite direction as the deviation. For example, when prices deviate and the ADR is *Cheap*, order imbalance from *single* and *pair* trades is, respectively, -24 and 8, on average. When prices converge, order imbalance from *single* and from *pair* trades is in the opposite direction of the deviation.

Because price deviations indicate price pressure, arbitrageurs should act as “cross-sectional market-makers” (Holden, 1995) and improve liquidity. First, arbitrageurs should affect the

²I use the terms ex-dividend and cum-dividend to refer to any corporate action which has a direct effect on the stock price.

position of the bid-ask spread because they trade against price pressure and thereby decrease the extent to which prices deviate from fundamentals. Second, arbitrageurs could affect the size of the spread. For example, the spread varies with the inventory of liquidity providers if liquidity providers are constrained by capital (Amihud and Mendelson, 1980), averse to risk (O'Hara and Oldfield, 1986), or face holding costs (Shen and Starr, 2002). If arbitrageurs decrease inventory imbalances, the spread should decrease immediately after *pair* trades.

I test whether this is indeed the case. I first investigate liquidity immediately after *pair* trades. Spreads and Hasbrouck (1993) pricing errors associated with and immediately after *pair* trades are lower than for *single* trades. Results from a panel VAR and IRFs estimated with 1-minute intervals indicate that a one-standard-deviation shock to absolute order imbalance from *pair* trades decreases deviations by almost five basis points and spreads by around half a basis point over the next five minutes. For comparison, results from a shock to absolute order imbalance from *single* trades has the opposite effect, increasing deviations and spreads by almost twenty and two basis points, respectively.

Next, I estimate an instrumental panel regression at the daily level. I use the ratio of *pair* to *single* trades as a proxy for arbitrage activity, and I use days between corporate actions as an instrument when explaining quoted spreads, effective spreads, and Hasbrouck (1993) pricing errors, estimated immediately after *pair* trades. I find that an increase in the ratio of *pair* to *single* trades decreases spreads and pricing errors. These results are statistically significant at least at the 5% level.

All in all, I find that price deviations indicate inventory concerns which are mitigated by arbitrageurs. In addition, higher arbitrage activity predicts higher liquidity, both in terms of the position (price deviations and pricing errors) and the size of the bid-ask spread. These findings shed light on how policy changes that increase impediments to arbitrage (such as short-sell bans or margin requirements) could negatively impact liquidity and ultimately increase firms' cost of capital (Amihud and Mendelson, 1986).

My paper relates to the empirical limits-of-arbitrage literature.³ But instead of investigating why price deviations persist and how illiquidity impacts arbitrage, I focus on (i) why price deviations arise (Foucault et al., 2017; Schultz and Shive, 2010) and (ii) how arbitrage impacts liquidity (Ben-David, Franzoni, and Moussawi, 2018; Choi, Getmansky, and Tookes, 2009; Foucault et al., 2017; Roll, Schwartz, and Subrahmanyam, 2007; Tomio, 2017). In contrast to previous papers, I find evidence that arbitrage improves market liquidity.

I add to these important contributions in three ways. First, in contrast to prior research I investigate joined dynamics between arbitrage and liquidity at high-frequency (1-minute intervals), which allows me to better address simultaneity issues.

Second, instead of estimating impediments to arbitrage I estimate arbitrage more directly using *pair* trades. A trading-based measure of arbitrage is more direct, because impediments to arbitrage can be low without actual arbitrage. *Pair* trades, on average, seem driven by arbitrage motivations and are negatively correlated to measures commonly used to estimate impediments to arbitrage (see the Online Appendix).

Third, I build upon previous work in the ADR literature, especially Gagnon and Karolyi (2010), who study price deviations in the ADR market, and Moulton and Wei (2009) and Werner and Kleidon (1996), who investigate differences in liquidity during and outside overlapping trading times. Unlike these papers I have access to tick-by-tick data for the home market, which allows me to precisely estimate price deviations during overlapping trading.

This paper is organized as follows. In Section 2 I discuss institutional details and data and variable construction, and I provide summary statistics. Section 3 investigates why price deviations arise and discusses identification. Section 4 investigates joined dynamics between trading, price deviations, and liquidity using intraday data. Section 5 reports results of an instrumental variable approach to estimate how arbitrage affects liquidity. Section 6 describes robustness tests. Section 7 concludes.

³ Among many significant contributions are Mitchell, Pulvino, and Stafford (2002); Lamont and Thaler (2003); De Jong, Rosenthal, and van Dijk (2009); Gagnon and Karolyi (2010).

2. Institutional details, data and variable construction, summary statistics

To investigate the impact of arbitrage on market liquidity, I focus on the American Depositary Receipts market (ADR). As Gagnon and Karolyi (2010) lay out, the ADR market is particularly suitable to studying arbitrage because institutions exist that facilitate arbitrage and because the feature of convertibility makes it possible to interpret price deviations at the time the position is opened as (almost) risk-free profits.

In other markets arbitrage risk might be even lower. For example, arbitrage between the same stock trading on different exchanges in the US would be even less risky. But smart order routing—enforced by Reg NMS—already ensures that price deviations can not persist. Because no such regulation exists across ADRs and their home market share and international markets are partially segmented (Bodurtha, Kim, and Lee, 1995) price deviations can persist, creating potential arbitrage opportunities.

2.1. The ADR market

The first American Depositary Receipt (ADR) was created by (a predecessor of) J.P. Morgan in 1927 as a negotiable receipt for a foreign security to overcome difficulties in settling trades of foreign securities in the US. J.P. Morgan guaranteed that the foreign security is held in deposit as long as the ADR is outstanding and that investors can convert the ADR to the foreign security and vice-versa. The exact contractual terms of the deposit are disclosed using SEC filing, form F-6.

I refer to, among others, Baruch, Karolyi, and Lemmon (2007), Gagnon and Karolyi (2009, 2010, 2013), and Karolyi (1998) for a comprehensive introduction to ADRs.

2.2. The mechanics of arbitrage

In contrast to textbook arbitrage, in the real world arbitrage is both capital intensive and risky (Liu and Longstaff, 2004; Shleifer and Vishny, 1997; Tuckman and Vila, 1992). The need for capital and the inherent risk explains why positive price deviations can persist. Arbitrageurs will only trade if expected profits are high enough.

To understand the underlying risk in more detail, it is crucial to review how arbitrageurs can close their position. First, arbitrageurs can wait until prices converge and trade in the opposite direction. Because arbitrageurs face holding costs (Tuckman and Vila, 1992) and because they do not know when prices will converge such a strategy is potentially very costly and risky. For dual-listed stocks, for example, it can take several weeks for prices to converge (De Jong et al., 2009).

Alternatively, arbitrageurs in the ADR market can close their position by converting the long position into the short position. After shares have settled, the share of the long position can be converted (within one business day and for less than five cents a share (Gagnon and Karolyi, 2010)) into the share of the short position either through a broker, a crossing platform (e.g. ADR Max, or ADR Navigator), or the actual depositary bank. The converted long position can be used to close down the short position, resulting in a risk-free profit equal the price deviation *at the time the position was opened*.

The advantage of converting shares is that it limits holding periods to at maximum a few days and provides an almost risk-free way to close the arbitrage position.

2.3. The mechanics of ADR arbitrage around ex-dates of corporate actions

Corporate actions, such as dividend payments, increase the impediments to arbitrage. The feature of convertibility is temporarily suspended around corporate actions (SEC F-6, 2001, I A.1). However, even though the books are closed for conversion, investors can still trade the ADR as on any other day. The only mechanism to close out an arbitrage position around corporate actions is to trade in the opposite direction, which works as on any other day as long as both the home-market share and the ADR have the same ex-date. But ex-dates often vary. An ADR's record date is often set following the record date of the home-market share (e.g., Section 4.6. of Vodafone, 2006). But differences in settling periods or market holidays in one (but not the other) market can create differences in ex-dates. In other words, it is common that the ADR or the home-market share is still cum dividend but the other is already ex-dividend.

During these days identifying mispricing is riskier because adjusting the cum-dividend price by the dividend is risky. Consistent with the difficulties in adjusting prices, previous literature finds that dividend capture trading in ADRs is more profitable than in comparable US stocks (Gorman, Mahajan, and Weigand, 2004).⁴ The ex-date price adjustment is unknown before the ex-date, often below the dividend payment, and can not always be explained by marginal tax rates (c.f. Lasfer (1995), Frank and Jagannathan (1998), or Henry and Koski (2017)). Factors complicating forecasting the ex-date price drop are: uncertainty in the dollar dividend payment paid out to ADR holders;⁵ investors discounting the dividend payment from the ex-date until the payment date (Lasfer, 1995); and complicated tax-law on foreign dividends, which requires payments of withholding taxes, potentially leads to double taxation of dividend payments, and favors long-term investors (Callaghan and Barry, 2003).

2.4. Data and sample

To construct my sample of ADRs and their respective home market shares, I use standard sources in the DR literature: Datastream, Bank of New York Complete Depositary Receipt Directory, and Deutsche Bank Depositary Receipts Services. Details about the sample construction can be found in Appendix A.

Initially, I identify 325 ADR/home market pairs for which the ADR is trading at the NYSE or Nasdaq. But because the analysis requires comparing contemporaneous prices across ADRs and home market shares, I drop all countries without overlapping trading times with the US. As in Werner and Kleidon (1996) and Moulton and Wei (2009), the overlapping trading time is defined as the time in which both the ADR and the home market are in their

⁴ Dividend capture involves buying shares cum-dividend and selling them on the ex-dividend date with the hope that the ex-date price drop is less than the dividend payment. In general, this strategy is not an arbitrage opportunity (to make it an arbitrage traders need to trade with different settlement dates, see Angel, 1998), but historically, it delivered positive returns. Both Gorman et al. (2004) and Henry and Koski (2017) report dividend capture returns of around 10 bp for US stocks. Gorman et al. (2004) report a return of over 40 bp for ADRs.

⁵ After receiving the dividend from the home-market stocks, in general weeks *after* the ex-date, the Depositary Bank needs to exchange the home market currency into U.S. dollars at an unknown exchange rate to pay the ADR holders. See Gorman et al. (2004) or Vodafone (2006, Section IV 4.1).

continuous trading session. For all matched pairs, I obtain tick-by-tick data on quotes and trades (time-stamped with at least millisecond precision) from the Thomson Reuters Tick History (TRTH) database from January 1996 (the earliest date available in TRTH) through December 2016. Similarly, I obtain tick-by-tick quotes for all currency pairs required to convert local prices into USD, the currency in which the ADR is quoted in, from TRTH.⁶

Quote and trade data is filtered as described in Appendix B. After filtering, the data contains almost 6 billion updates to the best bid and ask quotes and more than 4 billion trades, with nearly half on ADRs. Because of data availability and because of the filters, I only have sparse observations in the early years of the sample. Therefore, I focus on data starting February 2001, which has a secondary benefit of preventing issues stemming from the tick-size change on January, 29th, 2001. My final sample consists of 194 pairs across 22 exchanges.

2.5. Arbitrage (pair) and non-arbitrage (single) motivated trades

To construct my proxy for arbitrage and non-arbitrage trades I categorize all trades in the sample into simultaneous trades—defined as trades on both the host- and home-market within 1 second before 2007 and 0.1 seconds afterwards—and *single* trades, trades on only the host- or the home-market. Because in the early years of the sample there are few 1-minute intervals with simultaneous trades within 0.1 seconds, when using data before 2007 I define simultaneous trades as trades within one second. I ensure that these somewhat arbitrary choices do not affect the results. Results are robust to using data only after 2007 or 2011, and to excluding stock-days with less than 100 trades (Online Appendix Tables OA23, OA24, and OA25).

I then distinguish between simultaneous trades for which the host- and home-market trade are both either buyer- or seller-initiated (signing trades as in Lee and Ready, 1991)

⁶ The TRTH database is managed by the Securities Industry Research Center of Asia-Pacific (SIRCA) and is used in several recent studies (e.g. Fong, Holden, and Trzcinka (2017); Kahraman and Tookes (2017); Lau, Ng, and Zhang (2012); Lai, Ng, and Zhang (2014); Marshall, Nguyen, and Visaltanachoti (2011)).

and for which one is buyer- and the other is seller-initiated.⁷ As potential arbitrage trades, denoted *pair* trades, I only consider simultaneous trades for which one trade is buyer- and the other is seller-initiated.

This classification is imperfect. For example, even though trades occur simultaneously they might belong to two separate accounts, and therefore should be classified as *single* trades. Further, simultaneous trades can be wrongly classified as *pair* trades, for example, if Lee and Ready (1991) wrongly classifies the trades or when arbitrageurs use limit orders (Tomio, 2017). Despite these imperfections most arbitrage trades should show up as *pair* trades, and it is therefore mainly an empirical question whether *pair* trades proxies arbitrage well enough or just picks up noise.

2.6. Price deviations, Cheap, and Costly

I calculate spread-adjusted price-deviations ($|\Delta Price|$) from bid and ask prices, as the difference between the highest bid and the lowest ask price across the home and host market relative to the midpoint price of the home market. If this difference is negative, I set it to zero; in this case the host and home market are fairly priced.⁸

$$|\Delta Price|_{i,t} = \max \left(\frac{bid.home_{i,t} - ask.adr_{i,t}}{mid.home_{i,t}}, \frac{bid.adr_{i,t} - ask.home_{i,t}}{mid.home_{i,t}}, 0 \right) \quad (1)$$

where $mid.home_{i,t}$ is the mid-quote price of stock i at time t , and $bid.home_{i,t}$ ($ask.home_{i,t}$) is the bid (ask) of stock i at time t converted to USD using the prevailing bid (ask) of the respective currency pair (for example, EUR for Germany). Further, $bid.adr_{i,t}$ ($ask.adr_{i,t}$) is the bid (ask) at time t of the ADR trading in the US associated with stock i , adjusted for

⁷ A trade is classified as buyer- (seller-)initiated if it is closer to the ask (bid) of the prevailing quote. A trade at the midpoint of the quote is classified as buyer- (seller-)initiated if the previous price change is positive (negative). Ellis, Michaely, and O'Hara (2000), Lee and Radhakrishna (2000), Odders-White (2000), and Theissen (2001) provide evidence that this algorithm signs around 80% of all trades correctly for Nasdaq, NYSE, and German stocks.

⁸ For example, if the home- and host-stock are trading at the same bid and ask prices, $|\Delta Price|$ would be negative as long as the ask price is above the bid price. Because an arbitrageur would just not trade when "profits" are negative, I cap $|\Delta Price|$ at zero. Note that one cannot set up an arbitrage (using market-orders) by trading in the opposite way because this would require one to buy at the bid and sell at the ask price.

the respective bundling ratio as described in Appendix B. To avoid using stale quotes, I only consider quotes that are at most 300 seconds old.

Further, if the bid price of stock i (e.g., home-market) is higher than the ask price of the associated stock i' trading in the other market (e.g., host-market), I call stock i *Costly* and stock i' *Cheap*, i.e.:

$$Costly_{i,t} = \begin{cases} 1, & \text{if } bid_{i,t} > ask_{i',t} \\ 0, & \text{else} \end{cases} \quad Cheap_{i,t} = \begin{cases} 1, & \text{if } ask_{i,t} < bid_{i',t} \\ 0, & \text{else} \end{cases} \quad (2)$$

2.7. Market illiquidity and order imbalance

To measure illiquidity, I use proportional quoted spread (*PQSPR*), the difference in the ask and the bid prices scaled by the midpoint price, and effective spread (*PESPR*), twice the absolute difference between the logarithm of the trade price and the logarithm of the midpoint price of the prevailing quote.

To be able to also measure inventory effects resulting in bid and ask prices posted away from fundamentals, I estimate Hasbrouck (1993) pricing errors as in Rösch, Subrahmanyam, and van Dijk (2017, Section 1.3). In particular, for every stock-day I decompose midpoint prices into a random walk and stationary component using a five-lag Vector Autoregression (VAR) model with the following endogenous variables: (1) the logarithmic midpoint price return estimated from quotes prevailing trades, (2) the sign of the trade, (3) the sign of the trade times the number of shares traded, and (4) the sign of the trade times the square root of the number of shares traded. Using Equation (13) in Hasbrouck (1993), I estimate the pricing error associated with each trade. Because pricing errors could be below or above fundamentals, I focus on absolute pricing errors. To diminish the impact of outliers, I focus on the logarithmic transformation of *Hasbrouck*.

I further construct a measure of buying and selling pressure. First, I sign every trade in the host- and home-market using the Lee and Ready (1991) algorithm. Second, I calculate order imbalance as the difference between the number of buyer- and seller-initiated trades

(*OIB*) as in Chordia, Roll, and Subrahmanyam (2008).

To ensure that results are not driven by outliers, I cross-sectionally winsorize price deviation and illiquidity measures at the 1% and 99% level on each day.

2.8. Summary statistics

Table 1 presents cross-sectional summary statistics of minute-by-minute (Panel A) and daily (Panels B and C) time-series averages. In Panel A of Table 1, I average *PQSPR*, *PESPR*, absolute *OIB* for *single* and for *pair* trades, and maximum $|\Delta Price|$ within each one-minute interval during overlapping trading times, and I report cross-sectional summary statistics separately for home- and host-market stocks.

The average maximum price deviation is 0.80% ($|\Delta Price|$). For comparison, Gagnon and Karolyi (2010) use end-of-day data from 1993 to 2004 and find (cost-adjusted) average price deviations of 1.12%. The average home-market stock has a time series average *PQSPR* of around 0.57% and the respective ADR of 0.66%. Both the average home- and host-market stock has an absolute order imbalance of around five *single* trades and almost one *pair* trade, on average.

In Panels B and C, I report summary statistics from variables estimated at the stock-day level. In Panel B of Table 1, I estimate the ratio of *pair* to *single* trades on all days for which both the ADR and the home-market stock are either cum- or ex-dividend. On these days the average stock has a ratio of around 0.0712—i.e., for every 100 *single* trades there will be seven *pair* trades, on average. I also report the ratio estimated on days between corporate actions—i.e., when either the host or the home market is cum-dividend but the other is ex-dividend. On these days, the ratio drops from 7.12% to 3.21%, indicating a significant drop in arbitrage activity.

Panel B also reports cross-sectional summary statistics for the number of days between corporate actions across the 137 stock-pairs with at least one day between corporate actions. These days are frequent and common across exchanges, stocks, and time. For the average (median) stock, I observe 21 (11) days between corporate actions.

Panel C of Table 1 reports summary statistics for illiquidity and control variables.

3. Why do price deviations arise? Identification and methodology.

Why price deviations arise is intimately related to how arbitrage affects liquidity. If deviations arise as a result of differences in information, arbitrage might worsen liquidity (Foucault et al., 2017). For arbitrage to improve liquidity deviations need to arise due to price pressure. The end of this section discusses identification and methodology.

3.1. Why do price deviations arise?

I follow Foucault et al. (2017) and Schultz and Shive (2010) to investigate why price deviations arise. If for one particular stock i at time $t - 1$ price deviations are zero, but at time t deviations are positive, at least one bid or ask quote of at least one asset changed from time $t - 1$ to time t (this asset—either the ADR, the home market share, or the respective currency pair—is denoted the *First mover*). Similarly, if deviations are positive until time $\tau - 1 > t$, but zero at time τ , then at least one bid or ask quote of at least one asset changed (this asset is denoted the *Last mover*). In this case I say that the *First mover* creates a price deviation for stock i at time t , and the *Last mover* eliminates the deviation at time τ .⁹

Table 2 reports the number of price deviations by the *First* and *Last mover*.¹⁰ For each *First mover* I separately report the percentage of all deviations that are *toxic*. Following Foucault et al. (2017), I consider a deviation as “toxic” for market m if m was the *Last mover* and the other market was the *First mover*.

Table 2 reports 13,913,834 price deviations. Price movements in the home market create 4,092,945 of these price deviations, of which 45% are later eliminated because the price of the home market moves back and 24% are eliminated because of a price movement of the

⁹ In cases when the day opens with a price deviation, I consider the asset whose market opened last as the *First mover*. On the other hand, if a price deviation exists and either of the markets closes, in this section, I drop this price deviation from the analysis, as I do not know which asset closes down the price deviation. Both cases are infrequent and do not affect the main results in this section.

¹⁰ In the case that the currency pair moves simultaneously with the ADR or home market share, the *First mover* is considered to be the ADR or home market share.

ADR. The percentage of all toxic price deviations for the home market is 6.50% and for the ADR it is 7.03%.

These results indicate that price deviations arise less because of informational differences than of price pressure. To understand the origin of this price pressure and whether arbitrageurs trade against it, I investigate the sensitivity of order imbalances (*OIB*, difference in the number of buyer- and seller- initiated trades) to mispricings.

Within each stock-day I estimate *OIB* separately depending on whether the stock is during its overlapping trading times (*InOverlap*) and whether the stock is fairly priced, *Costly*, or *Cheap*. If price deviations arise because of price pressure, one would expect that when prices deviate *OIB* from non-arbitrage related trading should be in the same direction as the deviation, but potentially flips sign when prices converge. In this case pooling when prices deviate and when they converge might lead to the wrong conclusion that *OIB* is not sensitive to mispricings. To address this, I distinguish between periods within the day in which prices deviate and converge.

Empirically, I classify periods in which prices deviate as the time within each price deviation before the peak of the pricing gap using an indicator variable denoted *APeak*. For example, if $|\Delta Price|$ [from Eq. (1)] is zero at 10 and 11 o'clock but positive from 10 to 11 o'clock, I estimate when $|\Delta Price|$ peaked, for example, at 10:15. I then classify the time before the peak (from 10:00 to 10:15) as the time when prices deviate ($APeak = 0$) and the time after the peak (from 10:15 to 11:00) as the time when prices converge ($APeak = 1$).

To ensure that deviations reflect mispricings, I only consider stock-days on which the stock is in between corporate actions or trades fairly priced, *Cheap*, and *Costly* (but results are robust to using all stock-days, Table OA7).

I use panel regressions with two-dimensional stock and day fixed effects to explain *OIB* within each category by indicator variables, as below. I do not include *APeak* by itself as it is equal to one only when $Costly_{i,d,j}$ or $Cheap_{i,d,j}$ is equal to one and hence including it

would lead to linear dependence in the independent variables.

$$\begin{aligned}
 OIB_{i,d,j} = & FE_{i,d} + \beta_0 InOverlap_{i,d,j} + (\beta_1 + \gamma_1 APeak_{i,d,j}) Costly_{i,d,j} \\
 & + (\beta_2 + \gamma_2 APeak_{i,d,j}) Cheap_{i,d,j} + \epsilon_{i,d,j}
 \end{aligned} \tag{3}$$

Table 3 reports the results. In the home market, when prices deviate *OIB* from *single* trades is 4 when the stock is *Costly* and it is -4 when the stock is *Cheap* (p -values equal to 0.11 and 0.02). Results for the host market are stronger with an *OIB* of 8 and -24 when prices deviate and are, respectively, *Costly* and *Cheap* (p -values below 0.01). For both home and host market, *OIB* flips sign when prices converge.

In other words, order imbalance from *single* trades is in the same direction as the mispricing when prices deviate and flips sign when prices converge. *OIB* from *pair* trades is in the opposite direction of the deviation, regardless of whether prices deviate or converge. These results are consistent with the idea that price deviations arise due to price pressure visible in *single* trades and mitigated by *pair* trades.

3.2. Identification and methodology

Price deviations arise because of price pressure. Therefore, price deviations themselves are a measure of illiquidity and by definition arbitrageurs trading against these deviations supply liquidity and affect the position of the bid and ask spread.

Arbitrage might affect the size of the spread, too. Empirically, Comerton-Forde, Hendershott, Jones, Moulton, and Seasholes (2010) find that spreads are higher when inventories are imbalanced. Theoretically, inventories affect the size of the spread if liquidity providers face financing constraints, for example, if they are constrained by capital (Amihud and Mendelson, 1980), averse to risk (O'Hara and Oldfield, 1986), or face holding costs (Shen and Starr, 2002). Financing constraints implicitly raise marginal costs of providing liquidity in the direction of imbalances. Hence, if arbitrage decreases imbalances it decreases spreads, too. The size of the spread is also affected by the risk-aversion of liquidity providers, which

depends on the capital available for liquidity provision. If arbitrageurs increase this capital, risk-aversion and with it spreads set by liquidity providers might decrease (Foucault, Pagano, and Röell, 2013, p. 116).¹¹

To estimate the effect of arbitrage on liquidity, denoted b , one would ideally subtract outcome variables (liquidity, L) with and without arbitrage activity (A), i.e., $b = E[L|A = 1 \wedge C] - E[L|A = 0 \wedge C]$. With the structural model given by,

$$L = a + bA + cC + \epsilon$$

for a vector of size n of unknown covariates C , constants a, b , $(c_0, \dots, c_n) = c$, and some error term ϵ . Under certain assumptions, especially that regressors (A and C) are uncorrelated with errors (ϵ), the ordinary least square estimate of b , denoted $\hat{b}$, is an unbiased and consistent estimate of b , allowing $\hat{b}$ to be interpreted as the causal effect of arbitrage on liquidity. Previous considerations predict that

H0 if price deviations reflect price pressure, then b should be positive, with L inversely related to the position (empirically measured by $|\Delta Price|$ and Hasbrouck (1993) pricing errors) and the size of the spread ($PQSPR$ and $PESPR$).

In the following I discuss how unobserved heterogeneity, simultaneity, and other endogeneity issues, i.e., regressors being correlated with errors, affect causal inference, and how one can address these issues.

Unobserved heterogeneity is a concern, as it can lead to omitted-variable bias and preclude causal inference. For example, if time-varying funding liquidity influences arbitrage activity and market liquidity (as in Brunnermeier and Pedersen, 2008) this could cause an omitted-variable bias.¹² Even heterogeneity that could potentially be controlled for, such

¹¹ Spreads are also affected by competition (in terms of possible rents) (as in Moulton and Wei, 2009), and arbitrageurs might increase competition. As my main investigation assumes that arbitrageurs rely on market orders, I do not directly test whether arbitrageurs act themselves as local liquidity providers by quoting bid and ask prices and thereby increase competition.

¹² I control for the Ted spread, a measure of funding illiquidity, but the Ted spread is based on US interest

as price volatility, can preclude causal inference, if controls are endogenous (Lewbel, 2019, p.853). To address these issues, following the recommendation of Gormley and Matsa (2014), I estimate panel regressions with two-dimensional time-stock fixed effects, controlling for timevarying stock-level heterogeneity. Because results do not indicate any substantial differences between the host and home markets, for ease of exposition and to increase power, I pool both markets in one regression.¹³ To allow for correlations in the residuals across host- and home-market stocks, I cluster standard errors by symbol-pair.

Another empirical concern is that arbitrage activity and liquidity are likely simultaneously determined: higher liquidity increases arbitrage activity, affecting liquidity. This is especially a concern when using low-frequency measures, such as daily estimates. To address this concern, I focus on measuring liquidity immediately after arbitrage. Further, I estimate panel vector autoregressions (VARs) and impulse response functions (IRFs) at high frequency. VARs regress each variable on lagged versions of itself and on lagged versions of all other variables in the system. In particular, IRFs constructed from a VAR are commonly used to yield important information about the dynamics of jointly determined variables (cf. Roll et al., 2007). An IRF estimated from a VAR tracks the response on one variable from an impulse to another variable. By using the Cholesky decomposition to calculate orthogonalized impulse responses, an IRF also allows estimating contemporaneous effects. For details I refer to Doan (2010).

Because it is unclear at which frequency patterns of trading and price deviations evolve, it is also unclear at which frequency the VAR should be estimated. Ideally, one would distinguish between times when prices deviate and when they converge, as in Table 3, but this kind of trade time analysis is not suitable for panel regressions. Table 1 indicates that price deviations for the median stock persist on average for 6.41 minutes, indicating that

rates and might not fully capture changes in funding illiquidity of the home market.

¹³ Nevertheless, I verify that the main results are robust to estimating results separately for home- and host-market stocks (Tables OA27 and OA28) and to estimating panel regressions with variables from both home- and host-market stocks entering the regression simultaneously (Table OA29).

any analysis at a lower frequency than, for example, five minutes would miss most of the pattern. Because of a decline in the duration price-deviations persist (untabulated), and in order to capture meaningful variation for a large cross-section, I use 1-minute intervals (with 5-minutes as a robustness test; see Figure OA4). A frequency much higher than 1-minute is likely not feasible because, in contrast to most theoretical models, liquidity and pricing effects are not resolved by one trade.

In general, the empirical challenge is to address endogeneity of which unobserved heterogeneity and simultaneity are two examples. But endogeneity can also arise because of other issues, such as measurement error (Lewbel, 2019, p.852). I also estimate a two-stage regression, another common way to address endogeneity (cf. Foucault et al., 2017). This is possible if one finds a variable Z that is correlated with A (relevance) but not directly correlated with L (exclusion restriction), in which case Z is called an instrument. I propose using days between corporate actions as an instrument and discuss relevance and the exclusion restriction in Section 5. Because the instrument is only available at the stock-day level I can only estimate regressions at the daily level with two-dimensional stock-month fixed effects.

4. The intraday joint dynamics of trading, liquidity, and price deviations

In this section I investigate liquidity at and immediately after *single* and *pair* trades and estimate panel vector autoregressions and impulse response functions using 1-minute intervals.

4.1. Spreads and pricing errors associated with trades

To understand how *pair* trades affect liquidity I estimate daily average spreads associated with each trade and immediately afterwards, separately for *single* and *pair* trades. I only consider trades during overlapping trading times and days with at least one *pair* trade. I then estimate a panel regression with two-dimensional stock-day fixed effects explaining average spreads by an indicator variable, where spreads are estimated for *pair* trades ($Pair = 1$)

or for *single* trades ($Pair = 0$). To better estimate other inventory effects, I also estimate Hasbrouck (1993) pricing errors associated with and immediately after trades.

Table 4 reports the results. Quoted spreads, effective spreads, and pricing errors associated with *pair* trades are lower, respectively, by around 3, 2, and 8 basis points (bps) (all statistically significant with p -values below 0.01). After *pair* trades, quoted spreads, effective spreads, and pricing errors are also lower, respectively, by around 6, 0.5, and 8 bps (all statistically significant with p -values below 0.01).

In the Online Appendix (Table OA13) I re-estimate results using trades within mispricings only, an important covariate ensuring that the timing of *pair* and *single* trades does not endogenously differ from each other. (I cannot just control for mispricing because it is mechanically linked to spreads, as discussed in Section 4.2.) I again find that spreads and pricing errors are lower for *pair* than for *single* trades.

The estimate of $Pair$ explains the difference in liquidity associated with or following *pair* and *single* trades (respectively, $\Delta Pair[t]$ and $\Delta Pair[t + 1]$). By subtracting $\Delta Pair[t]$ from $\Delta Pair[t + 1]$, one can estimate the difference-in-difference (diff-in-diff) effect. Table 4 indicates that the diff-in-diff quoted spread is negative, i.e., quoted spreads decrease more after *pair* trades than after *single* trades (p -value below 0.01, untabulated). This provides initial evidence that *pair* trades decrease spreads. But the diff-in-diff effective spread is positive and the diff-in-diff pricing error is zero. These opposing results indicate several limitations with the analysis. For example, effective spreads associated with *pair* trades might be low because of large price improvements associated with *pair* trades (i.e., trades executed at better prices than quoted); in this case, effective spreads for the next trade might be higher and the diff-in-diff wrongly concludes that *pair* trades increase effective spreads. Another concern is that traders split up their orders into several trades. In this case, it is not easy to distinguish between liquidity associated with and immediately after a trade. I address these concerns in the following section by estimating liquidity and trades over fixed-time intervals.

4.2. Intraday vector autoregressions and impulse response functions

I estimate a panel vector autoregression (VAR) with five lags of the following vector of endogenous variables ($\mathbf{V}$) estimated within each 1-minute interval during overlapping trading times: the logarithm of 1 plus the absolute OIB from *Single* trades ($|OIB|_{single}$), maximum price deviations ($|\Delta Price|$), the logarithm of 1 plus the absolute OIB from *Pair* trades ($|OIB|_{pair}$), and average quoted spreads ($PQSPR$).¹⁴

$$\mathbf{V}_{i,m} = \mathbf{FE} + \sum_{l=1}^5 \beta \times \mathbf{V}_{i,m-l} + \epsilon_{i,m} \quad (4)$$

As discussed in Section 3.2, I include two-dimensional stock-day fixed effects and also include exchange-hour fixed effects to address liquidity seasonality within the day ($\mathbf{FE}$).

Panel A of Table 5 reports Granger causality tests: the sum of all lagged coefficients for each variable and the associated p -values, including whether this sum is statistically significantly different from zero and whether the lagged coefficients are jointly different from zero (in the Online Appendix I report estimates for each lagged coefficient instead of just their sum, Table OA14). The sum of all estimated lagged coefficients for a given explanatory variable can be interpreted as the long-run effect this variable has on the dependent variable and can also provide information regarding the direction of the relation (as in Chaboud, Chiquoine, Hjalmarsson, and Vega, 2014, p. 2063).

Results indicate Granger causality between all variables with directions as predicted, with two notable exceptions. Consistent with predictions, I find that an increase in $|OIB|_{pair}$ Granger causes lower $|\Delta Price|$ and $PQSPR$ and that an increase in $|OIB|_{single}$ Granger causes higher $|\Delta Price|$ and $PQSPR$. I discuss the economic magnitude of the estimated relations later, when discussing results from impulse response functions.

Surprisingly, an increase in $|\Delta Price|$ does not predict an increase in $|OIB|_{pair}$ with any

¹⁴ In most market-microstructure models price volatility plays a fundamental role. For brevity I exclude volatility from my main analysis but show that the main results are robust to its inclusion (Table OA15). Results are also robust to using ten lags (Table OA16).

statistical significance. This might indicate that not all deviations are arbitrage opportunities, as argued by the limits of arbitrage literature. Indeed, only large price deviations (defined as deviations one standard deviation above the daily average) Granger cause an increase in $|OIB|_{pair}$ (see Panel B of Table 5.) More surprisingly, an increase in $|\Delta Price|$ Granger causes an economically small but statistically significant decrease in $PQSPR$ (and vice versa). This is surprising because given that deviations indicate price pressure, both should be driven by the inventory imbalances of liquidity providers.

One concern is that $|\Delta Price|$ is mechanically negatively correlated with $PQSPR$. $|\Delta Price|$ (from Eq. 1) can be re-written to indicate that $|\Delta Price|$ is given by price deviations in midpoint prices in excess of half the quoted spread of the host- and the home-market stock. I therefore re-estimate the VAR using deviations from midpoint prices (ΔMid).

The main results are robust to this specification, an increase in $|OIB|_{pair}$ Granger causes lower ΔMid and $PQSPR$ and an increase in $|OIB|_{single}$ Granger causes higher $PQSPR$ but also lower ΔMid . In this specification, an increase in ΔMid Granger causes an increase in $PQSPR$, as one would expect (see Online Appendix, Table OA18).

The different finding of how $|OIB|_{single}$ affects price deviations is consistent with findings in Section 3.1 and further suggests the importance of distinguishing between when prices deviate and when they converge. One way to do so (without creating a look-ahead bias in determining the peak of any price deviation as in Section 3.1) is to interact $|OIB|_{single}$ with an indicator variable set to 1 if price deviations increased compared to the last interval ($Up|\Delta Price|_{i,t} = |\Delta Price|_{i,t} > |\Delta Price|_{i,t-1}$). If price deviations increased, the probability of a decrease in $|\Delta Price|$ increases because price deviations are mean-reverting, as discussed and estimated in Online Appendix Section OA3. Given this, I distinguish between periods when prices deviate and when prices converge, respectively, as periods with $Up|\Delta Price|$ equal to zero and equal to one.

Results (see Online Appendix, Table OA19) indicate that when $Up|\Delta Price|$ equals zero $|OIB|_{single}$ Granger causes an increase in $|\Delta Price|$. When $Up|\Delta Price|$ equals one $|OIB|_{single}$

Granger causes a decrease in $|\Delta Price|$. In other words, price deviations arise due to one-sided *single* trades, but *single* trades also contribute to bringing prices back in line. Regardless of whether prices deviate or converge, *pair* trades are bringing prices back in line. Results also show that only increasing price deviations ($Up|\Delta Price|$ equals one) Granger cause an increase in $|OIB|_{pair}$, confirming that only large price deviations attract arbitrageurs.

While the rest of the paper also uses Hasbrouck (1993) pricing errors to better capture inventory concerns, trading based measures are difficult to use when estimating VARs with 1-minute intervals, because not every 1-minute interval contains a trade. Instead, in Panel C of Table 5 I investigate how *pair* trades affect return reversals. The idea is that price pressure should lead to return reversal. If pair trades mitigate return reversal, the interaction of pair trades with lagged returns should be positive when explaining current returns.

Panel C of Table 5 shows that this is indeed the case. I estimate *return* as the 1-minute return from midpoints of bid and ask prices. The interaction of $|OIB|_{pair}$ with *return* Granger causes positive returns and therefore diminishes return reversal, visible in negative estimates of lagged *return* when explaining current *return*. A return of 10 basis points over the last 5-minutes reverts, on average, by around 3 basis points in the next minute, but does not revert at all if returns are accompanied by an order imbalance from pair trades of around 60 trades per minute.

While effective spreads also can only be estimated when trades occur, it is easier to substitute missing values. Effective spreads mainly differ from quoted spreads (estimated using quotes prevailing trades) because of price improvements, for example, because of hidden liquidity. Effective and quoted spreads might also differ when the size of the trade exceeds the size available at the best quoted price, but empirically this is rare.

I therefore replace missing effective spread estimates (*PESPR*) with quoted spreads (*PQSPR*). Of the around 150 million 1-minute intervals used in the sample, around half contain at least one trade and therefore non-missing *PESPR*.¹⁵

¹⁵ To ensure that the relatively high number of replacements does not drive the results, as robustness, I

Table 6 shows the results. Similar to when using *PQSPR*, I find that an increase in $|OIB|_{pair}$ Granger causes lower $|\Delta Price|$ and *PESPR* and that an increase in $|OIB|_{single}$ Granger causes higher $|\Delta Price|$ and *PESPR*.

Because Granger causality tests are based on only one equation of the VAR, they cannot provide a complete picture of the joined dynamics of all variables. A better approach is constructing impulse response functions (IRFs).¹⁶

To avoid spurious results in addition to using fixed effects, I purge seasonal effects from all variables by regressing them on a linear time-trend, a linear and squared time-trend within each day, and days of the week dummies. Following Roll et al. (2007), I then use the residuals from these regressions to estimate impulse response functions.

Figure 1 shows cumulative IRFs estimated from the panel VAR. For parsimony, Figure 1 only shows results when either *OIB* from *single* or *pair* trades is shocked and only responses on *PQSPR* and price deviations. Figure 1 also does not show confidence bands, because they are very tight and often overlap with the effect (as one might expect given the large amount of available observations). The full set of IRFs (together with bootstrapped confidence bands estimated based on 1000 runs) are reported in the Online Appendix (Figure OA1). Results from IRFs agree with results from the VAR.

The top and bottom of Figure 1 shows, respectively, responses to shocks to the absolute order imbalance from single (left side) and pair trades (right side) in the average quoted spread (*PQSPR*) and in the maximum price deviation within each 1-minute interval ($|\Delta Price|$). A one-standard-deviation shock to order imbalance from *pair* (*single*) trades is associated with a decrease (increase) in spreads of around half a (2) basis point and price deviations of almost 5 (20) basis points over the next 5 minutes. IRFs also indicate that

estimate the VAR using only days on which I can estimate *PESPR* for at least eight out of ten intervals (Online Appendix Table OA20). I also estimate the VAR replacing missing *PESPR* with lagged *PESPR* and using 5-minute intervals, which reduces intervals with missing *PESPR* to around one out of three (Online Appendix Tables OA21 and OA22). Results are robust.

¹⁶ It is not possible to construct IRFs from models with interaction terms such as those reported in Panel C of Table 5.

contemporaneously these effects are of the same direction but economically smaller.

Figure 2 shows that IRFs are similar when using *PESPR* instead of *PQSPR*. A one-standard-deviation shock to order imbalance from *pair* (*single*) trades is associated with a decrease (increase) in *PESPR* of around 0.2 (1.7) basis points, similar in magnitude to the effect on *PQSPR*, but with a much stronger contemporaneous effect. This indicates that *PESPR* reacts faster to changes in arbitrage activity, potentially because liquidity providers' willingness to offer price improvements changes.

Madhavan and Sofianos (1998) provide evidence suggesting that liquidity providers might be more willing to provide price improvements for trades that balance their inventories. As previous results indicate that *pair* trades are more likely to balance inventories, price improvements might be higher and *PESPR* might be lower for *pair* trades. Indeed, using the difference in quoted spreads estimated using quotes prevailing trades and effective spreads as a measure of price improvements confirms that *pair* (*single*) trades increases (decreases) price improvements, see Online Appendix Figure OA6.

5. Days between corporate actions as an instrument to determine the impact of arbitrage on liquidity

Results in Section 3 indicate that price deviations arise as a result of price pressure visible in one-sided *single* trades and that *pair* trades mitigate these shocks. Consistently, results in Section 4 show that an increase in *single* and *pair* trades predicts, respectively, an increase and decrease in deviations and spreads. Together, it seems reasonable to conjecture that the ratio of *pair* to *single* trades measures arbitrage activity and that high arbitrage activity improves liquidity, in terms of the position and the size of the spread.

As discussed in Section 3.2, finding an instrument provides another way to address the challenge of establishing causality between jointly determined variables. While it is challenging to motivate and statistically impossible to verify that both assumptions of relevance and exclusion hold, as a suitable candidate I propose an indicator variable that is 1 on days

between corporate actions—i.e, when either the host or the home market is cum-dividend but the other is ex-dividend.

First, impediments to arbitrage are higher on days between corporate actions, because, as discussed in Section 2.3, the expected holding period is longer than on other days and arbitrageurs introduce uncertainty when adjusting prices for the corporate action to compute their profits. Both increases risk (or cost, if this additional risk would be hedged away). As such, it is not surprising that during these days the ratio of *pair* to *single* trades is lower (see Table 1), indicating less arbitrage activity.

Second, to have a valid instrument, liquidity should only differ on these days because of changes in the impediments to arbitrage (exclusion restriction). Previous literature finds no significant change in liquidity around ex-dates for common stocks, i.e., assets not necessarily linked to other assets by arbitrage (Lasfer, 1995). But ADRs are special, not only because they are linked to their home-market share, but also because of foreign taxation of dividend incomes. For this reason many investors try to avoid dividend income from ADRs causing significantly more trading on cum- and ex-dates (Callaghan and Barry, 2003). The question is whether special trading of ADRs around ex-dates will cause variations in liquidity.

Illiquidity arises because of adverse selection risk or inventory holding costs. Clearly, ex-dates do not confer any information and therefore market-makers are not more likely to be adversely selected. But the increase in trading could increase inventory holding costs. It is important that a significant part of the abnormal trading volume around ex-dates is coming from nonstandard-settlement trades (Angel, 1998). These trades are negotiated and do not affect the limit order book and therefore also should not affect liquidity. Furthermore, non-fundamental demand pressure is supposed to be absorbed by arbitrageurs. It is only if arbitrageurs are not active that liquidity could potentially decrease.

In conclusion, it seems unlikely that ex-dates directly affect liquidity. Therefore, it seems even less reasonable to suspect that liquidity should be affected between corporate actions (other than by increasing the impediments to arbitrage). Rather, on days between corporate

actions local liquidity providers might be more risk-averse and inventory imbalances have a stronger effect on the position and the size of the spread because of less arbitrage activity and therefore less available capital for liquidity provision.

5.1. Trading on days between corporate actions

To further understand how days between corporate actions affect trading, I look at how the ratio of *pair* to *single* trades varies around these days. I perform an event study, with the event any day in between corporate actions. I compute the ratio per stock-day and then calculate the daily cross-sectional average ratio over all stocks. To calculate statistical significance, I first calculate the normal ratio per stock (estimated as the average ratio over 41 days before till 11 days before the event). I then test whether the average on any given day is statistically different from its normal ratio.

Figure 3 reports the results from 10 days before to 10 days after the event. Ten days before the event the ratio of *pair* to *single* trades is around 8%, similar to the average of 7.12% reported in Table 1, and indeed not statistically different from its normal ratio. The ratio is statistically significantly below its normal ratio, 7, 2, 1, and 0 days before the event. At around 5%, the ratio is lowest on event days. The drop before the event could be because during the event it is not possible to convert shares of one market into the other. Because only settled shares can be converted and US stocks settle T+3 during the sample period, one would expect an impact 1 and 2 days before.

Figure 3 also reports how the number of trades varies around the event. For each stock-pair, I estimate the number of trades on a given day as the sum of all *pair* and *single* trades during overlapping trading times in both the home- and the host-market. Results indicate that the number of trades is statistically significantly different only 1 and 2 days before the event. On event days trading is somewhat higher than usual, but not statistically significantly so.

5.2. Liquidity on days between corporate actions (reduced form)

Angrist and Krueger (2001) state that “Concerns about weak instruments can be mitigated most simply by looking at the reduced form equation, that is, the ordinary least squares regression of the dependent variable of interest on the instruments and exogenous variables.” (p. 79) Table 7 shows the results of estimating the reduced form explaining illiquidity by a dummy variable that is set to 1 on days between corporate actions. To minimize concerns of reverse causality and to directly estimate the impact of arbitrage on liquidity, I estimate illiquidity after *pair* trades. To focus on arbitrage motivated trades, I only consider *pair* trades during mispricings and drop days with less than 10 *pair* trades (results are robust to using all days with at least 5 or at least 20 *pair* trades, Tables OA34 and OA35). As discussed in Section 3.2, to control for unobserved heterogeneity I use two-dimensional stock-months fixed effects. In additional tests, I control for other, probably endogenously determined, variables commonly associated with liquidity (Foucault et al., 2017). To rule out that corporate actions by themselves affect liquidity, I also control for days in which both the ADR and home-market stock go ex-dividend together.

Table 7 reports how days between corporate actions affect quoted spreads (*PQSPR*), effective spreads (*PESPR*), and Hasbrouck (1993) pricing errors immediately after *pair* trades. Results indicate that on days between corporate actions, spreads are higher by around 1 basis point and pricing errors by around 5 basis points, with results statistically significant at the 1% threshold. Results and adjusted R^2 only marginally change when using additional control variables, mitigating concerns of an omitted variable bias (Oster, 2019).

5.3. Days between corporate actions as an instrument

In this section I estimate a two-stage panel regression with two-dimensional stock-month fixed effects as given below:

$$Pair/Single_{i,d} = FE + \beta_0 \times D_{i,d}^{EX} + \boldsymbol{\zeta}_0 \times \mathbf{Controls}_{i,d} + \eta_{i,d} \quad (5a)$$

$$Illiq_{i,d} = FE + \beta_1 \times \overline{Pair/Single_{i,d}} + \boldsymbol{\zeta}_1 \times \mathbf{Controls}_{i,d} + \epsilon_{i,d} \quad (5b)$$

where $Pair/Single_{i,d}$ is the ratio of *pair* trades to *single* trades for stock i on day d , and $D_{i,d}^{EX}$ is a dummy variable set to 1 on days between corporate actions (the instrument, and motivated in the previous sections). In the second equation $Illiq_{i,d}$ is a measure for illiquidity, quoted spreads ($PQSPR$), effective spreads ($PESPR$), or Hasbrouck (1993) pricing error immediately after *pair* trades. $\overline{Pair/Single_{i,d}}$ is the fitted value from the first equation (the first stage), $Controls_{i,d}$ are control variables as in Table 7, and FE are two-dimensional stock-month fixed effects.

Panels A and B of Table 8 show the results of first- and second-stage regressions, respectively. Panel A indicates that on days between corporate actions, the ratio of *pair* to *single* trades is abnormally low (as in Figure 3). Panel B indicates that if $Pair/Single_{i,d}$ increases $PQSPR$, $PESPR$, and pricing errors after *pair* trades decrease.

Results indicate that a 3% increase in $Pair/Single_{i,d}$ (the difference between the average and the median stock, see Table 1) decreases $PQSPR$, $PESPR$, and pricing errors, respectively, by around 0.8, 0.6, and 4 basis points. This indicates that arbitrage activity improves liquidity in terms of the position and the size of the spread.

6. Robustness

In this section I discuss additional selected robustness tests. I report further robustness tests in the Online Appendix, together with all results.

6.1. Robustness to measures of trading activity

When estimating intraday vector autoregressions and impulse response functions, I rely on order imbalance to estimate trade intensity, but when using daily data I use the number of trades. At the daily level I cannot measure trade intensity by order imbalances because price deviations often switch signs within the day; therefore, averaging order imbalances over the whole day would likely underestimate trade intensities. Intraday results are robust for using the number of trades (Table OA30).

Throughout the paper I also ignore the size of trades because it is common that large orders are split into many small trades. To ensure that results do not rely on this simplification, I report results using order imbalance estimated in USD. In this case, results are robust when considering the whole joint dynamics, i.e., when estimating IRFs (Figure OA5).

6.2. Robustness to measures of arbitrage activity

I rely on *pair* trades as my proxy for arbitrage trades and use this to construct a proxy of daily arbitrage activity, the ratio of *pair* to *single* trades. At the daily level, previous literature suggests several proxies for the impediments to arbitrage. For example, large or slow mean-reverting price deviations signal high impediments to arbitrage and low arbitrage activity (Roll et al., 2007). In the Online Appendix I report that the ratio of *pair* to *single* trades highly correlates with these less direct measures of arbitrage activity (Table OA5) and that results from instrumental variable regressions are robust to using any of these measures instead (see Section OA3.3).

7. Conclusion

Arbitrageurs enforce the law of one price by trading against mispricings, but whether by doing so arbitrageurs provide liquidity depends on the reason for the arbitrage opportunity to arise. In this paper, I study how arbitrage affects liquidity in the American Depositary Receipts market. The results from three empirical tests suggest that arbitrageurs provide liq-

uidity by trading against net market demand, or as Foucault et al. (2013) put it, arbitrageurs are “leaning against the wind” (p. 336).

Price deviations arise due to price pressure, with order imbalance from *pair* trades mitigating the price pressure from *single* trades. A positive shock to the number of *pair* trades decreases price deviations and bid-ask spreads over the next several minutes. And an increase in the ratio of *pair* trades to *single* trades increases liquidity.

One way to encourage arbitrage activity is to introduce portfolio margins, where the offsetting position between the home market stock and the associated ADR are incorporated in the margin requirements. Within the US, a similar concept is already approved by the SEC for index options.

Appendix A: Sample construction

This appendix describes details of the sample construction. I first retrieve all dead and alive American Depositary Receipts (ADRs) from Datastream, which are traded at the New York Stock Exchange (NYSE) or Nasdaq. I identify the home market share, associated to any of the ADRs, using Datastream and verify the match using data from adrbnymellon.com and adr.db.com. Both websites offer a list of DRs and an ISIN code for the home market share.

As the analysis requires intraday data, I use the Thomson Reuters Tick History (TRTH) database. I filter out any DR for which I could not establish the RIC (the primary identifier in TRTH) for either the DR or the home-market stock. Upon request, Datastream provides a RIC field; however this field is empty for around 50% of all DRs. In the case of a missing RIC field for the ADR or for the home market shares, I use the TRTH API to search for a RIC code by ISIN. For every ISIN the RIC from the major exchange of the home market country is chosen. In this way I identify 351 ADRs and their respective home market share.

To avoid non-synchronous prices, I only focus on countries with overlapping trading times to the US. This excludes all Asian countries (China, Hong Kong, India, Indonesia, Japan, Philippines, South Korea, and Taiwan) as well as Australia and New Zealand. I am left with 238 pairs from the following 27 countries: Argentina (12 pairs), Austria (1), Belgium (2), Brazil (27), Chile (13), Colombia (1), Denmark (1), Finland (1), France (13), Germany (12), Greece (2), Ireland (3), Israel (9), Italy (4), Luxembourg (2), Mexico (23), Netherlands (12), Norway (3), Peru (3), Portugal (1), Spain (8), Sweden (6), Switzerland (5), Turkey (1), United Kingdom (58), Russian Federation (3), and South Africa (12).

Of these 238 stocks I filter out 30 (and with it five countries drop), because I could not align prices of the home market with prices of the ADR, as described in more detail in Appendix B. Of the remaining 208 stocks, I have 194 stocks with at least one year of data from 2001 to 2016 (the sample period for the main analysis). In the online appendix I report summary statistics for the main variables used in this paper separately for each of the 22 countries (Table OA4).

Appendix B: Data filters

This appendix describes the quote and trade data filters, based on all available data from 1996 to 2016. I discard nonpositive bid and ask quotes (in total 196,061 quotes), quotes where the ask is lower or equal to the bid quote (193,936,018 quotes), and quotes outside the continuous trading session (814,278,837 quotes).¹⁷ Further, I remove outliers (144,258 quotes). An outlier is defined as a bid (ask) quote that differs by more than 10% of the average of the ten surrounding bid (ask) quotes. In a similar way trade prices are filtered. Of the in total 4.45 and 2.23 billion host- and home-market trades, I discard trades that fall outside the continuous trading session in the US (13,631,796 trades) and on the respective home market (50,938,498 trades). I also discard trades with a nonpositive price (3,190,264 trades) or a price that is more than 10% different from the trade price of the ten surrounding trades for ADRs (1,529 trades) and home-market shares (13,542,154 trades). Further, I discard trades of more than 100,000 shares for ADRs (170,142 trades) and in the home market (7,060,211 trades), because large trades are often negotiated before they get reported.

To make prices comparable between the home-market stock and the ADR, bid and ask quotes of the ADR are converted according to the bundling ratio, which I got from either Datastream, adrbnymellon.com, or adr.db.com. Bundling ratios can be time-varying, but all sources only report the latest bundling ratio. To adjust the bundling ratio over time, I get all corporate actions from TRTH for all ADRs and home-market shares. Changes in the bundling ratio can occur, for example because of solo stock-splits. To verify the accuracy of the bundling ratio, I plot the daily average currency-adjusted mid-quote ratios for each stock in the sample (unreported). If the ratio does not vary around one and does not resemble a step function, the stock is dropped from the sample. As such, 30 stocks are dropped from the sample because prices of the home market could not be aligned to prices of the ADR.

For 82 ADRs the bundling ratio changed over the sample, with a maximum of three

¹⁷ Most quotes outside continuous trading times are from the ADR. ADR prices seem to update when the home market share is traded, regardless of whether the ADR is in its continuous trading session.

changes for four ADRs. For 59 ADRs the bundling ratio changed once over the sample.

To further ensure that stocks are mapped properly and prices are adjusted correctly, I drop any stock-day if (adjusted) price deviations ΔQTE are higher than 100% (of the mid-quote of the home market—in total, 273 stock-days) or higher than USD 5 (1,285 stock-days). These very large price deviations are driven by two stocks from Argentina with 718 and 536 days, with price deviations above USD 5.

- Amihud, Y. and H. Mendelson (1980). Dealership market: Market-making with inventory. *Journal of Financial Economics* 8, 31 – 53.
- Amihud, Y. and H. Mendelson (1986). Asset pricing and the bid-ask spread. *Journal of Financial Economics* 17, 223–249.
- Angel, J. J. (1998). Nonstandard-settlement transactions. *Financial Management* 27, 31–46.
- Angrist, J. D. and A. B. Krueger (2001). Instrumental variables and the search for identification: From supply and demand to natural experiments. *Journal of Economic Perspectives* 15, 69–85.
- Baruch, S., G. A. Karolyi, and M. L. Lemmon (2007). Multimarket trading and liquidity: Theory and evidence. *Journal of Finance* 62, 2169–2200.
- Ben-David, I., F. Franzoni, and R. Moussawi (2018). Do ETFs increase volatility? *Journal of Finance* 73, 2471–2535.
- Bodurtha, Jr, J. N., D.-S. Kim, and C. M. Lee (1995). Closed-end country funds and u.s. market sentiment. *Review of Financial Studies* 8, 879–918.
- Brunnermeier, M. K. and L. H. Pedersen (2008). Market Liquidity and Funding Liquidity. *Review of Financial Studies* 22, 2201–2238.
- Callaghan, S. R. and C. B. Barry (2003). Tax Induced Trading of Equity Securities: Evidence from the ADR Market. *Journal of Finance* 58, 1583–1612.
- Chaboud, A., B. Chiquoine, E. Hjalmarsson, and C. Vega (2014). Rise of the machines: Algorithmic trading in the foreign exchange market. *Journal of Finance* 69, 2045–2084.
- Choi, D., M. Getmansky, and H. Tookes (2009). Convertible bond arbitrage, liquidity externalities, and stock prices. *Journal of Financial Economics* 91, 227–251.
- Chordia, T., R. Roll, and A. Subrahmanyam (2008). Liquidity and market efficiency. *Journal of Financial Economics* 87, 249–268.
- Comerton-Forde, C., T. Hendershott, C. M. Jones, P. C. Moulton, and M. S. Seasholes (2010). Time Variation in Liquidity: The Role of Market-Maker Inventories and Revenues. *Journal of Finance* 65, 295–331.
- De Jong, A., L. Rosenthal, and M. van Dijk (2009). The Risk and Return of Arbitrage in Dual-Listed Companies. *Review of Finance* 13, 495–520.
- Doan, T. A. (2010). RATS Handbook for Vector Autoregressions. *working paper, Estima*.
- Ellis, K., R. Michaely, and M. O’Hara (2000). The accuracy of trade classification rules: Evidence from nasdaq. *Journal of Financial and Quantitative Analysis* 35, 529–551.
- Fong, K., W. C. Holden, and C. Trzcinka (2017). What Are the Best Liquidity Proxies for Global Research? *Review of Finance* 21, 1355–1401.

- Foucault, T., R. Kozhan, and W. W. Tham (2017). Toxic Arbitrage. *Review of Financial Studies* 30, 1053–1094.
- Foucault, T., M. Pagano, and A. Röell (2013). *Market Liquidity Theory, Evidence, and Policy*. New York: Oxford University Press.
- Frank, M. and R. Jagannathan (1998). Why do stock prices drop by less than the value of the dividend? evidence from a country without taxes. *Journal of Financial Economics* 47, 161 – 188.
- Gagnon, L. and A. G. Karolyi (2010). Multi-market trading and arbitrage. *Journal of Financial Economics* 97, 53–80.
- Gagnon, L. and A. G. Karolyi (2013). *Do International Cross-listings Still Matter? in Evidence on Financial Globalization and Crises*. London: Elsevier.
- Gagnon, L. and G. A. Karolyi (2009). Information, Trading Volume, and International Stock Return Comovements: Evidence from Cross-Listed Stocks. *Journal of Financial and Quantitative Analysis* 44, 953–986.
- Gorman, L., A. Mahajan, and R. Weigand (2004). The ex-dividend day behaviour of american depositary receipts. *Journal of Multinational Financial Management* 14, 1–18.
- Gormley, T. A. and D. A. Matsa (2014). Common errors: How to (and not to) control for unobserved heterogeneity. *Review of Financial Studies* 27, 617–661.
- Gromb, D. and D. Vayanos (2010). Limits of arbitrage: The state of the theory. *Annual Review of Financial Economics* 2, 251–275.
- Hasbrouck, J. (1993). Assessing the quality of a security market: A new approach to transaction-cost measurement. *Review of Financial Studies* 6, 191–212.
- Hendershott, T. and A. J. Menkveld (2014). Price pressures. *Journal of Financial Economics* 114, 405 – 423.
- Henry, T. R. and J. L. Koski (2017). Ex-dividend profitability and institutional trading skill. *Journal of Finance* 72, 461–494.
- Holden, C. W. (1995). Index arbitrage as cross-sectional market making. *Journal of Futures Markets* 15, 423–455.
- Kahraman, B. and H. E. Tookes (2017). Trader leverage and liquidity. *Journal of Finance* 72, 1567–1610.
- Karolyi, G. A. (1998). Why do companies list shares abroad?: A survey of the evidence and its managerial implications. *Financial Markets, Institutions & Instruments*, 1–60.
- Lai, S., L. Ng, and B. Zhang (2014). Does PIN Affect Equity Prices Around the World? *Journal of Financial Economics* 114, 178 – 195.

- Lamont, O. a. and R. H. Thaler (2003). Can the Market Add and Subtract? Mispricing in Tech Stock Carve-outs. *Journal of Political Economy* 111, 227–268.
- Lasfer, M. A. (1995). Ex-day behavior: Tax or short-term trading effects. *Journal of Finance* 50, 875–897.
- Lau, S., L. Ng, and B. Zhang (2012). Information environment and equity risk premium volatility around the world. *Management Science* 58, 1322–1340.
- Lee, C. M. and B. Radhakrishna (2000). Inferring investor behavior: Evidence from TORQ data. *Journal of Financial Markets* 3, 83–111.
- Lee, C. M. and M. J. Ready (1991). Inferring trade direction from intraday data. *Journal of Finance* 46, 733–746.
- Lewbel, A. (2019). The identification zoo: Meanings of identification in econometrics. *Journal of Economic Literature* 57, 835–903.
- Liu, J. and F. A. Longstaff (2004). Losing money on arbitrage: Optimal dynamic portfolio choice in markets with arbitrage opportunities. *Review of Financial Studies* 17, 611–641.
- Madhavan, A. and G. Sofianos (1998). An empirical analysis of NYSE specialist trading. *Journal of Financial Economics* 48, 189 – 210.
- Marshall, B. R., N. H. Nguyen, and N. Visaltanachoti (2011). Commodity Liquidity Measurement and Transaction Costs. *Review of Financial Studies* 25, 599–638.
- Mitchell, M., T. Pulvino, and E. Stafford (2002). Limited Arbitrage in Equity Markets. *Journal of Finance* 57, 551–584.
- Moulton, P. C. and L. Wei (2009). A tale of two time zones: The impact of substitutes on cross-listed stock liquidity. *Journal of Financial Markets* 12, 570–591.
- Odders-White, E. R. (2000). On the occurrence and consequences of inaccurate trade classification. *Journal of Financial Markets* 3, 259–286.
- O’Hara, M. and G. S. Oldfield (1986). The microeconomics of market making. *Journal of Financial and Quantitative Analysis* 21, 361–376.
- Oster, E. (2019). Unobservable selection and coefficient stability: Theory and evidence. *Journal of Business & Economic Statistics* 37(2), 187–204.
- Roll, R. W., E. S. Schwartz, and A. Subrahmanyam (2007). Liquidity and The Law of One Price: The Case of the Cash/Futures Basis. *Journal of Finance* 62, 2201–2234.
- Rösch, D. M., A. Subrahmanyam, and M. A. van Dijk (2017). The Dynamics of Market Efficiency. *Review of Financial Studies* 30, 1151–1187.
- Schultz, P. and S. Shive (2010). Mispricing of dual-class shares: Profit opportunities, arbitrage, and trading. *Journal of Financial Economics* 98, 524–549.

- SEC F-6 (2001). Registration Statement Under the Securities Act of 1933 for Depositary Shares Evidenced by American Depositary Receipts. <https://www.sec.gov/files/formf-6.pdf> [Online; accessed 04-March-2019].
- Shen, P. and R. M. Starr (2002). Market-makers' supply and pricing of financial market liquidity. *Economics Letters* 76, 53 – 58.
- Shleifer, A. and R. W. Vishny (1997). The Limits of Arbitrage. *Journal of Finance* 52, 35–55.
- Theissen, E. (2001). A test of the accuracy of the lee/ready trade classification algorithm. *Journal of International Financial Markets, Institutions and Money* 11, 147 – 165.
- Tomio, D. (2017). Arbitraging Liquidity. *working paper, Darden School of Business*.
- Tuckman, B. and J.-L. Vila (1992). Arbitrage with holding costs: A utility-based approach. *Journal of Finance* 47, 1283–1302.
- Vodafone (2006). Deposit agreement. <https://www.sec.gov/Archives/edgar/data/839923/000101915506000120/vodafonepagmtexab.htm> [Online; accessed 11-September-2018].
- Werner, I. M. and A. W. Kleidon (1996). UK and US trading of British cross-listed stocks: An intraday analysis of market integration. *Review of Financial Studies* 9, 619–664.

Table 1

Cross-sectional summary statistics of time-series averages, 2001 - 2016

This table reports cross-sectional averages, standard deviations, and 5, 50, and 95 percentiles of minute-by-minute (Panel A) and daily (Panels B and C) time-series averages by stock. Panel A reports the average 1-minute maximum price deviation, computed for every stock as the tick-by-tick difference in the highest bid and the lowest ask price across the home- and host-market relative to the mid-quote price of the home market ($|\Delta Price|$, as in Eq. 1); average proportional quoted spread ($PQSPR$); average proportional effective spread ($PESPR$); and absolute order imbalance (the number of buyer minus seller initiated trades) for single ($|OIB|_{single}$) and pair trades ($|OIB|_{pair}$). Single trades for the host (home) market are trades without a simultaneous trade in the home (host) market. Pair trades are simultaneous trades of both the host- and the home-market share with opposite signs. Panel B reports the average number of minutes for a price deviation to persist, conditional on price deviations that arise and vanish within the same day ($INARB$). And the average daily ratio of *pair* to *single* trades estimated using all days on which both the host (the ADR) and the home market stock are either cum- or ex-dividend (*outside ex-days*) and between corporate actions (*between ex-days*), i.e., when either the host or the home-market is cum-dividend but the other is ex-dividend. # *between ex-days*, provides cross-sectional summary statistics of the total number of days between corporate actions. Panel C reports cross-sectional summary statistics of daily liquidity and control variables, the daily average of the absolute pricing error (*Hasbrouck*, following the estimation of Hasbrouck, 1993), volatility, defined as the annualized volatility calculated from 5-minute mid-quote returns (*Volatility*), average trade size (*Trsize*), number of trades (*Trades*), number of quote updates (*Quotes*), the absolute order imbalance (the number of buyer minus seller initiated trades) (*OIB*), *size* calculated as the market cap for each stock at the beginning of each year, and *price* calculated as the end-of-day price from Datastream. Both *size* and *price* are in USD. All variables (except *price*, and *size*) are measured during the overlapping trading times only, i.e. when both the home- and host-market share are trading. All variables are scaled as denoted in the table. All price deviation and spread measures are cross-sectionally winsorized each day at the 1% and 99% level. Data to compute *size* and *price* is from Datastream all other data underlying the computations is from TRTH.

		avg	stddev	p5	median (p50)	p95
Panel A: 1-minute data						
$max(\Delta Price)$ [%]		0.80	2.16	0.03	0.21	5.74
PQSPR [%]	Home	0.57	1.03	0.06	0.20	2.39
	Host	0.66	1.07	0.05	0.23	2.70
PESPR [%]	Home	0.27	0.28	0.05	0.15	0.94
	Host	0.29	0.34	0.03	0.13	1.01
$ OIB _{single}$	Home	5.35	5.89	0.06	3.65	16.78
	Host	4.68	7.73	0.03	1.56	16.64
$ OIB _{pair}$		0.80	1.23	0.0002	0.21	3.11
Panel B: Daily Arbitrage						
$INARB$ [minutes]		12.41	18.54	0.51	6.41	39.27
<i>Pair/Single</i> outside ex-days[%]		7.12	10.95	0.01	4.05	25.60
<i>Pair/Single</i> between ex-days[%]		3.21	5.94	0.00	1.01	13.43
# between ex-days		21	28	1	11	74

Table 1 continued

		avg	stddev	p5	median (p50)	p95
Panel C: Daily Liquidity and control variables						
Hasbrouck [%]	Home	0.12	0.24	0.01	0.04	0.49
	Host	0.12	0.22	0.01	0.03	0.65
Volatility	Home	0.23	0.08	0.13	0.19	0.38
	Host	0.22	0.08	0.12	0.19	0.35
Trsize [ten thousands]	Home	0.32	0.49	0.02	0.17	1.12
	Host	0.04	0.02	0.02	0.04	0.08
Trades [ten thousands]	Home	0.16	0.21	0.002	0.09	0.51
	Host	0.23	0.58	0.001	0.06	0.82
Quotes [ten thousands]	Home	0.18	0.18	0.004	0.15	0.48
	Host	0.28	0.31	0.01	0.19	0.88
OIB [thousands]	Home	0.20	0.29	0.01	0.12	0.83
	Host	0.17	0.29	0.003	0.08	0.65
Size [billion USD]		41.30	234.96	0.29	7.91	101.05
Price [USD]		42.34	258.63	0.55	8.81	67.63

Table 2

Average number of daily price deviations and reasons for why they arise, 2001 - 2016

This table presents the total number of price deviations (*# Price deviations*) by the asset that moves to create the deviation (*First mover*) and by the asset that moves to eliminate it (*Last mover*). The first column indicates the asset that moves to create the price deviation: either the home-market share (*Home*), the host-market share (*Host*), both the home- and the host-market share (*Both*), or the respective currency pair (*Forex*). The second column (*#Price deviations*) indicates the total number of price deviations across all stocks and days in this category. The third column (*%Toxic*) indicates the percentage of all price deviations that are toxic, i.e., when one share moves to create the price deviation and later the other moves back to eliminate it (for example, if the *Home*-market share is the first mover *%Toxic* is defined as the number of price deviations starting in the *Home*-market and ending in the *Host* market as a percentage of all price deviations). The rest of the columns, *Home*, *Host*, *Both*, and *Forex*, indicate the percentage of all price deviations that get eliminated because of a movement in the respective asset. All data underlying the computations are from TRTH.

First mover	#Price deviations	%Toxic	Last mover:			
			%Home	%Host	%Both	%Forex
<i>Home</i>	4,092,945	6.50	44.65	23.90	16.39	15.06
<i>Host</i>	4,235,914	7.03	21.33	47.09	16.12	15.47
<i>Both</i>	2,484,622		27.36	29.35	29.74	13.55
<i>Forex</i>	3,100,353		20.62	22.54	13.03	43.81

Table 3

Order imbalances during price deviations, 2001 - 2016

This table reports results from stock-day fixed effect panel regressions to estimate the sensitivity of order imbalances (OIB) from *single* and *pair* trades to mispricings:

$$OIB_{i,d,j} = FE_{i,d} + \beta_0 InOverlap_{i,d,j} + (\beta_1 + \gamma_1 APeak_{i,d,j}) Costly_{i,d,j} + (\beta_2 + \gamma_2 APeak_{i,d,j}) Cheap_{i,d,j} + \epsilon_{i,d,j}$$

with $OIB_{i,d,j}$ the order imbalance—the number of seller- minus buyer-initiated trades—for *single* or *pair* trades of stock i (host- or home-market) on day d and j one the following periods: outside overlapping trading times, or during overlapping trading times and stock i is fairly priced, *Costly*, or *Cheap* (see Eq. 2). When the stock is *Costly* or *Cheap* I further distinguish between whether the stock is before ($APeak = 0$) or after ($APeak = 1$) the peak in the price gap. Single trades for the host (home) market are trades without a simultaneous trade in the home (host) market. Pair trades are simultaneous trades of both the host- and the home-market share with opposite signs. $OIB_{i,d,j}$ is explained by four indicator variables ($InOverlap_{i,d,j}$, $Costly_{i,d,j}$, $Cheap_{i,d,j}$, and $APeak_{i,d,j}$) set equal to one if j falls within the respective period and zero otherwise. Whether a stock is *Costly* or *Cheap* can only be determined when $InOverlap_{i,d,j}$ is equal to one. Whether $APeak$ is equal to one can only be determined when $Costly_{i,d,j}$ or $Cheap_{i,d,j}$ is equal to one and hence I can not include $APeak$ by itself as an explanatory variable. $FE_{i,d}$ are two dimensional stock-day fixed effects. Standard errors are clustered by stock-pair, and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. p -values are in parentheses below the coefficients. The sample only contains days on which the stock is in between corporate actions or trades fairly priced, *Cheap*, and *Costly*. All data underlying the computations are from TRTH.

	single trades		pair trades	
	OIB_{home}	OIB_{host}	OIB_{home}	OIB_{host}
<i>Costly</i>	3.68 (0.11)	8.32*** (0.00)	-7.74*** (0.00)	-4.37*** (0.00)
× <i>APeak</i>	-26.5*** (0.00)	-28.4*** (0.00)	-3.14*** (0.00)	-4.90*** (0.00)
<i>Cheap</i>	-3.84** (0.02)	-23.5*** (0.00)	4.18*** (0.00)	7.93*** (0.00)
× <i>APeak</i>	26.1*** (0.00)	39.07*** (0.00)	5.5*** (0.00)	2.7*** (0.00)
<i>InOverlap</i>	8.89** (0.02)	-7.14*** (0.00)		
<i>StockDayFE</i>	Yes	Yes	Yes	Yes
Adj. R^2 [%]	14.73	9.92	16.98	17.11
Observations	928,079	928,079	768,570	768,570

Table 4

Liquidity at and after single and pair trades, 2001 - 2016

This table presents results of the following fixed-effect panel regressions.

$$Illi_{i,d,j} = FE + \beta_1 \times Pair_{i,d,j} + \epsilon_{i,d}$$

with $Illi_{i,d,j}$ quoted spreads, effective spreads, or the logarithm of the absolute Hasbrouck (1993) pricing error associated with *pair* ($PairTrade = 1$) or *single* ($PairTrade = 0$) trades. For each trade I estimate spreads and pricing errors associated with the trade ($PQSPR_{i,t}$, $PESPR_{i,t}$, and $\ln|Has|_{i,t}$) and associated with the next trade ($PQSPR_{i,t+1}$, $PESPR_{i,t+1}$, $\ln|Has|_{i,t+1}$). I then average spreads and pricing errors within each stock-day, separately for *pair* and *single* trades. $FE_{i,d}$ are two-dimensional stock-day fixed effects. I drop stock-days without any pair trade. $PQSPR$ and $PESPR$ are measured in per cent. Standard errors are clustered by stock-pair, and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. p -values are in parentheses below the coefficients. All data underlying the computations are from TRTH.

	$PQSPR_{i,t}$	$PQSPR_{i,t+1}$	$PESPR_{i,t}$	$PESPR_{i,t+1}$	$\ln Has_{i,t} $	$\ln Has_{i,t+1} $
$PairTrade_{i,t}$	-0.025*** (0.00)	-0.057*** (0.00)	-0.016*** (0.00)	-0.005*** (0.00)	-0.079*** (0.00)	-0.079*** (0.00)
$Stock \times Day$ FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2 [%]	5.41	2.96	11.19	19.29	94.87	94.77
Observations	1,164,020	1,164,020	1,164,020	1,164,020	1,150,588	1,150,154

Table 5Intraday Vector autoregressions and Granger causality (*PQSPR*), 2001 - 2016

This table reports results from panel vector autoregressions using two-dimensional stock-day and exchange-hour fixed effects. Panels A and B report vector autoregression results estimated using absolute order imbalance from single trades and from pair trades, price deviations, and proportional quoted spreads as endogenous variables with five lags.

$$LHS_{i,t} = FE + \sum_{l=1}^5 \beta_{1,l} |OIB|_{single,i,t-l} + \sum_{l=1}^5 \beta_{2,l} |\Delta Price|_{i,t-l} + \sum_{l=1}^5 \beta_{3,l} |OIB|_{pair,i,t-l} + \sum_{l=1}^5 \beta_{4,l} PQSPR_{i,t-l} + \epsilon_{i,t}$$

where $|OIB|_{single}$ and $|OIB|_{pair}$ is estimated as the logarithm of 1 plus the absolute difference in buyer- and seller-initiated single and pair trades during 1-minute interval t , respectively; in Panel A $|\Delta Price|$ is the maximum price deviation within interval t (from Eq. 1); in Panel B $|\Delta Price|$ is the maximum price deviation within interval t (from Eq. 1) if it is above 1 standard deviation of the stock-day average and zero otherwise; and $PQSPR$ is average proportional quoted spread within interval t . Panel C reports results explaining mid-point returns by lagged mid-point returns, $|OIB|_{single}$, $|OIB|_{pair}$, and interaction terms. All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. For a detailed description of these variables I refer to Table 1. In all cases the table reports the sum of all lagged variables for each endogenous variable. p -values are in parentheses below the coefficients, based on a χ^2 test for testing the null hypothesis that the sum of the lagged coefficients is equal to zero when explaining the column variable. The next row reports the p -value of testing joined significance of the lagged values. Standard errors are clustered by stock-pair, and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. All data underlying the computations are from TRTH.

	Panel A: <i>PQSPR</i>				Panel C:
	$PQSPR_{i,t}$	$ \Delta Price _{i,t}$	$ OIB _{single,i,t}$	$ OIB _{pair,i,t}$	$return_{i,t}$
$\sum_{l=1}^5 OIB _{pair,i,t-l} \times return_{i,t-l}$					0.075*** (0.00) (0.00)
$\sum_{l=1}^5 OIB _{pair,i,t-l}$	-0.0011*** (0.00) (0.00)	-0.015*** (0.00) (0.00)	0.076*** (0.00) (0.00)	0.252*** (0.00) (0.00)	0.000*** (0.00) (0.00)
$\sum_{l=1}^5 OIB _{single,i,t-l}$	0.0036*** (0.00) (0.00)	0.035*** (0.00) (0.00)	0.292*** (0.00) (0.00)	0.054*** (0.00) (0.00)	0.000*** (0.00) (0.00)
$\sum_{l=1}^5 \Delta Price _{i,t-l}$	-0.0002*** (0.00) (0.00)	0.699*** (0.00) (0.00)	0.001*** (0.00) (0.00)	0.000 (0.66) (0.00)	0.000 (0.26) (0.29)
$\sum_{l=1}^5 PQSPR_{i,t-l}$	0.949*** (0.00) (0.00)	-0.012*** (0.00) (0.00)	-0.023*** (0.00) (0.00)	-0.002*** (0.00) (0.00)	
$\sum_{l=1}^5 return_{i,t-l}$					-0.303*** (0.00) (0.00)
<i>Stock</i> $\times$ <i>Day</i> FE	Yes	Yes	Yes	Yes	Yes
<i>Exchange</i> $\times$ <i>Hour</i> FE	Yes	Yes	Yes	Yes	Yes
Adj. R^2 [%]	98.02	89.23	60.54	35.13	2.31
Observations	155,759,235	155,759,235	155,759,235	155,759,235	154,908,431

Table 5 continued

	Panel B: large $ \Delta Price $			
	$PQSPR_{i,t}$	$ \Delta Price _{i,t}$	$ OIB _{single,i,t}$	$ OIB _{pair,i,t}$
$\sum_{l=1}^5 OIB _{pair,i,t-l}$	-0.0011*** (0.00) (0.00)	-0.0035*** (0.00) (0.00)	0.076*** (0.00) (0.00)	0.25*** (0.00) (0.00)
$\sum_{l=1}^5 OIB _{single,i,t-l}$	0.0036*** (0.00) (0.00)	0.0078*** (0.00) (0.00)	0.29*** (0.00) (0.00)	0.054*** (0.00) (0.00)
$\sum_{l=1}^5 \Delta Price _{i,t-l}$	-0.00028*** (0.00) (0.00)	0.72*** (0.00) (0.00)	0.0011*** (0.00) (0.00)	0.00032** (0.03) (0.00)
$\sum_{l=1}^5 PQSPR_{i,t-l}$	0.95*** (0.00) (0.00)	-0.0072*** (0.00) (0.00)	-0.023*** (0.00) (0.00)	-0.0019*** (0.00) (0.00)
<i>Stock</i> $\times$ <i>Day</i> FE	Yes	Yes	Yes	Yes
<i>Exchange</i> $\times$ <i>Hour</i> FE	Yes	Yes	Yes	Yes
Adj. R^2 [%]	98.02	70.68	60.54	35.13
Observations	155,759,235	155,759,235	155,759,235	155,759,235

Table 6Intraday Vector autoregressions and Granger causality (*PESPR*), 2001 - 2016Compared to Panel A of Table 5 I use effective spread (*PESPR*) and use quoted spread (*PQSPR*) only in case of missing trades within any interval.

This table reports results from a panel vector autoregression using two dimensional stock-day and exchange-hour fixed effects. The vector autoregression is estimated using absolute order imbalance from single trades and from pair trades, price deviations, and proportional effective spreads as endogeneous variables with five lags.

$$LHS_{i,t} = FE + \sum_{l=1}^5 \beta_{1,l} |OIB|_{single,i,t-l} + \sum_{l=1}^5 \beta_{2,l} |\Delta Price|_{i,t-l} + \sum_{l=1}^5 \beta_{3,l} |OIB|_{pair,i,t-l} + \sum_{l=1}^5 \beta_{4,l} PESPR_{i,t-l} + \epsilon_{i,t}$$

where $|OIB|_{single}$ and $|OIB|_{pair}$ is estimated as the logarithm of 1 plus the absolute difference in buyer- and seller-initiated single and pair trades during 1-minute interval t , respectively; $|\Delta Price|$ is the maximum price deviation within interval t (from Eq. 1); and *PESPR* is average proportional effective spread within interval t . All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. For a detailed description of these variables I refer to Table 1. In all cases the table reports the sum of all lagged variables for each endogenous variable. p -values are in parentheses below the coefficients, based on a χ^2 test for testing the null hypothesis that the sum of the lagged coefficients is equal to zero when explaining the column variable. The next row reports the p -value of testing joined significance of the lagged values. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. All data underlying the computations are from TRTH.

	$PESPR_{i,t}$	$ \Delta Price _{i,t}$	$ OIB _{single,i,t}$	$ OIB _{pair,i,t}$
$\sum_{l=1}^5 OIB _{pair,i,t-l}$	-0.0042*** (0.00) (0.00)	-0.015*** (0.00) (0.00)	0.076*** (0.00) (0.00)	0.252*** (0.00) (0.00)
$\sum_{l=1}^5 OIB _{single,i,t-l}$	0.011*** (0.00) (0.00)	0.035*** (0.00) (0.00)	0.29*** (0.00) (0.00)	0.054*** (0.00) (0.00)
$\sum_{l=1}^5 \Delta Price _{i,t-l}$	0.00013 (0.16) (0.00)	0.699*** (0.00) (0.00)	0.001*** (0.00) (0.00)	-0.000018 (0.74) (0.00)
$\sum_{l=1}^5 PESPR_{i,t-l}$	0.936*** (0.00) (0.00)	-0.015*** (0.00) (0.00)	-0.02*** (0.00) (0.00)	-0.0012*** (0.00) (0.00)
<i>Stock</i> $\times$ <i>Day</i> FE	Yes	Yes	Yes	Yes
<i>Exchange</i> $\times$ <i>Hour</i> FE	Yes	Yes	Yes	Yes
Adj. R^2 [%]	96.88	89.23	60.53	35.13
Observations	155,759,235	155,759,235	155,759,235	155,759,235

Table 7

Illiquidity during days between corporate actions, 2001 - 2016

This table presents results of the following fixed-effect panel regressions to explain how liquidity varies on days between corporate actions, i.e., when either the host or the home-market is cum-dividend but the other is ex-dividend ($Between_{i,d}$).

$$Illiq_{i,d} = FE + \beta \times Between_{i,d} + \zeta \times \mathbf{Controls}_{i,d} + \epsilon_{i,d}$$

where illiquidity ($Illiq_{i,d}$) of home- or host-market stock i on day d is measured by the average proportional quoted spread ($PQSPR$), proportional effective spread ($PESPR$), or the logarithm of the average absolute pricing error ($\ln|Has|$) immediately after *pair* trades. Averages are taken only after trades that occur when the stock is mispriced (i.e., when the stock is *Cheap*, *Costly*, or in between corporate actions) and days with less than ten *pair* trades are dropped from the analysis. And where **Controls** is a vector of control variables: *CorpAct* is a dummy variable set to one on days on which both the home and the host-market go ex-dividend; *OIB* is the order imbalance, measured as the difference between the number of buyer and seller initiated trades; *TED* is the Ted spread, the difference between 3-Month USD LIBOR and Treasury Bills; and the other variables are as defined in Table 1. All stock specific variables (except *price*) are measured during the overlapping trading time, i.e. when both the home- and host-market are in their continuous trading session. Standard errors are clustered by stock-pair, and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. *p*-values are in parentheses below the coefficients. Data to compute *price* are from Datastream, *Ted* is from the Federal Reserve Bank of St. Louis, and all other data underlying the computations are from TRTH.

	$PQSPR_{i,d}$		$PESPR_{i,d}$		$\ln Has _{i,d}$	
$Between_{i,d}$	0.010*** (0.00)	0.010*** (0.00)	0.008*** (0.00)	0.009*** (0.00)	0.052*** (0.00)	0.058*** (0.00)
$CorpAct_{i,d}$		-0.005 (0.42)		-0.000 (0.82)		-0.009 (0.24)
$\log(price)_{i,d}$		-0.115*** (0.00)		-0.105*** (0.00)		-0.487*** (0.00)
$Vola_{i,d}$		0.123*** (0.00)		0.172*** (0.00)		1.248*** (0.00)
$OIB_{i,d}$		0.000 (0.38)		0.000 (0.97)		-0.000** (0.04)
$Trades_{i,d}$		-0.002** (0.03)		-0.001 (0.23)		-0.204*** (0.00)
$Quotes_{i,d}$		-0.001 (0.28)		-0.001* (0.09)		0.022* (0.09)
$Trsize_{i,d}$		0.002 (0.93)		-0.002 (0.76)		-0.074** (0.03)
Ted_d		0.010*** (0.00)		0.010*** (0.00)		0.080*** (0.00)
$Stock \times Month$ FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2 [%]	12.57	12.51	34.63	34.59	80.47	81.22
Observations	407,684	398,822	407,684	398,822	406,274	397,418

Table 8

Instrumental variable regressions to estimate arbitrage on illiquidity, 2001 - 2016

This table presents the first- (Panel A) and second-stage (Panel B) regressions of instrumenting the ratio of pair to single trades when explaining illiquidity by a dummy variable which is one on days between corporate actions, i.e., when either the host or the home market is cum-dividend but the other is ex-dividend. In Panel B, each column presents the results of the following panel regression with two dimensional stock-month fixed effects:

$$Illiq_{i,d} = FE + \beta_1 \times \overline{Pair/Single}_{i,d} + \zeta_1 \times \mathbf{Controls}_{i,d} + \epsilon_{i,d}$$

where illiquidity ($Illiq_{i,d}$) of home- or host-market stock i on day d is measured by the average proportional quoted spread ($PQSPR$), proportional effective spread ($PESPR$), or the logarithm of the average absolute pricing error ($\ln|Has|$) immediately after *pair* trades. Averages are taken only after trades that occur when the stock is mispriced (i.e., when the stock is *Cheap*, *Costly*, or in between corporate actions) and days with less than ten *pair* trades are dropped from the analysis. $\overline{Pair/Single}_{i,d}$ is the fitted value from first stage regressions reported in Panel A. And **Controls** is a vector of control variables. For a description of these variables I refer to Tables 1 and 7. All stock-specific variables (except *price*) are measured during the overlapping trading time, i.e. when both the home- and host-market are in their continuous trading session. Standard errors are clustered by stock-pair, and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. *p*-values are in parentheses below the coefficients. In Panel B R^2 statistics are not reported, because they cannot be properly interpreted in two stage regressions. Data to compute *price* are from Datastream, *Ted* is from the Federal Reserve Bank of St. Louis, and all other data underlying the computations are from TRTH.

	Panel A: 1st Stage		Panel B: 2nd Stage					
	$\overline{Pair/Single}_{i,d}$		$PQSPR_{i,d}$		$PESPR_{i,d}$		$\ln Has _{i,d}$	
<i>Between</i>	-0.037*** (0.00)	-0.042*** (0.00)						
$\overline{Pair/Single}_{i,d}$			-0.268** (0.02)	-0.244** (0.02)	-0.216** (0.03)	-0.203** (0.03)	-1.392*** (0.00)	-1.369*** (0.00)
$CorpAct_{i,d}$		0.001 (0.85)		-0.004 (0.46)		-0.000 (0.91)		-0.009 (0.35)
$\log(price)_{i,d}$		0.082*** (0.00)		-0.095*** (0.00)		-0.089*** (0.00)		-0.375*** (0.00)
$Vola_{i,d}$		0.747*** (0.00)		0.306*** (0.00)		0.324*** (0.00)		2.277*** (0.00)
$OIB_{i,d}$		0.000** (0.04)		0.000* (0.09)		0.000 (0.13)		0.000 (0.47)
$Trades_{i,d}$		0.091*** (0.01)		0.021* (0.07)		0.018* (0.07)		-0.079** (0.01)
$Quotes_{i,d}$		-0.018* (0.05)		-0.005* (0.09)		-0.005* (0.09)		-0.002 (0.73)
$Trsize_{i,d}$		0.090** (0.02)		0.023 (0.25)		0.016 (0.16)		0.046 (0.31)
Ted_d		-0.011*** (0.00)		0.008*** (0.00)		0.007*** (0.00)		0.066*** (0.00)
<i>Stock</i> × <i>Month</i> FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2 [%]	56.29	57.16	NA	NA	NA	NA	NA	NA
Observations	407,682	398,820	407,682	398,820	407,682	398,820	406,272	397,416

Fig. 1. Intraday Impulse response functions from shocks to *single* and *pair* trades, 2001- 2016

This figure shows cumulative impulse response functions (IRF) from a panel vector autoregression (VAR) estimated like Panel A of Table 5, as described in more detail in the main text. For a description of the VAR and these variables I refer to Table 1 and Table 5. IRFs in the first row show *PQSPR* in response to a Cholesky 1-standard-deviation shock to the logarithm of 1 plus the absolute order imbalances (*OIB*) in minute 0 (the contemporaneous effect) to minute 5, with the left and right figure showing responses to a shock to *OIB* from, respectively, single and pair trades. The second row shows the maximum price deviation within each minute (from Eq. 1, $|\Delta Price|$) in response to a shock to order imbalances. All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. Effects are statistically significant with very tight bootstrapped confidence bands estimated based on 1000 runs (plotted in Figure OA1 of the Online Appendix). All data underlying the computations are from TRTH.

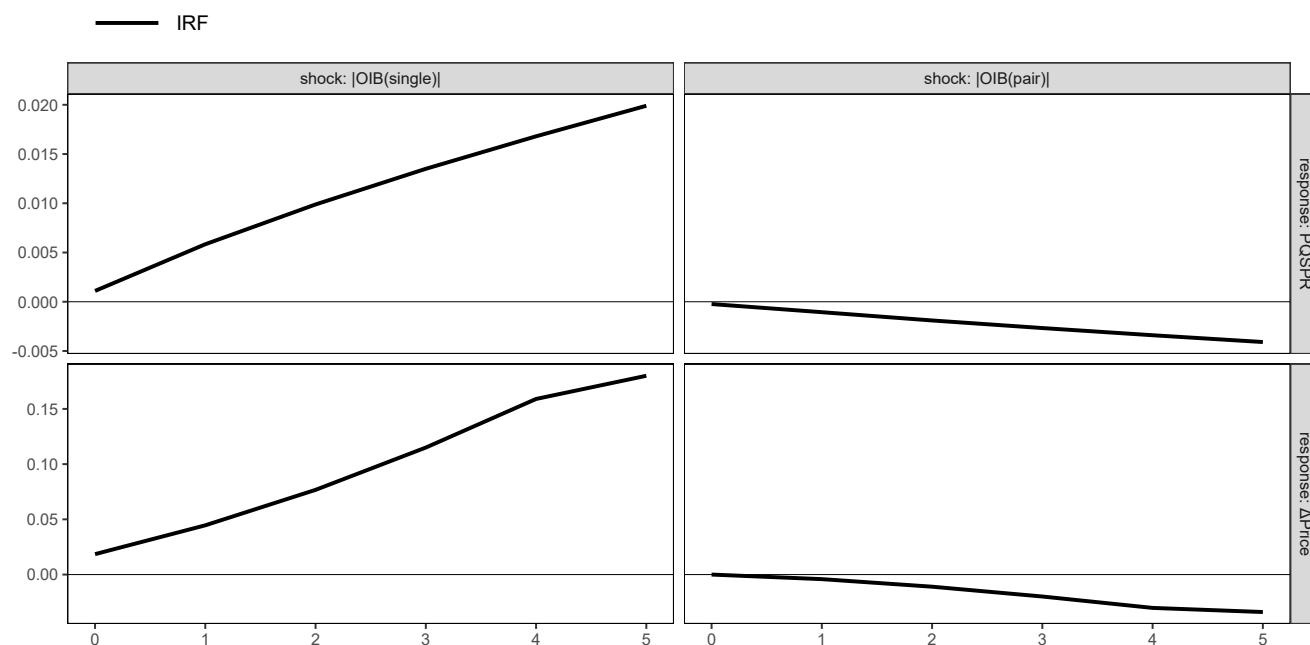


Fig. 2. Intraday Impulse response functions from shocks to *single* and *pair* trades (PESPR) 2001- 2016

This figure shows cumulative impulse response functions (IRF) from a panel vector autoregression (VAR) estimated like Table 6, as described in more detail in the main text. For a description of the VAR and these variables I refer to Table 1 and Table 6. IRFs in the first row show *PESPR* in response to a Cholesky 1-standard-deviation shock to the logarithm of 1 plus the absolute order imbalances (*OIB*) in minute 0 (the contemporaneous effect) to minute 5, with the left and right figure showing responses to a shock to *OIB* from, respectively, single and pair trades. The second row shows the maximum price deviation within each minute (from Eq. 1, $|\Delta Price|$) in response to a shock to order imbalances. All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. Effects are statistically significant with very tight bootstrapped confidence bands estimated based on 1000 runs (plotted in Figure OA1 of the Online Appendix). All data underlying the computations are from TRTH.

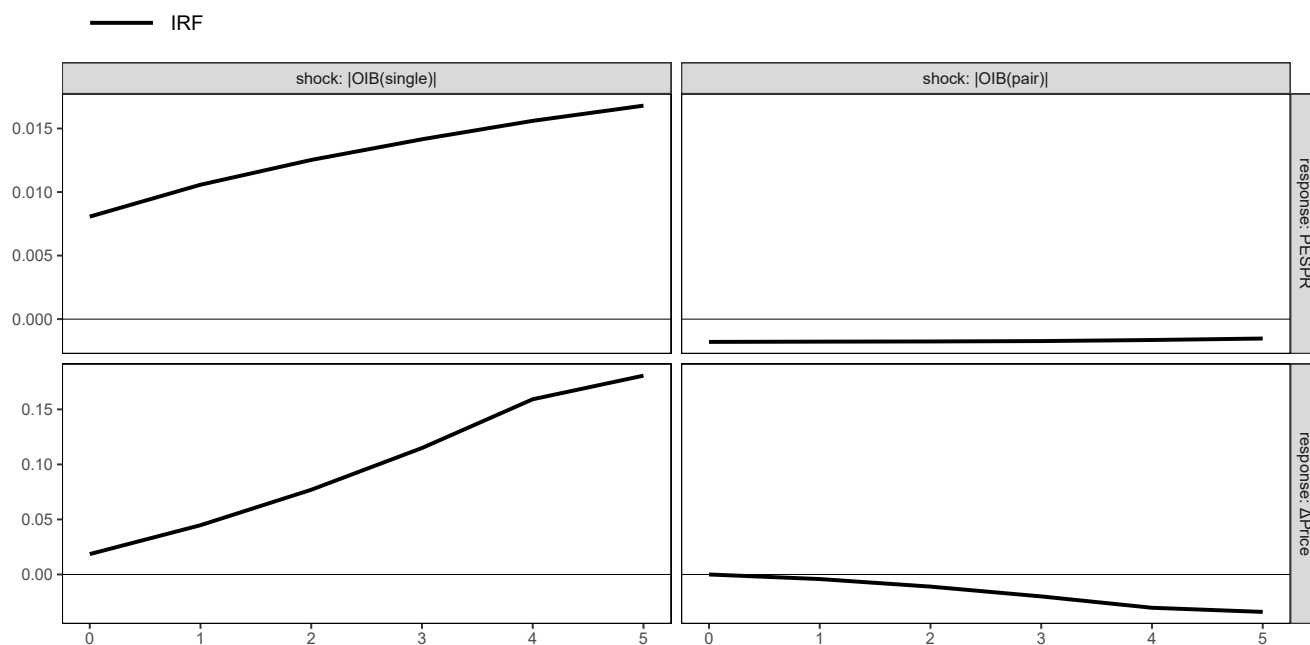
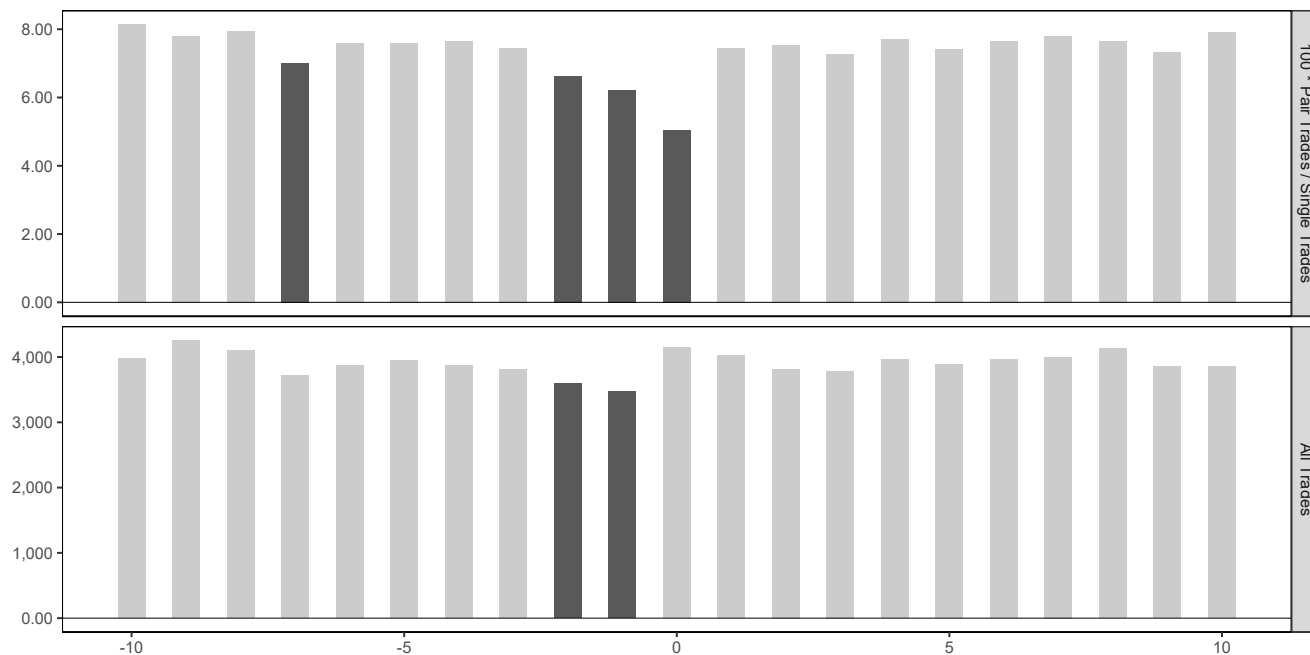


Fig. 3. Trading around days when either the host or the home-market is cum-dividend but the other is not, 2001- 2016

This figure shows the ratio of *pair* trades to *single* trades and the total number of host- and home-market *pair* and *single* trades around days in between corporate actions (*Event*), i.e., when either the host or the home-market is cum-dividend but the other is ex-dividend. *Single* trades for the host (home) market are trades without a simultaneous trade in the home (host) market. *Pair* trades are simultaneous trades of both the host- and the home-market share with opposite signs. The figure shows the cross-sectional average ratio of *pair* to *single* trades and total trades on ten days before to ten days after the event. On the event, the ratio and total trades is first averaged over all days in between corporate actions. Statistical significance at 1% is indicated in bold, of whether the ratio or total trades is different from its normal average (estimated from 41 days before till 11 days before the event). All data underlying the computations are from TRTH.



Online Appendix

accompanying the paper:

“The impact of arbitrage on market liquidity”

This Online Appendix reports several robustness tests. Sections OA1, OA2, and OA3.3 discuss robustness tests regarding the main empirical findings of the paper, namely, that price deviations arise because of non-fundamental demand pressure, that an increase in arbitrage lowers price deviations and spreads in the next minutes, and that an increase in arbitrage activity on a given day improves liquidity.

In the Online Appendix I also report results using alternative ways to measure arbitrage activity, the average number of minutes for a price deviation to persist, conditional on price deviations that arise and vanish within the same day (*INARB*); the maximum price deviation within a day (based on bid and ask prices, mid-prices, or trade prices); and the speed at which price deviations mean revert. These measures are introduced in more detail in Section OA3. Section OA4 lists all robustness tests.

OA1. Additional results investigating why price deviations arise

In addition to providing robustness tests of results reported in Tables 2 and 3 of the paper, I also report results based on abnormal trading. For each host- and home-market stock I compute the whole sample number of trades separately for whether the trade occurred when the stock was fairly priced, *Costly*, or *Cheap*, whether the trade is buyer or seller initiated, and whether the trade occurred before or after the peak in the price gap within each price deviation (to ensure that the size of the trade does not systematically vary across periods, I show that results are robust to only including trades of less than 500 shares or measuring abnormal trading in USD, Tables OA11 and OA12).

For each stock i , I then calculate abnormal trading within each period j as the number of trades in period j (denoted $\sum_d Trades_{i,d,j}$) in excess of when the stock is fairly priced (denoted $\sum_d Trades_{i,d,j,Fair}$), scaled by all trades:

$$ATrades_{i,j} = \frac{\sum_d Trades_{i,d,j}}{\sum_d \sum_j Trades_{i,d,j}} - \frac{\sum_d Trades_{i,d,j,Fair}}{\sum_d \sum_j Trades_{i,d,j}}$$

Panels A, B, and C of Table OA9 report, respectively, the results for abnormal trading of

single, *pair*, and *quote* trades within each category. When the stock is *Costly* and prices deviate, results in Panel A indicate abnormally low seller (-3.98%) and high buyer initiated trading (3.49%) in the host-market. When prices converge, seller initiated trading is statistically indifferent from trading when the stock is fairly priced, but buyer initiated trading switches signs and now indicates abnormally low trading (-3.55%). When the host-market stock is *Costly*, the home-market stock is *Cheap*, and for the home-market the number of buyer initiated trades is abnormally low when prices deviate (-2.58%) but abnormally high when prices converge (4.10%) accompanied by abnormally low seller initiated trades (-8.61%). These results are statistically significant at the 1% level.

Panel B shows that when the stock is *Costly* buyer initiated *pair* trades are either abnormally low or statistically indifferent to trading when the stock is fairly priced, but seller initiated *pair* trades are abnormally high both when prices deviate and converge and both as a percentage of all host trades (1.60% and 2.43%) and home-trades (1.76% and 2.70%). While these percentages are lower than the effect on the percentage of *single* trades, the percentage of *pair* trades as a total of all trades is much lower. For example, when stocks are fairly priced around 80% of all trades are *single* trades and around 8% are *pair* trades (the remaining 12% are simultaneous trades but not of opposite signs), when stocks are fairly priced around 50% of all *single* and *pair* trades are buyer initiated and 50% are seller initiated. Results in Table OA9 indicate, that when prices deviate and the stock is *Costly* the percentage of seller initiated *pair* trades in the host-market increases from around 4% to 5.6% ($= 4\% + 1.6\%$), i.e., almost by one-half.

These results provide further evidence that price deviations arise due to abnormal trading activity in one direction visible in *single* trades; that prices converge both due to *single* trades and due to *pair* trades in both the host- and home-market; and that *pair* trades are against the price deviation both when prices deviate and when prices converge.

OA2. Additional results estimating intraday dynamics

Figure OA1 reports all impulse response functions (IRFs) instead of reporting just the ones of main interest, as in Figure 1 of the paper. If price deviations arise because of non-fundamental

demand pressure, high price deviations and high spreads should be similar affected by arbitrage. I therefore also estimate VARs and IRFs separately using only three endogeneous variables, OIB_{single} , $Liquidity$, and OIB_{pair} , with $Liquidity$ either $|\Delta Price|$ or $PQSPR$, see Figures OA2 and OA3, respectively.

These results are robust to various other design choices, as listed in Section OA4. For these additional tests, I only report IRFs if the VAR substantially differs from the VAR reported in the paper.

OA3. Robustness to alternative measures for arbitrage activity

In this section I use other commonly used proxies for arbitrage activity, show that these measures are highly correlated with the ratio of pair-to-single trades, and that previous results are robust to using any of these other measures.

OA3.1. Alternative measures for arbitrage activity

As in Section 2.6 of the paper I estimate price-deviations from bid and ask quote prices, I then consider the maximum deviation within each stock-day, denoted ΔQTE , as a measure for the impediments to arbitrage.

Large price deviations signal high impediments to arbitrage and low arbitrage activity. If the market is reasonably efficient so that “prices reflect information to the point where the marginal benefits of acting ... do not exceed the marginal costs” (Fama, 1991, p. 1575), price deviations reflect underlying frictions that impede arbitrage, such as short-selling restrictions, risk, or capital constraints. Such an interpretation is common in the literature and several studies provide empirical justification for it.¹⁸

Previous literature suggest an alternative proxy for the impediments to arbitrage: the speed at which price deviations revert back to their long-term mean (Roll et al., 2007).

¹⁸ Among others, De Jong et al. (2009); Gagnon and Karolyi (2010); Hu, Pan, and Wang (2013); Lamont and Thaler (2003); Mitchell et al. (2002); Roll et al. (2007).

$$dX_t = \theta(\mu - X_t)dt + \sigma dW_t \quad (6)$$

To estimate the speed of mean-reversion, θ , from an Ornstein-Uhlenbeck process given in Equation 6, I estimate Equation 7 per stock-pair-day using ordinary least square:

$$\Delta X_{i,d,s} = \alpha_{i,d} + \beta_{i,d} X_{i,d,s-1} + \epsilon_{i,d,s} \quad (7)$$

with $X_{i,d,s}$ the signed, dollar price deviation for stock-ADR pair i , on day d , in second s , i.e., $X_{i,d,s} = mid.adr_{i,d,s} - mid.home_{i,d,s}$, with *mid* the mid-point price as in Eq 1. Standard theory implies that the speed of mean reversion, θ , is given by $-\beta$.

OA3.2. Correlations

Table OA5 reports pairwise pooled Spearman rank correlations between the ratio of *pair* to *single* trades (*Pair/Single*), the maximum price-deviation within each day ($max(\Delta QTE)$), the speed at which price deviations revert back to their long-term mean (θ), and quoted and effective spreads.

Quoted and effective spreads are strongly correlated with each other. Correlations are also strong between measures of arbitrage activity and spreads. The strong positive correlation between price deviations in quotes and quoted spreads is surprising. Mechanically, an increase in quoted spreads in either the home market share or the ADR would lower price deviations. But the finding supports the argument that these price deviations measure impediments to arbitrage: higher illiquidity should be associated with higher impediments to arbitrage.

Correlation between *Pair/Single* and $max(\Delta QTE)$ is -42%, between *Pair/Single* and θ is 55%, and between $max(\Delta QTE)$ and θ is -30%. All correlations are statistically significant at the 1% level. Because θ is an inverse measure for the impediments to arbitrage these correlations indicate that impediments to arbitrage are negatively correlated to a more direct measure of arbitrage activity (*Pair/Single*).

OA3.3. Additional results estimating IVs

Tables OA36 and OA37 show that results reported in Table 8 of the paper are robust to using alternative measures of arbitrage activity, namely the maximum price deviation within a day and the speed at which price deviations mean-revert.

OA4. List of all additional results

- The following three tables adopt the analysis from Foucault et al. (2017) to the ADR market.
 - Table OA1 shows the effect of arbitrageurs' relative speed and arbitrage mix on illiquidity.
 - Table OA2 shows the effect of arbitrageurs' relative speed and arbitrage mix on illiquidity separately for home-market stocks and ADRs.
 - Table OA3 estimates the effect of arbitrageurs' relative speed on illiquidity using an instrument variable regression.
- Table OA4 shows summary statistics by home-market.
- Table OA5 shows correlations across measures of arbitrage activity and liquidity.
- The following show robustness tests to why deviations arise (Section 3 in the paper)
 - Table OA6 shows that the results reported in Table 2 of the paper are robust to only using the largest price deviation from each stock-day.
 - Table OA7 shows that results reported in Table 3 of the paper are robust to including stock-days even if the stock is fairly priced, *Costly*, or *Cheap* the whole day
 - Table OA8 shows whether OIB sensitivity to mispricings (similar to Table 3 of the paper) is affected by impediments to arbitrage
 - Table OA9 investigates abnormal buyer and seller, host- and home-market trading when prices are *Cheap* or *Costly* and prices deviate or converge.

- * Table OA10 shows that results in Table OA9 are robust to measuring simultaneous trades as trades within one second.
 - * Table OA11 shows that results in Table OA9 are robust to only including trades of less than 500 shares.
 - * Table OA12 shows that results in Table OA9 are robust to measuring abnormal trading in USD.
- The following tables and figures show robustness tests related to Section 4 of the paper. I first compare results using Vector Autoregressions (VARs) and Granger causality. If these tests deviate from these reported in the paper I also estimate Impulse Response Functions (IRFs).
 - Table OA13 estimates spreads and pricing errors around *pair* and *single* trades only at times when the stock is *Costly* or *Cheap*.
 - Figure OA1 reports IRFs as Figure 1 of the paper, but reports all results
 - Figure OA2 reports IRFs as Figure 1 of the paper, but the VAR is estimated using only order imbalances from *single* and *pair* trades and price deviations
 - Figure OA3 reports IRFs as Figure 1 of the paper, but the VAR is estimated using only order imbalances from *single* and *pair* trades and quoted spreads
 - Table OA15 shows VAR estimated with volatility
 - Table OA16 shows VAR estimated with 10 lags
 - Table OA17 shows VAR estimated setting price deviations to zero if they are not above 0.5 standard deviations of the stock-day average.
 - Table OA18 shows VAR estimated with midpoint price deviations
 - Table OA19 shows VAR estimated with an indicator whether price deviations increased
 - Table OA20 shows VAR estimated using PESPR and using only stock-days on which I can estimate effective spread (*PESPR*) for at least 8 out of 10 intervals.

- Table OA21 shows VAR estimated using effective quoted spreads (*PESPR*) and replacing missing *PESPR* observations with lagged *PESPR*
- Table OA22 shows VAR estimated using *PESPR* and 5-minute intervals.
- Table OA23 shows VAR estimated with data only after 2007
- Table OA24 shows VAR estimated with data only after 2011
- Table OA25 shows VAR estimated with data using days with at least 100 trades
- Table OA26 shows VAR estimated from 5-minute intervals and Figure OA4 shows the corresponding IRFs
- Table OA27 shows VAR estimated with using only home-market stocks
- Table OA28 shows VAR estimated with using only host-market stocks
- Table OA29 shows VAR estimated with using quoted spreads and *single* trades from both home- and host-market stocks simultaneously as endogenous variables
- Table OA30 shows VAR estimated with using the number of trades
- Table OA31 shows VAR estimated with using order imbalances in USD and Figure OA5 shows corresponding IRFs.
- Table OA32 shows VAR estimated with using price improvements and Figure OA6 shows corresponding IRFs.
- The following tables show robustness tests related to Section 5 of the paper.
 - Table OA33 investigates trading around days between corporate actions using event-fixed effect panel regressions.
 - Tables OA34 and OA35 show that results of Table 8 of the paper are robust to using stock-days with, respectively, at least five and at least twenty pair trades.
 - Table OA36 shows that results in Table 8 of the paper are robust to using the daily maximum price deviations as an inverse measure of arbitrage activity.

- Table OA37 shows that results in Table 8 of the paper are robust to using the speed of mean-reversion as an alternative measure of arbitrage activity.

- [OA14] De Jong, A., L. Rosenthal, and M. van Dijk (2009). The Risk and Return of Arbitrage in Dual-Listed Companies. *Review of Finance* 13, 495–520.
- [OA2] Fama, E. F. (1991). Efficient capital markets: II. *Journal of Finance* 46, 1575–1617.
- [OA18] Foucault, T., R. Kozhan, and W. W. Tham (2017). Toxic Arbitrage. *Review of Financial Studies* 30, 1053–1094.
- [OA21] Gagnon, L. and A. G. Karolyi (2010). Multi-market trading and arbitrage. *Journal of Financial Economics* 97, 53–80.
- [OA5] Hu, G. X., J. Pan, and J. Wang (2013). Noise as Information for Illiquidity. *Journal of Finance* 68, 2341–2382.
- [OA34] Lamont, O. a. and R. H. Thaler (2003). Can the Market Add and Subtract? Mispricing in Tech Stock Carve-outs. *Journal of Political Economy* 111, 227–268.
- [OA43] Mitchell, M., T. Pulvino, and E. Stafford (2002). Limited Arbitrage in Equity Markets. *Journal of Finance* 57, 551–584.
- [OA48] Roll, R. W., E. S. Schwartz, and A. Subrahmanyam (2007). Liquidity and The Law of One Price: The Case of the Cash/Futures Basis. *Journal of Finance* 62, 2201–2234.

Table OA1

Arbitrageurs' relative speed, arbitrage mix and liquidity in the ADR market, 2001 - 2016

This table presents results of regressions explaining illiquidity by arbitrageurs' relative speed (π), arbitrage mix (ϕ), and price deviations (σ). Each column presents the results of the following panel regression with two dimensional stock-month fixed effects:

$$Illiq_{i,d} = FE + a_1\pi_{i,d} + a_2\phi_{i,d} + a_3\alpha_{i,d} + a_4\sigma_{i,d} + \zeta \times \mathbf{Controls}_{i,d} + \epsilon_{i,d}$$

where illiquidity ($Illiq_{i,d}$) of home- or host-market stock i on day d is measured by the proportional quoted spread (Panel A: $PQSPR$) and proportional effective spread (Panel B: $PESPR$). $\pi_{i,d}$ is the number of toxic price deviations that end with a trade divided by the number of toxic price deviations; $\phi_{i,d}$ is the number of toxic price deviations divided by the number of all price deviations; $\alpha_{i,d}$ is the number of price deviations divided by the number of trades; $\sigma_{i,d}$ is the average deviation in mid-quote prices. And where **Controls** is a vector of control variables: *Vola* is the 5-minute mid-quote return volatility; *Trsize* is the average trade size (in 10,000 shares); *Quotes* is the number of quote updates (in 10,000); *Ted* is the Ted spread, the difference between 3-Month USD LIBOR and Treasury Bills. All regressions are estimated with a linear time-trend (unreported). All stock specific variables are measured during the overlapping trading time, i.e. when both the home- and host-market are in their continuous trading session. Standard errors are clustered by stock and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. p -values are in parentheses below the coefficients. *Ted* is from the Federal Reserve Bank of St. Louis and all other data underlying the computations are from TRTH.

	Panel A: $PQSPR_{i,d}$		Panel B: $PESPR_{i,d}$	
$\pi_{i,d,Toxic}$	-0.008*** (0.00)	-0.007*** (0.00)	-0.003* (0.08)	-0.003* (0.09)
$\pi_{i,d,notToxic}$		-0.006*** (0.01)		0.006 (0.14)
$\phi_{i,d}$	0.021 (0.17)	0.025 (0.15)	-0.010 (0.53)	-0.001 (0.96)
$\alpha_{i,d}$	0.006*** (0.00)	0.006*** (0.00)	0.002 (0.18)	0.002 (0.17)
$\sigma_{i,d,Toxic}$	0.104*** (0.00)	0.082*** (0.01)	0.103*** (0.00)	0.075*** (0.00)
$\sigma_{i,d}$		0.048** (0.01)		0.056*** (0.00)
$Vola_{i,d}$	0.141** (0.02)	0.140** (0.02)	0.036 (0.18)	0.034 (0.18)
$Trsize_{i,d}$	-0.005 (0.77)	0.000 (0.99)	-0.054 (0.14)	-0.049 (0.18)
$Quotes_{i,d}$	-0.006 (0.11)	-0.005 (0.11)	-0.001 (0.68)	-0.001 (0.67)
Ted_d	0.024*** (0.00)	0.025*** (0.00)	0.030*** (0.00)	0.030*** (0.00)
Adj. R^2 [%]	75.49	74.69	41.21	40.41
Observations	513,837	511,777	512,611	510,615

Table OA2
Arbitrageurs' relative speed, arbitrage mix and liquidity in the ADR market, 2001 - 2016

Compared to Table OA1 I separately estimate the regression for home-market stocks (Panel A) and ADRs (Panel B).

This table presents results of how liquidity varies on days between corporate actions, i.e., when either the host or the home-market is cum-dividend but the other is ex-dividend (*BetweenCorpAct_{i,d}*). Each column presents the results of the following panel regression with day and stock-pair fixed effects:

$$Illiq_{i,d} = FE + a_0t + a_1\pi_{i,d} + a_2\phi_{i,d} + a_3\alpha_{i,d} + a_4\sigma_{i,d} + a_5Vola_{i,d} + a_6Trsize_{i,d} + a_7Quotes_{i,d} + a_8Ted_{i,d} + \epsilon_{i,d}$$

where illiquidity (*Illiq_{i,d}*) of home- or host-market stock *i* on day *d* is measured by the proportional quoted spread (Panel A: *PQSPR*), proportional effective spread (Panel B: *PESPR*), and the difference in PQSPR during and outside overlapping trading times (Panel C: *δPQSPR*), as defined in Table 1. And where **Controls** is a vector of control variables. *CorpAct* is a dummy variable set to one on days on which both the home and the host-market go ex-dividend; *log(price)* is the logarithm of the end-of-day closing price (in USD); *Vola* is the 5-minute mid-return volatility; *Quotes* is the number of updates to the (NBBO) quote (in 10,000); *Trsize* is the average trade size (in 10,000 shares); *Ted* is the Ted spread, the difference between 3-Month USD LIBOR and Treasury Bills. All stock specific variables (except *price*) are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. Standard errors are clustered by stock and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. *p*-values are in parentheses below the coefficients. Data to compute *price* are from Datastream, *Ted* is from the Federal Reserve Bank of St. Louis, and all other data underlying the computations are from TRTH.

	Panel A: Home-market				Panel B: ADR			
	<i>PQSPR_{i,d}</i>	<i>PESPR_{i,d}</i>			<i>PQSPR_{i,d}</i>	<i>PESPR_{i,d}</i>		
$\pi_{i,d,Toxic}$	-0.005*** (0.00)	-0.005*** (0.00)	0.001 (0.87)	0.001 (0.88)	-0.008*** (0.00)	-0.007*** (0.00)	-0.005*** (0.00)	-0.004*** (0.00)
$\pi_{i,d,notToxic}$		-0.011*** (0.00)		0.002 (0.73)		-0.014*** (0.00)		-0.006*** (0.01)
$\phi_{i,d}$	0.035 (0.18)	0.030 (0.38)	0.000 (0.99)	0.008 (0.82)	0.038** (0.01)	0.053*** (0.00)	-0.005 (0.68)	0.009 (0.46)
$\alpha_{i,d}$	0.015*** (0.01)	0.015*** (0.01)	0.009 (0.12)	0.009 (0.11)	0.004** (0.01)	0.004*** (0.01)	0.001 (0.49)	0.001 (0.58)
$\sigma_{i,d,Toxic}$	0.106** (0.01)	0.096** (0.03)	0.118*** (0.00)	0.111*** (0.00)	0.081*** (0.00)	0.045* (0.05)	0.056*** (0.00)	0.004 (0.85)
$\sigma_{i,d}$		0.028** (0.04)		0.018* (0.05)		0.066* (0.05)		0.094*** (0.00)
$Vola_{i,d}$	1.618*** (0.00)	1.602*** (0.00)	1.503*** (0.00)	1.489*** (0.00)	0.117** (0.01)	0.116** (0.01)	0.012 (0.27)	0.010 (0.28)
$Trsize_{i,d}$	-0.025 (0.20)	-0.019 (0.40)	-0.076** (0.04)	-0.071* (0.05)	0.105** (0.01)	0.099** (0.02)	-0.013 (0.64)	-0.020 (0.44)
$Quotes_{i,d}$	-0.309*** (0.00)	-0.306*** (0.00)	-0.364*** (0.00)	-0.363*** (0.00)	-0.006* (0.09)	-0.006* (0.09)	0.002 (0.19)	0.002 (0.19)
Ted_d	-0.014** (0.03)	-0.013** (0.04)	-0.002 (0.68)	-0.002 (0.75)	0.033*** (0.00)	0.033*** (0.00)	0.033*** (0.00)	0.032*** (0.00)
Adj. R^2	0.7692	0.7516	0.3683	0.3547	0.7513	0.7577	0.7401	0.7648
Obs.	257,862	256,218	257,157	255,561	255,928	255,512	255,407	255,007

Table OA3
Arbitrageurs' relative speed, arbitrage mix and liquidity in the ADR market, 2001 - 2016

Compared to Table OA1 I estimate the regression using an instrumental variable regression.

This table presents results of regressions explaining illiquidity by arbitrageurs' relative speed (π), arbitrage mix (ϕ), and price deviations (σ). Each column presents the results of the following panel regression with two dimensional stock-month fixed effects:

$$Illiq_{i,d} = FE + a_1\pi_{i,d} + a_2\phi_{i,d} + a_3\alpha_{i,d} + a_4\sigma_{i,d} + \zeta \times \mathbf{Controls}_{i,d} + \epsilon_{i,d}$$

where illiquidity ($Illiq_{i,d}$) of home- or host-market stock i on day d is measured by the proportional quoted spread (Panel A: $PQSPR$) and proportional effective spread (Panel B: $PESPR$). $\pi_{i,d}$ is the number of toxic price deviations that end with a trade divided by the number of toxic price deviations; $\phi_{i,d}$ is the number of toxic price deviations divided by the number of all price deviations; $\alpha_{i,d}$ is the number of price deviations divided by the number of trades; $\sigma_{i,d}$ is the average deviation in mid-quote prices. And where **Controls** is a vector of control variables: $Vola$ is the 5-minute mid-return volatility; $Trsize$ is the average trade size (in 10,000 shares); $Quotes$ is the number of updates to the (NBBO) quote (in 10,000); Ted is the Ted spread, the difference between 3-Month USD LIBOR and Treasury Bills. The regression is estimated using instrument variable approach, with $\widehat{\pi_{i,d}}$ the fitted value from explaining $\pi_{i,d}$ by a dummy variable set to 1 for ADRs trading on the NYSE after the introduction of the NYSE-Hybrid program (after 2007-01-24). All regressions are estimated with a linear time-trend (unreported). All stock specific variables (except *price*) are measured during the overlapping trading time, i.e. when both the home- and host-market are in their continuous trading session. Standard errors are clustered by stock and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. p -values are in parentheses below the coefficients. Data to compute *price* are from Datastream, *Ted* is from the Federal Reserve Bank of St. Louis, and all other data underlying the computations are from TRTH.

	Panel A: $PQSPR_{i,d}$		Panel B: $PESPR_{i,d}$	
$\widehat{\pi_{i,d,Toxic}}$	-0.009 (0.77)	-0.011 (0.77)	-0.096 (0.40)	-0.106 (0.35)
$\pi_{i,d,notToxic}$		-0.006** (0.05)		0.010 (0.14)
$\phi_{i,d}$ 0.020	0.024 (0.20)	-0.012 (0.19)	-0.008 (0.50)	(0.70)
$\alpha_{i,d}$	0.006*** (0.00)	0.006*** (0.00)	0.001 (0.45)	0.001 (0.43)
$\sigma_{i,d,Toxic}$	0.105*** (0.00)	0.082*** (0.01)	0.104*** (0.00)	0.076*** (0.00)
$\sigma_{i,d}$		0.048** (0.01)		0.056*** (0.00)
$Vola_{i,d}$	0.140** (0.02)	0.139** (0.02)	0.034 (0.19)	0.032 (0.19)
$Trsize_{i,d}$	-0.005 (0.77)	0.000 (1.00)	-0.060* (0.10)	-0.056 (0.13)
$Quotes_{i,d}$	-0.005 (0.11)	-0.005 (0.11)	-0.001 (0.65)	-0.001 (0.64)
Ted_d	0.024*** (0.00)	0.024*** (0.00)	0.029*** (0.00)	0.029*** (0.00)
Obs.	502,625	500,574	501,421	499,431

Table OA4
Summary statistics, 2001 - 2016

This table reports cross-sectional averages of time-series average by stock separately for each country in the sample. For each country: *Stocks* is the number of home-market/ADR pairs; *Days* is the number of days in the sample; *Corpack* is the number of days between corporate actions; *IN* is the number of minutes for a price deviation to persist, conditional on price deviations that arise and vanish within the same day; *pair/single* is the one hundred times the ratio of *pair* to *single* trades in the host and the market; θ is the speed at which price deviations mean revert; *max* is the maximum price deviation computed from quotes Illiquidity measures are reported separately for the home-market stock and the ADR: the daily autocorrelation from mid-quote prices (θ); the daily time-weighted average proportional quoted spread (*PQSPR*); and the difference in quoted spread between and outside overlapping trading times. For a more detailed description of these variables I refer to Table 1 of the paper. All variables (except, $\delta PQSPR$) are measured during the overlapping trading times only, i.e. when both the home- and host-market share are trading. All illiquidity and price deviations are in percent. All price deviation and spread measures are cross-sectionally winsorized each day at the 1% level. Data to compute all data underlying the computations is from Datastream and TRTH.

Country	Stocks	Days	Arbitrage				<i>AC</i>		<i>PQSPR</i>		$\delta PQSPR$	
			Corpack	In <i>pair/single</i>	θ	max	Home	ADR	Home	ADR	Home	ADR
Argentina	13	2575	101	41.167	0.05	0.004	11.93	-0.104	-0.305	1.605	1.306	
Belgium	2	2418	43	0.904	23.07	0.058	0.21	-0.029	-0.008	0.084	0.106	-0.009 -0.075
Brazil	24	2072	547	25.221	6.72	0.024	3.70	-0.083	-0.283	0.525	0.386	
Chile	10	1868	394	20.946	0.49	0.006	1.14	-0.118	-0.255	0.841	0.753	
France	12	2313	311	2.817	5.46	0.034	0.94	-0.091	-0.101	0.169	0.468	-0.005 -0.080
Germany	12	2180	181	3.617	7.18	0.028	0.74	-0.109	-0.106	0.387	0.872	0.004 -0.238
Greece	2	1707	11	4.158	0.57	0.023	1.95	-0.110	-0.232	0.268	0.785	0.017 0.342
Ireland	3	2916	6	12.014	0.48	0.009	7.34	-0.055	-0.132	0.657	0.322	-0.040 0.027
Israel	8	417	50	3.920	5.09	0.013	0.79	-0.101	-0.121	0.345	0.932	0.023 0.024
Italy	3	3663	6	3.690	6.59	0.029	0.31	-0.081	-0.219	0.126	0.162	-0.001 -0.030
Luxembourg	1	3335	2	2.090	17.26	0.041	0.38	-0.023	-0.195	0.218	0.097	-0.003 0.014
Mexico	19	1816	171	15.573	7.26	0.014	1.49	-0.107	-0.206	1.483	1.023	
Netherlands	12	2424	84	2.211	9.66	0.048	0.48	-0.083	-0.116	0.165	0.213	-0.010 -0.043
Peru	2	894	13	25.836	0.56	0.005	1.51	-0.097	-0.383	1.925	1.094	
Portugal	1	2354	10	19.137	5.82	0.011	1.20	-0.145	-0.069	0.126	0.245	-0.006 0.014
Russia	2	1468	18	4.897	4.11	0.019	2.15	-0.171	-0.080	0.175	0.425	0.011 0.103
South Africa	8	2174	216	5.730	3.00	0.016	0.94	-0.098	-0.219	0.514	0.543	-0.249 0.104
Spain	6	2274	383	5.277	5.09	0.028	1.05	-0.109	-0.171	0.283	0.537	-0.004 0.024
Sweden	6	511	4	6.162	2.36	0.033	0.88	-0.155	-0.212	0.678	2.808	0.035 -0.442
Switzerland	5	2592	104	2.136	2.51	0.053	0.25	-0.082	-0.171	0.065	0.130	-0.001 -0.032
Turkey	1	1943	6	5.631	1.71	0.015	0.93	-0.069	-0.258	0.298	0.150	0.013 0.042
U.K.	42	2206	201	8.002	13.85	0.040	0.87	-0.086	-0.163	0.540	0.645	-0.065 -0.142

Table OA5

Pooled Spearman rank correlations of daily arbitrage activity and illiquidity, 2001 - 2016

This table reports pooled Spearman rank correlations between the following daily measures: the ratio of pair to single trades (*Pair/Single*), the maximum price deviation ($\max(\Delta QTE)$), the speed at which price deviations mean-revert (θ), home- and host-market proportional quoted (*PQSPR*) and effective spread (*PESPR*). For a description of these variables I refer to Table 1. All measures are computed during the overlapping trading time, i.e. when both the home- and host-market are in their continuous trading session. All correlations are significant at the 1% level. All data underlying the computations are from TRTH.

				<i>PQSPR</i>		<i>PESPR</i>	
	<i>Pair/Single</i>	$\max(\Delta QTE)$	θ	<i>Home</i>	<i>Host</i>	<i>Home</i>	<i>Host</i>
<i>Pair/Single</i>	100%						
$\max(\Delta QTE)$	-42%	100%					
θ	55%	-30%	100%				
<i>PQSPR</i> _{Home}	-61%	59%	-57%	100%			
<i>PQSPR</i> _{Host}	-63%	56%	-61%	74%	100%		
<i>PESPR</i> _{Home}	-46%	62%	-52%	81%	66%	100%	
<i>PESPR</i> _{Host}	-56%	59%	-58%	73%	97%	68%	100%

Table OA6

Average number of daily price deviations and reasons for why they arise, 2001 - 2016

Compared to Table 2 of the paper I only consider the largest price deviation from each stock-day.

This table presents the total number of price deviations (*# Price deviations*) by the asset that moves to create the deviation (*First mover*) and by the asset that moves to eliminate it (*Last mover*) (inspired by Schultz and Shive (2010)). Each I only consider the maximum price deviation within each stock-day. The first column indicates the asset that moves to create the price deviation: either the home-market share (*Home*), the host-market share (*Host*), both the home- and the host-market share (*Both*), or the respective currency pair (*Forex*). The second column (*#Price deviations*) indicates the total number of price deviations across all stocks and days in this category. The third column (*%Toxic*) indicates the percentage of all price deviations that are toxic, when one share moves to create the price deviation and later the other moves back to eliminate it (for example, if the *Home*-market share is the first mover *%Toxic* is defined as the number of price deviations starting in the *Host*-market and ending in the *Home* market as a percentage of all price deviations). The rest of the columns *Home*, *Host*, *Both*, and *Forex* indicate the percentage of all price deviations that get eliminated because of a movement in the respective asset. All data underlying the computations are from TRTH.

First mover	#Price deviations	%Toxic	Last mover:			
			%Home	%Host	%Both	%Forex
<i>Home</i>	104,192	11.20	44.76	32.74	16.58	5.92
<i>Host</i>	137,530	10.21	27.25	47.52	16.02	9.21
<i>Both</i>	61,330		27.04	29.65	36.06	7.26
<i>Forex</i>	31,140		26.80	33.49	19.23	20.48

Table OA7
Order imbalances and liquidity during price deviations, 2001 - 2016

Compared to Table 3 of the paper I include all stock-days, even if the stock was fairly priced, *Cheap*, or *Costly* during the whole day.

This table reports results from stock-day fixed effect panel regressions to estimate the sensitivity of order imbalances (OIB) from “single” and “pair” trades to mispricings (Panel A) and sensitivity conditional on whether OIB is estimated after the peak in the price gap (Panel B: *APeak*):

$$OIB_{i,d,j} = FE_{i,d} + \beta_0 InOverlap_{i,d,j} + (\beta_1 + \gamma_1 A_{i,d,j}) Costly_{i,d,j} + (\beta_2 + \gamma_2 A_{i,d,j}) Cheap_{i,d,j} + \epsilon_{i,d,j}$$

with $OIB_{i,d,j}$ the order imbalance—the number of seller minus buyer initiated trades—for single or pair trades of stock i (host- or home-market) on day d and j one the following four periods: outside overlapping trading times, or during overlapping trading times and stock i is fairly priced, *Costly*, or *Cheap* (see Eq. 2). In Panel B, j further refers to whether the stock is before or after the peak in the price gap when the stock is *Costly* or *Cheap* (*APeak*). Single trades for the host (home) market are trades without a simultaneous trade in the home (host) market. Pair trades are simultaneous trades of both the host- and the home-market share with opposite signs. $OIB_{i,d,j}$ is explained by three dummy variables ($InOverlap_{i,d,j}$, $Costly_{i,d,j}$, and $Cheap_{i,d,j}$) set equal to one if j falls within the respective period and zero otherwise. In Panel A, $A_{i,d,j}$ is zero. In Panel B, $A_{i,d,j}$ is set to one if period j falls within *AfterPeak* and zero otherwise. Whether a stock is *Costly* or *Cheap* can only be determined when $InOverlap_{i,d,j}$ is equal to one. Whether a stock is *APeak* can only be determined when $Costly_{i,d,j}$ or $Cheap_{i,d,j}$ is equal to one. $FE_{i,d}$ are two dimensional stock-day fixed effects. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. p -values are in parentheses below the coefficients. The sample only contains days on which the stock is in between corporate actions or trades fairly priced, *Cheap*, and *Costly*. All data underlying the computations are from TRTH.

	Panel A:				Panel B:			
	single trades		pair trades		single trades		pair trades	
	OIB_{home}	OIB_{host}	OIB_{home}	OIB_{host}	OIB_{home}	OIB_{host}	OIB_{home}	OIB_{host}
<i>Costly</i>	-20.90*** (0.00)	-5.40*** (0.00)	-15.00*** (0.00)	-14.44*** (0.00)	2.27 (0.23)	6.85*** (0.00)	-8.35*** (0.00)	-7.02*** (0.00)
$\times APeak$					-26.67*** (0.00)	-24.82*** (0.00)	-0.08 (0.93)	-1.48* (0.06)
<i>Cheap</i>	15.35*** (0.00)	1.63 (0.43)	14.63*** (0.00)	14.88*** (0.00)	-4.78*** (0.00)	-17.77*** (0.00)	6.85*** (0.00)	8.53*** (0.00)
$\times APeak$					26.72*** (0.00)	34.10*** (0.00)	1.97** (0.02)	-0.39 (0.67)
<i>InOverlap</i>	6.85* (0.06)	-6.53*** (0.00)			7.09* (0.05)	-6.49*** (0.00)		
<i>StockDayFE</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2 [%]	15.49	14.43	8.81	8.94	18.36	13.18	18.78	18.90
Observations	1,168,041	1,168,041	779,557	779,557	1,602,434	1,602,434	1,237,946	1,237,946

Table OA8
Order imbalances during price deviations conditional on when trading occurs, 2001 - 2016

This table reports results from the following stock-day fixed effect panel regressions to estimate the sensitivity of order imbalances (OIB) from “single” and “pair” trades to mispricings, conditional on whether OIB is estimated after the peak in the price gap (Panel A: *AfterPeak*), by the maximum price deviation (Panel B: $\max(\Delta QTE)$), and on days between corporate actions i.e., when either the host or the home-market is cum-dividend but the other is ex-dividend (Panel C: *Between*):

$$OIB_{i,d,j} = FE_{i,d} + InOverlap_{i,d,j} [\beta_0 + \gamma_0 A_{i,d,j} + (\beta_1 + \gamma_1 A_{i,d,j}) Costly_{i,d,j} + (\beta_2 + \gamma_2 A_{i,d,j}) Cheap_{i,d,j}] + \epsilon_{i,d,j}$$

with $OIB_{i,d,j}$ the order imbalance—the number of seller minus buyer initiated trades—for single or pair trades of stock i (host- or home-market) on day d and j one the following four periods: outside overlapping trading times, or during overlapping trading times and stock i is fairly priced, *Costly*, or *Cheap* (see Eq. 2). In Panel A, j further refers to whether the stock is before or after the peak in the price gap when the stock is *Costly* or *Cheap* (*AfterPeak*). Single trades for the host (home) market are trades without a simultaneous trade in the home (host) market. Pair trades are simultaneous trades of both the host- and the home-market share with opposite signs. As in Table 3, $OIB_{i,d,j}$ is explained by three dummy variables ($InOverlap_{i,d,j}$, $Costly_{i,d,j}$, and $Cheap_{i,d,j}$) set equal to one if j falls within the respective period and zero otherwise. In addition, A varies in Panels A, B, and C. In Panel A, $A_{i,d,j}$ is set to one if period j falls within *AfterPeak* and zero otherwise. In Panel B, $A_{i,d,j}$ is set to the daily maximum price deviation $\max(\Delta QTE)$. In Panel C, $A_{i,d,j}$ is set to one if stock i on day d is *Between* corporate actions and zero otherwise. Whether a stock is *Costly* or *Cheap* can only be determined when $InOverlap_{i,d,j}$ is equal to one. Whether a stock is *AfterPeak* can only be determined when $Costly_{i,d,j}$ or $Cheap_{i,d,j}$ is equal to one. $FE_{i,d}$ are two dimensional stock-day fixed effects. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. p -values are in parentheses below the coefficients. The sample only contains periods with at least ten trades and stock-days in which the stock is either in between corporate actions or trades fairly priced, *Cheap*, and *Costly*. All data underlying the computations are from TRTH.

	Panel A: <i>AfterPeak</i>		Panel B: $\max(\Delta QTE)$		Panel C: <i>Between</i>	
	OIB single	OIB pair	OIB single	OIB pair	OIB single	OIB pair
In-Overlap	-0.34 (0.90)		0.81 (0.73)		-0.59 (0.81)	
$\times \max(\Delta QTE)$			-2.69** (0.02)			
$\times Between$					-27.58 (0.24)	
$\times Costly$	8.75*** (0.00)	-10.98*** (0.00)	-18.71*** (0.00)	-22.03*** (0.00)	-19.17*** (0.00)	-21.61*** (0.00)
$\times AfterPeak$	-39.36*** (0.00)	-4.27*** (0.00)				
$\times \max(\Delta QTE)$			-0.67 (0.25)	0.71*** (0.01)		
$\times Between$					9.48 (0.37)	26.76** (0.02)
$\times Cheap$	-18.58*** (0.00)	10.89*** (0.00)	15.09*** (0.00)	22.13*** (0.00)	14.78*** (0.00)	21.67*** (0.00)
$\times AfterPeak$	46.67*** (0.00)	4.42*** (0.00)				
$\times \max(\Delta QTE)$			-0.40 (0.23)	-0.77*** (0.00)		
$\times Between$					-0.76 (0.92)	-18.26*** (0.00)
Stock $\times$ Day FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2 [%]	12.75	16.17	14.37	4.74	14.34	4.72
Observations	1,391,074	1,135,406	1,052,680	797,545	1,052,783	797,757

Table OA9
Trading during price deviations, 2001 - 2016

This table reports abnormal trading in the host- and home-market stock when the share is *Costly* or *Cheap* (see Eq. 2). For each stock-pair and over all days on which I observe the stock fairly priced, *Costly*, and *Cheap*, I sum up all host- and home-market trades separately for whether the trade occurred when the stock was fairly priced, *Costly*, or *Cheap*, whether the trade is buyer or seller initiated, and whether the trade occurred before or after the peak in the price gap within each price deviation. When summing up trades I distinguish between single (Panel A) and pair trades (Panel B). Single trades for the host (home) market are trades without a simultaneous trade in the home (host) market. Pair trades are simultaneous trades of both the host- and the home-market share with opposite signs. For each stock i , I calculate abnormal *pair* trades (and similarly, *single* trades) within each period j as the number of *pair* trades in period j (denoted $\sum_d Pair_{i,d,j}$) in excess of when the stock is fairly priced (denoted $\sum_d Pair_{i,d,j,Fair}$), scaled by all buy and sell trades within the respective period:

$$APair_{i,j} = \frac{\sum_d Pair_{i,d,j}}{\sum_d Trades_{i,d,j,Buy/Sell}} - \frac{\sum_d Pair_{i,d,j,Fair}}{\sum_d Trades_{i,d,j,Fair,Buy/Sell}}$$

The table reports the cross-sectional mean of $ASingle_{i,j}$ (Panel A) and $APairs_{i,j}$ (Panel B). Cross-sectional t -statistics are in parentheses. Statistical significance at the 1% level is indicated by #. All data underlying the computations are from TRTH.

Price	Panel A: single trades			Panel B: pair trades	
	Initiated Peak	Host [%]	Home [%]	Host [%]	Home [%]
Costly Seller	Before	-3.98# (-6.25)	-2.31# (-3.14)	1.60# (9.01)	1.76# (8.03)
	After	0.26 (0.50)	3.83# (5.62)	2.43# (13.33)	2.70# (12.86)
	Buyer Before	3.49# (5.66)	0.27 (0.38)	-0.43# (-3.12)	-0.25 (-1.56)
	After	-3.55# (-6.72)	-8.40# (-13.12)	-0.01 (-0.13)	0.14 (1.30)
Cheap Seller	Before	2.41# (2.64)	1.23 (1.83)	-0.43 (-2.40)	-0.42# (-3.16)
	After	-4.59# (-7.88)	-8.61# (-13.49)	0.13 (1.45)	0.03 (0.33)
	Buyer Before	-2.99# (-3.18)	-2.58# (-4.22)	1.36# (7.55)	1.83# (9.12)
	After	1.26 (2.22)	4.10# (6.31)	2.19# (11.38)	2.93# (13.09)

Table OA10
Trading during price deviations, 2001 - 2016

Compared to Table OA9 of the paper I define simultaneous trades as trades within one second.

This table reports abnormal trading in the host- and home-market stock when the share is *Costly* or *Cheap* (see Eq. 2). For each stock-pair and over the whole sample, I sum up all host- and home-market trades separately for whether the trade occurred when the stock was fairly priced, *Costly*, or *Cheap*, whether the trade is buyer or seller initiated, and whether the trade occurred before or after the peak in the price gap within each price deviation. When summing up trades I distinguish between single (Panel A), pair (Panel B), and quote trades (Panel C). Single trades for the host (home) market are trades without a simultaneous trade in the home (host) market. Pair trades are simultaneous trades of both the host- and the home-market share with opposite signs. Quote trades are trades that occurred at the prevailing ask or bid quote. For each stock i , I calculate abnormal trading within each period j as the number of trades in period j (denoted $\sum_d Trades_{i,d,j}$) in excess of when the stock is fairly priced (denoted $\sum_d Trades_{i,d,j,Fair}$), scaled by all trades:

$$ATrades_{i,j} = \frac{\sum_d Trades_{i,d,j}}{\sum_d \sum_j Trades_{i,d,j}} - \frac{\sum_d Trades_{i,d,j,Fair}}{\sum_d \sum_j Trades_{i,d,j}}$$

The table reports the cross-sectional mean of $ATrades_{i,j}$ and the cross-sectional t -statistics are in parentheses. Statistical significance at the 1% level is indicated by #. All data underlying the computations are from TRTH.

Price	Initiator	Peak	Panel A: single trades		Panel B: pair trades		Panel C: quote trades	
			Host [%]	Home [%]	Host [%]	Home [%]	Host [%]	Home [%]
Costly	Seller	Before	-5.38# (-8.74)	-4.96# (-6.36)	4.96# (10.59)	4.17# (13.02)	1.22 (2.31)	3.75# (4.90)
		After	-3.34# (-6.59)	-2.59# (-3.27)	6.49# (15.32)	6.86# (17.66)	4.51# (9.13)	8.65# (12.60)
	Buyer	Before	0.36 (0.57)	-1.62 (-2.43)	-0.76# (-2.76)	-0.23 (-0.66)	2.36# (4.07)	-2.40# (-3.48)
		After	-5.49# (-10.78)	-9.53# (-13.78)	-0.25 (-1.42)	0.20 (0.82)	-1.62# (-4.63)	-8.30# (-15.26)
Cheap	Seller	Before	-0.67 (-0.71)	-0.65 (-1.00)	-0.63 (-2.09)	-0.60 (-2.63)	3.43# (4.34)	-2.61# (-3.72)
		After	-6.04# (-10.64)	-9.53# (-14.66)	-0.11 (-0.64)	0.11 (0.42)	-1.51# (-3.34)	-8.60# (-14.07)
	Buyer	Before	-3.80# (-4.32)	-5.26# (-8.78)	3.57# (9.08)	4.83# (11.63)	0.00 (0.01)	2.83# (3.79)
		After	-2.32# (-3.82)	-2.62# (-3.70)	5.65# (16.48)	7.68# (16.13)	4.56# (9.88)	9.11# (13.09)

Table OA11
Trading during price deviations, 2001 - 2016

Compared to Table OA9 of the paper I only look at trades below 500 shares.

This table reports abnormal trading in the host- and home-market stock when the share is *Costly* or *Cheap* (see Eq. 2). For each stock-pair and over the whole sample, I sum up all host- and home-market trades separately for whether the trade occurred when the stock was fairly priced, *Costly*, or *Cheap*, whether the trade is buyer or seller initiated, and whether the trade occurred before or after the peak in the price gap within each price deviation. When summing up trades I distinguish between single (Panel A), pair (Panel B), and quote trades (Panel C). Single trades for the host (home) market are trades without a simultaneous trade in the home (host) market. Pair trades are simultaneous trades of both the host- and the home-market share with opposite signs. Quote trades are trades that occurred at the prevailing ask or bid quote. For each stock i , I calculate abnormal trading within each period j as the number of trades in period j (denoted $\sum_d Trades_{i,d,j}$) in excess of when the stock is fairly priced (denoted $\sum_d Trades_{i,d,j,Fair}$), scaled by all trades:

$$ATrades_{i,j} = \frac{\sum_d Trades_{i,d,j}}{\sum_d \sum_j Trades_{i,d,j}} - \frac{\sum_d Trades_{i,d,j,Fair}}{\sum_d \sum_j Trades_{i,d,j}}$$

The table reports the cross-sectional mean of $ATrades_{i,j}$ and the cross-sectional t -statistics are in parentheses. Statistical significance at the 1% level is indicated by #. All data underlying the computations are from TRTH.

Price	Initiator	Peak	Panel A: single trades		Panel B: pair trades		Panel C: quote trades	
			Host [%]	Home [%]	Host [%]	Home [%]	Host [%]	Home [%]
Costly	Seller	Before	-3.18# (-3.61)	-1.88 (-2.36)	0.96# (6.15)	1.99# (8.13)	0.39 (0.54)	2.55# (2.71)
		After	0.89 (1.68)	2.23# (3.57)	1.72# (10.30)	2.84# (11.94)	4.71# (9.21)	6.35# (9.33)
	Buyer	Before	3.67# (3.61)	-0.36 (-0.43)	-0.33# (-2.73)	-0.31 (-1.72)	3.22# (3.81)	-0.76 (-0.89)
		After	-3.26# (-5.94)	-7.10# (-11.49)	-0.03 (-0.47)	0.20 (1.04)	-1.04# (-3.00)	-6.57# (-9.24)
Cheap	Seller	Before	2.91# (4.04)	1.58 (2.13)	-0.34 (-1.79)	-0.46# (-3.55)	3.48# (5.38)	-0.33 (-0.47)
		After	-4.42# (-6.80)	-6.54# (-9.96)	0.03 (0.45)	0.09 (0.86)	-1.12 (-2.03)	-6.08# (-12.86)
	Buyer	Before	-1.79 (-1.79)	-2.77# (-3.91)	0.89# (6.47)	1.57# (7.83)	0.66 (0.83)	0.41 (0.52)
		After	2.29# (3.60)	1.60 (2.53)	1.50# (9.42)	2.96# (12.37)	4.50# (9.67)	5.57# (7.47)

Table OA12
Trading in USD during price deviations, 2001 - 2016

Compared to Table OA9 of the paper I report abnormal trading volumes in USD.

This table reports abnormal trading in the host- and home-market stock when the share is *Costly* or *Cheap* (see Eq. 2). For each stock-pair and over the whole sample, I sum up all host- and home-market trading volume in USD separately for whether the trade occurred when the stock was fairly priced, *Costly*, or *Cheap*, whether the trade is buyer or seller initiated, and whether the trade occurred before or after the peak in the price gap within each price deviation. When summing up trading volume I distinguish between single (Panel A) and pair (Panel B) trades. Single trades for the host (home) market are trades without a simultaneous trade in the home (host) market. Pair trades are simultaneous trades of both the host- and the home-market share with opposite signs. For each stock i , I calculate abnormal trading within each period j as the trading volume in period j (denoted $\sum_d Trades_{i,d,j}$) in excess of when the stock is fairly priced (denoted $\sum_d Trades_{i,d,j,Fair}$), scaled by all trading volume:

$$ATrades_{i,j} = \frac{\sum_d Trades_{i,d,j}}{\sum_d \sum_j Trades_{i,d,j}} - \frac{\sum_d Trades_{i,d,j,Fair}}{\sum_d \sum_j Trades_{i,d,j}}$$

The table reports the cross-sectional mean of $ATrades_{i,j}$ and the cross-sectional t -statistics are in parentheses. Statistical significance at the 1% level is indicated by #. All data underlying the computations are from TRTH.

Price	Initiator	Panel A: single trades			Panel B: pair trades	
		Peak	Host [%]	Home [%]	Host [%]	Home [%]
Costly	Seller	Before	-3.67# (-4.25)	-3.43# (-3.35)	1.53# (6.91)	1.41# (5.12)
		After	0.42 (0.73)	4.46# (5.20)	2.45# (12.45)	2.44# (10.03)
	Buyer	Before	3.80# (4.62)	1.12 (1.09)	-0.41# (-2.86)	-0.04 (-0.30)
		After	-2.84# (-5.15)	-8.96# (-11.39)	-0.12 (-1.19)	0.33# (2.63)
Cheap	Seller	Before	2.99# (2.69)	2.29 (2.21)	-0.48# (-2.95)	-0.15 (-0.85)
		After	-4.28# (-6.11)	-9.84# (-11.95)	0.05 (0.46)	0.05 (0.46)
	Buyer	Before	-3.76# (-3.29)	-3.70# (-3.86)	1.66# (6.06)	1.77# (5.52)
		After	1.03 (1.48)	5.91# (7.25)	2.66# (8.11)	2.59# (9.57)

Table OA13

Liquidity at and after single and pair trades, 2001 - 2016

Compared to Table 4 of the paper I estimate spreads and pricing errors of trades only within mispricings, i.e., when the stock is in between corporate actions or *Costly* or *Cheap*.

This table presents results of the following fixed-effect panel regressions.

$$Illiq_{i,d,j} = FE + \beta_1 \times Pair_{i,d,j} + \epsilon_{i,d}$$

with $Illiq_{i,d,j}$ quoted spreads, effective spreads, or the logarithm of the absolute Hasbrouck (1993) pricing error associated with *pair* ($PairTrade = 1$) or *single* ($PairTrade = 0$) trades. For each trade I estimate spreads and pricing errors associated with the trade ($PQSPR_{i,t}$, $PESPR_{i,t}$, and $\ln|Has|_{i,t}$) and associated with the next trade ($PQSPR_{i,t+1}$, $PESPR_{i,t+1}$, $\ln|Has|_{i,t+1}$). I then average spreads and pricing errors within each stock-day, separately for *pair* and *single* trades. $FE_{i,d}$ are two dimensional stock-day fixed effects. I drop stock-days without any pair trade and trades outside mispricings. $PQSPR$ and $PESPR$ are measured in per cent. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. p -values are in parentheses below the coefficients. All data underlying the computations are from TRTH.

	$PQSPR_{i,t}$	$PQSPR_{i,t+1}$	$PESPR_{i,t}$	$PESPR_{i,t+1}$	$\ln Has_{i,t} $	$\ln Has_{i,t+1} $
$PairTrade_{i,t}$	-0.016*** (0.01)	-0.041*** (0.00)	-0.008** (0.02)	0.003* (0.06)	-0.076*** (0.00)	-0.076*** (0.00)
$Stock \times Day$ FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2 [%]	5.06	2.47	17.13	8.54	94.27	94.25
Observations	1,085,396	1,085,396	1,085,396	1,085,396	1,075,530	1,075,170

Table OA14
Intraday Vector autoregressions and Granger causality, 2001 - 2016

Compared to Panel A of Table 5 I report estimates for all coefficients and not just their sums.

This table reports results from a panel vector autoregression using two dimensional stock-day and exchange-hour fixed effects. The vector autoregression is estimated using absolute order imbalance from single trades and from pair trades, price deviations, and proportional quoted spreads as endogenous variables with five lags.

$$LHS_{i,t} = FE + \sum_{l=1}^5 \beta_{1,l} \ln|OIB|_{single,i,t-l} + \sum_{l=1}^5 \beta_{2,l} |\Delta Price|_{i,t-l} + \sum_{l=1}^5 \beta_{3,l} \ln|OIB|_{pair,i,t-l} + \sum_{l=1}^5 \beta_{4,l} PQSPR_{i,t-l} + \epsilon_{i,t}$$

where $\ln|OIB|_{single}$ and $\ln|OIB|_{pair}$ is estimated as the logarithm of one plus the absolute difference in buyer and seller initiated single and pair trades during 1-minute interval t , respectively; $|\Delta Price|$ is the maximum price deviation within interval t (from Eq. 1); and $PQSPR$ is average proportional quoted spread within interval t . All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. For a detailed description of these variables I refer to Table 1. In all cases the table reports the sum of all lagged variables for each endogenous variable. p -values are in parentheses below the coefficients, based on a χ^2 test for testing the null hypothesis that the sum of the lagged coefficients is equal to zero when explaining the column variable. The next row reports the p -value of testing joined significance of the lagged values. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. All data underlying the computations are from TRTH.

	$PQSPR_{i,t}$	$ \Delta Price _{i,t}$	$\ln OIB _{single,i,t}$	$\ln OIB _{pair,i,t}$
$\ln OIB_{pair,i,t-1}$	-0.0007*** (0.00)	-0.006*** (0.00)	0.030*** (0.00)	0.091*** (0.00)
$\ln OIB_{pair,i,t-2}$	-0.00023*** (0.00)	-0.005*** (0.00)	0.014*** (0.00)	0.0522*** (0.00)
$\ln OIB_{pair,i,t-3}$	-0.0001*** (0.00)	-0.0051*** (0.00)	0.012*** (0.00)	0.041*** (0.00)
$\ln OIB_{pair,i,t-4}$	-0.00002 (0.54)	-0.0048*** (0.00)	0.010*** (0.00)	0.035*** (0.00)
$\ln OIB_{pair,i,t-5}$	0.00001 (0.78)	0.0058*** (0.00)	0.010*** (0.00)	0.0325*** (0.00)
$\ln OIB_{single,i,t-1}$	0.0046*** (0.00)	0.0123*** (0.00)	0.105*** (0.00)	0.022*** (0.00)
$\ln OIB_{single,i,t-2}$	-0.0007*** (0.00)	0.011*** (0.00)	0.0606*** (0.00)	0.011*** (0.00)
$\ln OIB_{single,i,t-3}$	0.00003 (0.50)	0.012*** (0.00)	0.048*** (0.00)	0.0081*** (0.00)
$\ln OIB_{single,i,t-4}$	-0.00014*** (0.00)	0.013*** (0.00)	0.041*** (0.00)	0.007*** (0.00)
$\ln OIB_{single,i,t-5}$	-0.0002*** (0.00)	-0.014*** (0.00)	0.038*** (0.00)	0.007*** (0.00)
$ \Delta Price _{i,t-1}$	-0.0007***	0.748***	0.0009***	0.00006

Table OA14 continued

	$PQSPR_{i,t}$	$ \Delta Price _{i,t}$	$\ln OIB _{single,i,t}$	$\ln OIB _{pair,i,t}$
	(0.00)	(0.00)	(0.00)	(0.38)
$ \Delta Price _{i,t-2}$	0.0007***	0.0023	-0.0003***	-0.0001***
	(0.00)	(0.21)	(0.00)	(0.00)
$ \Delta Price _{i,t-3}$	-0.0001***	-0.0007	0.00002	-0.000006
	(0.00)	(0.58)	(0.55)	(0.11)
$ \Delta Price _{i,t-4}$	0.000009	-0.006***	0.00003	-0.000006
	(0.29)	(0.00)	(0.38)	(0.12)
$ \Delta Price _{i,t-5}$	-0.00002	-0.045***	0.0004***	0.00002
	(0.12)	(0.00)	(0.00)	(0.13)
$PQSPR_{i,t-1}$	0.907***	-0.0008	-0.070***	-0.011***
	(0.00)	(0.83)	(0.00)	(0.00)
$PQSPR_{i,t-2}$	-0.072***	-0.001***	0.025***	0.0054***
	(0.00)	(0.00)	(0.00)	(0.00)
$PQSPR_{i,t-3}$	0.059***	-0.0026**	0.0032***	0.00036*
	(0.00)	(0.01)	(0.00)	(0.10)
$PQSPR_{i,t-4}$	0.0124***	0.028***	0.0047***	0.0009***
	(0.00)	(0.00)	(0.00)	(0.00)
$PQSPR_{i,t-5}$	0.043***	-0.0269***	0.014***	0.0019***
	(0.00)	(0.00)	(0.00)	(0.00)
<i>Stock</i> × <i>Day</i> FE	Yes	Yes	Yes	Yes
<i>Exchange</i> × <i>Hour</i> FE	Yes	Yes	Yes	Yes
Adj. R^2 [%]	98.02	89.23	60.54	35.13
Observations	155,759,235	155,759,235	155,759,235	155,759,235

Table OA15
Intraday Vector autoregressions and Granger causality, 2001 - 2016

Compared to Table 5 I include squared midpoint returns ($Return^2$) as a measure for volatility as an endogeneous variable.

This table reports results from a panel vector autoregression using two dimensional stock-day and exchange-hour fixed effects. The vector autoregression is estimated using volatility, absolute order imbalance from single trades and from pair trades, price deviations, and proportional quoted spreads as endogeneous variables with five lags.

$$LHS_{i,t} = FE + \sum_{l=1}^5 \beta_{1,l} \ln|OIB|_{single,i,t-l} + \sum_{l=1}^5 \beta_{2,l} |\Delta Price|_{i,t-l} + \sum_{l=1}^5 \beta_{3,l} \ln|OIB|_{pair,i,t-l} + \sum_{l=1}^5 \beta_{4,l} PQSPR_{i,t-l} + \sum_{l=1}^5 \beta_{5,l} Return^2_{i,t-l}$$

where $\ln|OIB|_{single}$ and $\ln|OIB|_{pair}$ is estimated as the logarithm of one plus the absolute difference in buyer and seller initiated single and pair trades during 1-minute interval t , respectively; $|\Delta Price|$ is the maximum price deviation within interval t (from Eq. 1); and $PQSPR$ is average proportional quoted spread within interval t . All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. For a detailed description of these variables I refer to Table 1. In all cases the table reports the sum of all lagged variables for each endogenous variable. p -values are in parentheses below the coefficients, based on a χ^2 test for testing the null hypothesis that the sum of the lagged coefficients is equal to zero when explaining the column variable. The next row reports the p -value of testing joined significance of the lagged values. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. All data underlying the computations are from TRTH.

	$PQSPR_{i,t}$	$ \Delta Price _{i,t}$	$\ln OIB _{single,i,t}$	$\ln OIB _{pair,i,t}$	$Return^2_{i,t}$
$\sum_{l=1}^5 OIB _{pair,i,t-l}$	-0.0011*** (0.00) (0.00)	-0.016*** (0.00) (0.00)	0.075*** (0.00) (0.00)	0.250*** (0.00) (0.00)	0.0000 (0.46) (0.04)
$\sum_{l=1}^5 OIB _{single,i,t-l}$	0.004*** (0.00) (0.00)	0.035*** (0.00) (0.00)	0.292*** (0.00) (0.00)	0.054*** (0.00) (0.00)	0.000*** (0.00) (0.00)
$\sum_{l=1}^5 \Delta Price _{i,t-l}$	0.000*** (0.00) (0.00)	0.699*** (0.00) (0.00)	0.001*** (0.00) (0.00)	0.000 (0.68) (0.00)	0.000*** (0.00) (0.00)
$\sum_{l=1}^5 PQSPR_{i,t-l}$	0.949*** (0.00) (0.00)	-0.012*** (0.00) (0.00)	-0.023*** (0.00) (0.00)	-0.002*** (0.00) (0.00)	0.000*** (0.00) (0.00)
$\sum_{l=1}^5 Return^2_{i,t-l}$	-0.556*** (0.00) (0.01)	2.177 (0.12) (0.05)	0.174* (0.06) (0.00)	0.037 (0.20) (0.16)	-0.009*** (0.00) (0.00)
<i>Stock</i> $\times$ <i>Day</i> FE	Yes	Yes	Yes	Yes	Yes
<i>Exchange</i> $\times$ <i>Hour</i> FE	Yes	Yes	Yes	Yes	Yes
Adj. R^2 [%]	98.02%	89.22%	60.54%	35.06%	-0.21%
Observations	154,908,431	154,908,431	154,908,431	154,908,431	154,908,431

Table OA16
Intraday Vector autoregressions and Granger causality, 2001 - 2016

Compared to Table 5 I use ten lags.

This table reports results from a panel vector autoregression using two dimensional stock-day and exchange-hour fixed effects. The vector autoregression is estimated using absolute order imbalance from single trades and from pair trades, price deviations, and proportional quoted spreads as endogenous variables with ten lags.

$$LHS_{i,t} = FE + \sum_{l=1}^{10} \beta_{1,l} \ln|OIB|_{single,i,t-l} + \sum_{l=1}^{10} \beta_{2,l} |\Delta Price|_{i,t-l} + \sum_{l=1}^{10} \beta_{3,l} \ln|OIB|_{pair,i,t-l} + \sum_{l=1}^{10} \beta_{4,l} PQSPR_{i,t-l} + \epsilon_{i,t}$$

where $\ln|OIB|_{single}$ and $\ln|OIB|_{pair}$ is estimated as the logarithm of one plus the absolute difference in buyer and seller initiated single and pair trades during 1-minute interval t , respectively; $|\Delta Price|$ is the maximum price deviation within interval t (from Eq. 1); and $PQSPR$ is average proportional quoted spread within interval t . All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. For a detailed description of these variables I refer to Table 1. In all cases the table reports the sum of all lagged variables for each endogenous variable. p -values are in parentheses below the coefficients, based on a χ^2 test for testing the null hypothesis that the sum of the lagged coefficients is equal to zero when explaining the column variable. The next row reports the p -value of testing joint significance of the lagged values. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. All data underlying the computations are from TRTH.

	$PQSPR_{i,t}$	$ \Delta Price _{i,t}$	$ OIB _{single,i,t}$	$ OIB _{pair,i,t}$
$\sum_{l=1}^{10} OIB _{pair,i,t-l}$	-0.001** (0.00) (0.00)	-0.018*** (0.00) (0.00)	0.078*** (0.00) (0.00)	0.321*** (0.00) (0.00)
$\sum_{l=1}^{10} OIB _{single,i,t-l}$	0.003*** (0.00) (0.00)	0.039*** (0.00) (0.00)	0.383*** (0.00) (0.00)	0.064*** (0.00) (0.00)
$\sum_{l=1}^{10} \Delta Price _{i,t-l}$	0.000*** (0.00) (0.00)	0.740*** (0.00) (0.00)	0.001** (0.00) (0.00)	0.000 (0.66) (0.00)
$\sum_{l=1}^{10} PQSPR_{i,t-l}$	0.955*** (0.00) (0.00)	-0.010*** (0.00) (0.00)	-0.020*** (0.00) (0.00)	-0.001*** (0.00) (0.00)
<i>Stock</i> $\times$ <i>Day</i> FE	Yes	Yes	Yes	Yes
<i>Exchange</i> $\times$ <i>Hour</i> FE	Yes	Yes	Yes	Yes
Adj. R^2 [%]	98.04	89.33	60.74	35.09
Observations	151,511,874	151,511,874	151,511,874	151,511,874

Table OA17
Intraday Vector autoregressions and Granger causality, 2001 - 2016

Compared to Table 5 I set $|\Delta Price|$ to zero if it is small, i.e., not above 0.5 standard deviations of the stock-day average.

This table reports results from a panel vector autoregression using two dimensional stock-day and exchange-hour fixed effects. The vector autoregression is estimated using absolute order imbalance from single trades and from pair trades, price deviations, and proportional quoted spreads as endogenous variables with five lags.

$$LHS_{i,t} = FE + \sum_{l=1}^5 \beta_{1,l} \ln|OIB|_{single,i,t-l} + \sum_{l=1}^5 \beta_{2,l} |\Delta Price|_{i,t-l} + \sum_{l=1}^5 \beta_{3,l} \ln|OIB|_{pair,i,t-l} + \sum_{l=1}^5 \beta_{4,l} PQSPR_{i,t-l} + \epsilon_{i,t}$$

where $\ln|OIB|_{single}$ and $\ln|OIB|_{pair}$ is estimated as the logarithm of one plus the absolute difference in buyer and seller initiated single and pair trades during 1-minute interval t , respectively; $|\Delta Price|$ is the maximum price deviation within interval t (from Eq. 1) if it is above 0.5 standard deviations of the stock-day average and zero otherwise; and $PQSPR$ is average proportional quoted spread within interval t . All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. For a detailed description of these variables I refer to Table 1. In all cases the table reports the sum of all lagged variables for each endogenous variable. p -values are in parentheses below the coefficients, based on a χ^2 test for testing the null hypothesis that the sum of the lagged coefficients is equal to zero when explaining the column variable. The next row reports the p -value of testing joined significance of the lagged values. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. All data underlying the computations are from TRTH.

	$PQSPR_{i,t}$	$ \Delta Price _{i,t}$	$ OIB _{single,i,t}$	$ OIB _{pair,i,t}$
$\sum_{l=1}^5 OIB _{pair,i,t-l}$	-0.0011*** (0.00) (0.00)	-0.0084*** (0.00) (0.00)	0.076*** (0.00) (0.00)	0.25*** (0.00) (0.00)
$\sum_{l=1}^5 OIB _{single,i,t-l}$	0.0036*** (0.00) (0.00)	0.018*** (0.00) (0.00)	0.29*** (0.00) (0.00)	0.054*** (0.00) (0.00)
$\sum_{l=1}^5 \Delta Price _{i,t-l}$	-0.00013*** (0.00) (0.00)	0.74*** (0.00) (0.00)	0.00088*** (0.00) (0.00)	0.00014* (0.10) (0.00)
$\sum_{l=1}^5 PQSPR_{i,t-l}$	0.95*** (0.00) (0.00)	-0.011*** (0.00) (0.00)	-0.023*** (0.00) (0.00)	-0.002*** (0.00) (0.00)
<i>Stock</i> $\times$ <i>Day</i> FE	Yes	Yes	Yes	Yes
<i>Exchange</i> $\times$ <i>Hour</i> FE	Yes	Yes	Yes	Yes
Adj. R^2 [%]	98.02%	78.51%	60.54%	35.13%
Observations	155,759,235	155,759,235	155,759,235	155,759,235

Table OA18
Intraday Vector autoregressions and Granger causality, 2001 - 2016

Compared to Panel A of Table 5 I use price deviations estimated from midpoint prices.

This table reports results from panel vector autoregressions using two dimensional stock-day and exchange-hour fixed effects. The vector autoregression is estimated using absolute order imbalance from single trades and from pair trades, midpoint price deviations, and proportional quoted spreads as endogenous variables with five lags.

$$LHS_{i,t} = FE + \sum_{l=1}^5 \beta_{1,l} \ln|OIB|_{single,i,t-l} + \sum_{l=1}^5 \beta_{2,l} \Delta Mid_{i,t-l} + \sum_{l=1}^5 \beta_{3,l} \ln|OIB|_{pair,i,t-l} + \sum_{l=1}^5 \beta_{4,l} PQSPR_{i,t-l} + \epsilon_{i,t}$$

where $\ln|OIB|_{single}$ and $\ln|OIB|_{pair}$ is estimated as the logarithm of one plus the absolute difference in buyer and seller initiated single and pair trades during 1-minute interval t , respectively; ΔMid is the midpoint price deviation at time t ; and $PQSPR$ is average proportional quoted spread within interval t . All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. For a detailed description of these variables I refer to Table 1. In all cases the table reports the sum of all lagged variables for each endogenous variable. p -values are in parentheses below the coefficients, based on a χ^2 test for testing the null hypothesis that the sum of the lagged coefficients is equal to zero when explaining the column variable. The next row reports the p -value of testing joint significance of the lagged values. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. All data underlying the computations are from TRTH.

	$PQSPR_{i,t}$	$\Delta Mid_{i,t}$	$ OIB _{single,i,t}$	$ OIB _{pair,i,t}$
$\sum_{l=1}^5 OIB _{pair,i,t-l}$	-0.001*** (0.00) (0.00)	-0.001*** (0.00) (0.00)	0.076*** (0.00) (0.00)	0.252*** (0.00) (0.00)
$\sum_{l=1}^5 OIB _{single,i,t-l}$	0.004*** (0.00) (0.00)	-0.002*** (0.00) (0.00)	0.293*** (0.00) (0.00)	0.054*** (0.00) (0.00)
$\sum_{l=1}^5 \Delta Mid_{i,t-l}$	0.002*** (0.00) (0.00)	0.955*** (0.00) (0.00)	0.003*** (0.00) (0.00)	0.000 (0.94) (0.00)
$\sum_{l=1}^5 PQSPR_{i,t-l}$	0.949*** (0.00) (0.00)	-0.002** (0.03) (0.00)	-0.023*** (0.00) (0.00)	-0.002*** (0.00) (0.00)
<i>Stock</i> $\times$ <i>Day</i> FE	Yes	Yes	Yes	Yes
<i>Exchange</i> $\times$ <i>Hour</i> FE	Yes	Yes	Yes	Yes
Adj. R^2 [%]	98.03	99.97	60.54	35.14
Observations	155,759,235	155,759,235	155,759,235	155,759,235

Table OA19

Intraday Vector autoregressions and Granger causality, 2001 - 2016

Compared to Panel A of Table 5 I estimate panel regressions including an indicator variable whether price deviations increased, $Up\Delta_{i,t} = |\Delta Price|_{i,t} > |\Delta Price|_{i,t-1}$.

This table reports results from panel vector autoregressions using two dimensional stock-day and exchange-hour fixed effects. The vector autoregression is estimated using absolute order imbalance from single trades and from pair trades, price deviations, and proportional quoted spreads as endogenous variables with five lags.

$$LHS_{i,t} = FE + \sum_{l=1}^5 \beta_{0,l} Up\Delta_{i,t-l} + \sum_{l=1}^5 (\beta_{1,l} + \gamma_{1,l} Up|\Delta Price|_{i,t-l}) \ln|OIB|_{single,i,t-l} + \sum_{l=1}^5 \beta_{2,l} \Delta_{i,t-l} \\ + \sum_{l=1}^5 (\beta_{3,l} + \gamma_{2,l} Up\Delta_{i,t-l}) \ln|OIB|_{pair,i,t-l} + \sum_{l=1}^5 \beta_{4,l} PQSPR_{i,t-l} + \epsilon_{i,t}$$

where $\ln|OIB|_{single}$ and $\ln|OIB|_{pair}$ is estimated as the logarithm of one plus the absolute difference in buyer and seller initiated single and pair trades during 1-minute interval t , respectively; $|\Delta Price|$ is the maximum price deviation within interval t (from Eq. 1); and $PQSPR$ is average proportional quoted spread within interval t . All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. For a detailed description of these variables I refer to Table 1. In all cases the table reports the sum of all lagged variables for each endogenous variable. Results with $Up\Delta_{i,t}$ (or its interaction with order imbalances) on the *LHS* are not reported. p -values are in parentheses below the coefficients, based on a χ^2 test for testing the null hypothesis that the sum of the lagged coefficients is equal to zero when explaining the column variable. The next row reports the p -value of testing joined significance of the lagged values. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. All data underlying the computations are from TRTH.

	$PQSPR_{i,t}$	$ \Delta Price _{i,t}$	$ OIB _{single,i,t}$	$ OIB _{pair,i,t}$
$\sum_{l=1}^5 OIB _{pair,i,t-l}$	-0.0013*** (0.00) (0.00)	-0.013** (0.01) (0.07)	0.065*** (0.00) (0.00)	0.236*** (0.00) (0.00)
$\sum_{l=1}^5 OIB _{pair,i,t-l} \times Up\Delta_{i,t-l}$	0.0004*** (0.00) (0.00)	0.0000 (1.00) (0.00)	0.0292*** (0.00) (0.00)	0.0369*** (0.00) (0.00)
$\sum_{l=1}^5 OIB _{single,i,t-l}$	0.003*** (0.00) (0.00)	0.051*** (0.00) (0.00)	0.296*** (0.00) (0.00)	0.045*** (0.00) (0.00)
$\sum_{l=1}^5 OIB _{single,i,t-l} \times Up\Delta_{i,t-l}$	0.001** (0.04) (0.00)	-0.061*** (0.00) (0.00)	-0.0198*** (0.00) (0.00)	0.0253*** (0.00) (0.00)
$\sum_{l=1}^5 \Delta Price _{i,t-l}$	-0.0002*** (0.00) (0.00)	0.699*** (0.00) (0.00)	0.000 (0.75) (0.00)	-0.0002*** (0.00) (0.00)
$\sum_{l=1}^5 PQSPR_{i,t-l}$	0.949*** (0.00) (0.00)	-0.010*** (0.00) (0.00)	-0.021*** (0.00) (0.00)	-0.002*** (0.00) (0.00)

Table OA19 continued

	$PQSPR_{i,t}$	$ \Delta Price _{i,t}$	$ OIB _{single,i,t}$	$ OIB _{pair,i,t}$
$\sum_{l=1}^5 Up\Delta_{i,t-l}$	-0.001	0.129***	0.115***	0.016***
	(0.25)	(0.00)	(0.00)	(0.00)
	(0.00)	(0.00)	(0.00)	(0.00)
<i>Stock</i> $\times$ <i>Day</i> FE	Yes	Yes	Yes	Yes
<i>Exchange</i> $\times$ <i>Hour</i> FE	Yes	Yes	Yes	Yes
Adj. R^2 [%]	98.02%	89.24%	60.58%	35.20%
Observations	155,759,235	155,759,235	155,759,235	155,759,235

Table OA20

Intraday Vector autoregressions and Granger causality (*PESPR*), 2001 - 2016

Compared to Table 6 I only use stock-days on which I can estimate effective spread (*PESPR*) for at least 8 out of 10 intervals.

This table reports results from a panel vector autoregression using two dimensional stock-day and exchange-hour fixed effects. The vector autoregression is estimated using absolute order imbalance from single trades and from pair trades, price deviations, and proportional effective spreads as endogeneous variables with five lags.

$$LHS_{i,t} = FE + \sum_{l=1}^5 \beta_{1,l} |OIB|_{single,i,t-l} + \sum_{l=1}^5 \beta_{2,l} |\Delta Price|_{i,t-l} + \sum_{l=1}^5 \beta_{3,l} |OIB|_{pair,i,t-l} + \sum_{l=1}^5 \beta_{4,l} PESPR_{i,t-l} + \epsilon_{i,t}$$

where $|OIB|_{single}$ and $|OIB|_{pair}$ is estimated as the logarithm of 1 plus the absolute difference in buyer- and seller-initiated single and pair trades during 1-minute interval t , respectively; $|\Delta Price|$ is the maximum price deviation within interval t (from Eq. 1); and *PESPR* is average proportional effective spread within interval t . All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. For a detailed description of these variables I refer to Table 1. In all cases the table reports the sum of all lagged variables for each endogenous variable. p -values are in parentheses below the coefficients, based on a χ^2 test for testing the null hypothesis that the sum of the lagged coefficients is equal to zero when explaining the column variable. The next row reports the p -value of testing joined significance of the lagged values. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. All data underlying the computations are from TRTH.

	$PESPR_{i,t}$	$ \Delta Price _{i,t}$	$ OIB _{single,i,t}$	$ OIB _{pair,i,t}$
$\sum_{l=1}^5 OIB _{pair,i,t-l}$	-0.00041* (0.06) (0.00)	-0.0023* (0.06) (0.00)	0.09*** (0.00) (0.00)	0.23*** (0.00) (0.00)
$\sum_{l=1}^5 OIB _{single,i,t-l}$	0.00071 (0.10) (0.00)	0.0051** (0.02) (0.00)	0.26*** (0.00) (0.00)	0.07*** (0.00) (0.00)
$\sum_{l=1}^5 \Delta Price _{i,t-l}$	-0.000045 (0.85) (0.01)	0.9*** (0.00) (0.00)	0.024*** (0.00) (0.00)	0.0024 (0.35) (0.27)
$\sum_{l=1}^5 PESPR_{i,t-l}$	0.91*** (0.00) (0.00)	0.0045 (0.24) (0.00)	-0.17*** (0.00) (0.00)	-0.026*** (0.00) (0.00)
<i>Stock</i> $\times$ <i>Day</i> FE	Yes	Yes	Yes	Yes
<i>Exchange</i> $\times$ <i>Hour</i> FE	Yes	Yes	Yes	Yes
Adj. R^2 [%]	90.09%	97.90%	35.05%	28.79%
Observations	45,397,580	45,397,580	45,397,580	45,397,580

Table OA21
Intraday Vector autoregressions and Granger causality, 2001 - 2016

Compared to Panel A of Table 5 I use effective spreads (*PESPR*) instead of quoted spreads and use lagged *PESPR* in case of missing trades within any interval.

This table reports results from panel vector autoregressions using two dimensional stock-day and exchange-hour fixed effects. The vector autoregression is estimated using absolute order imbalance from single trades and from pair trades, price deviations, and proportional quoted spreads as endogenous variables with five lags.

$$LHS_{i,t} = FE + \sum_{l=1}^5 \beta_{1,l} \ln|OIB|_{single,i,t-l} + \sum_{l=1}^5 \beta_{2,l} |\Delta Price|_{i,t-l} + \sum_{l=1}^5 \beta_{3,l} \ln|OIB|_{pair,i,t-l} + \sum_{l=1}^5 \beta_{4,l} PESPR_{i,t-l} + \epsilon_{i,t}$$

where $\ln|OIB|_{single}$ and $\ln|OIB|_{pair}$ is estimated as the logarithm of one plus the absolute difference in buyer and seller initiated single and pair trades during 1-minute interval t , respectively; $|\Delta Price|$ is the maximum price deviation within interval t (from Eq. 1); and *PESPR* is average proportional effective spread within interval t . All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. For a detailed description of these variables I refer to Table 1. In all cases the table reports the sum of all lagged variables for each endogenous variable. p -values are in parentheses below the coefficients, based on a χ^2 test for testing the null hypothesis that the sum of the lagged coefficients is equal to zero when explaining the column variable. The next row reports the p -value of testing joint significance of the lagged values. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. All data underlying the computations are from TRTH.

	$PESPR_{i,t}$	$ \Delta Price _{i,t}$	$ OIB _{single,i,t}$	$ OIB _{pair,i,t}$
$\sum_{l=1}^5 OIB _{pair,i,t-l}$	-0.00014* (0.07) (0.00)	-0.016*** (0.00) (0.00)	0.076*** (0.00) (0.00)	0.25*** (0.00) (0.00)
$\sum_{l=1}^5 OIB _{single,i,t-l}$	-0.00033* (0.10) (0.00)	0.035*** (0.00) (0.00)	0.29*** (0.00) (0.00)	0.054*** (0.00) (0.00)
$\sum_{l=1}^5 \Delta Price_{i,t-l}$	-0.000038*** (0.01) (0.01)	0.7*** (0.00) (0.00)	0.0011*** (0.00) (0.00)	-0.000013 (0.81) (0.00)
$\sum_{l=1}^5 PESPR_{i,t-l}$	0.9*** (0.00) (0.00)	0.0042 (0.19) (0.09)	-0.053*** (0.00) (0.00)	-0.0067*** (0.00) (0.00)
<i>Stock</i> × <i>Day</i> FE	Yes	Yes	Yes	Yes
<i>Exchange</i> × <i>Hour</i> FE	Yes	Yes	Yes	Yes
Adj. R^2 [%]	95.07%	89.23%	60.53%	35.13%
Observations	155,759,235	155,759,235	155,759,235	155,759,235

Table OA22

Intraday Vector autoregressions and Granger causality (*PESPR*), 2001 - 2016

Compared to Table 6 I estimate the VAR using 5-minute intervals.

This table reports results from a panel vector autoregression using two dimensional stock-day and exchange-hour fixed effects. The vector autoregression is estimated using absolute order imbalance from single trades and from pair trades, price deviations, and proportional effective spreads as endogeneous variables with five lags.

$$LHS_{i,t} = FE + \sum_{l=1}^5 \beta_{1,l} |OIB|_{single,i,t-l} + \sum_{l=1}^5 \beta_{2,l} |\Delta Price|_{i,t-l} + \sum_{l=1}^5 \beta_{3,l} |OIB|_{pair,i,t-l} + \sum_{l=1}^5 \beta_{4,l} PESPR_{i,t-l} + \epsilon_{i,t}$$

where $|OIB|_{single}$ and $|OIB|_{pair}$ is estimated as the logarithm of 1 plus the absolute difference in buyer- and seller-initiated single and pair trades during 5-minute interval t , respectively; $|\Delta Price|$ is the maximum price deviation within interval t (from Eq. 1); and $PESPR$ is average proportional effective spread within interval t . All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. For a detailed description of these variables I refer to Table 1. In all cases the table reports the sum of all lagged variables for each endogenous variable. p -values are in parentheses below the coefficients, based on a χ^2 test for testing the null hypothesis that the sum of the lagged coefficients is equal to zero when explaining the column variable. The next row reports the p -value of testing joined significance of the lagged values. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. All data underlying the computations are from TRTH.

	$PESPR_{i,t}$	$ \Delta Price _{i,t}$	$ OIB _{single,i,t}$	$ OIB _{pair,i,t}$
$\sum_{l=1}^5 OIB _{pair,i,t-l}$	-0.002** (0.01) (0.00)	-0.0063 (0.15) (0.00)	0.093*** (0.00) (0.00)	0.16*** (0.00) (0.00)
$\sum_{l=1}^5 OIB _{single,i,t-l}$	0.0093*** (0.00) (0.00)	0.044*** (0.00) (0.00)	0.17*** (0.00) (0.00)	0.068*** (0.00) (0.00)
$\sum_{l=1}^5 \Delta Price _{i,t-l}$	0.00048 (0.18) (0.00)	0.42*** (0.00) (0.00)	0.0027*** (0.00) (0.00)	-0.00014 (0.38) (0.50)
$\sum_{l=1}^5 PESPR_{i,t-l}$	0.83*** (0.00) (0.00)	-0.041*** (0.00) (0.00)	-0.044*** (0.00) (0.00)	-0.0034*** (0.00) (0.00)
<i>Stock</i> $\times$ <i>Day</i> FE	Yes	Yes	Yes	Yes
<i>Exchange</i> $\times$ <i>Hour</i> FE	Yes	Yes	Yes	Yes
Adj. R^2 [%]	93.18%	85.32%	69.93%	56.41%
Observations	28,743,738	28,743,738	28,743,738	28,743,738

Table OA23
Intraday Vector autoregressions and Granger causality, 2007 - 2016

Compared to Panel A of Table 5 I only use data starting from 2007.

This table reports results from panel vector autoregressions using two dimensional stock-day and exchange-hour fixed effects. The vector autoregression is estimated using absolute order imbalance from single trades and from pair trades, price deviations, and proportional quoted spreads as endogeneous variables with five lags.

$$LHS_{i,t} = FE + \sum_{l=1}^5 \beta_{1,l} \ln|OIB|_{single,i,t-l} + \sum_{l=1}^5 \beta_{2,l} |\Delta Price|_{i,t-l} + \sum_{l=1}^5 \beta_{3,l} \ln|OIB|_{pair,i,t-l} + \sum_{l=1}^5 \beta_{4,l} PQSPR_{i,t-l} + \epsilon_{i,t}$$

where $\ln|OIB|_{single}$ and $\ln|OIB|_{pair}$ is estimated as the logarithm of one plus the absolute difference in buyer and seller initiated single and pair trades during 1-minute interval t , respectively; $|\Delta Price|$ is the maximum price deviation within interval t (from Eq. 1); and $PQSPR$ is average proportional quoted spread within interval t . All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. For a detailed description of these variables I refer to Table 1. In all cases the table reports the sum of all lagged variables for each endogenous variable. p -values are in parentheses below the coefficients, based on a χ^2 test for testing the null hypothesis that the sum of the lagged coefficients is equal to zero when explaining the column variable. The next row reports the p -value of testing joined significance of the lagged values. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. All data underlying the computations are from TRTH.

	$PQSPR_{i,t}$	$ \Delta Price _{i,t}$	$ OIB _{single,i,t}$	$ OIB _{pair,i,t}$
$\sum_{l=1}^5 OIB _{pair,i,t-l}$	-0.001*** (0.00) (0.00)	-0.017*** (0.00) (0.00)	0.069*** (0.00) (0.00)	0.257*** (0.00) (0.00)
$\sum_{l=1}^5 OIB _{single,i,t-l}$	0.003*** (0.00) (0.00)	0.036*** (0.00) (0.00)	0.305*** (0.00) (0.00)	0.056*** (0.00) (0.00)
$\sum_{l=1}^5 \Delta Price _{i,t-l}$	0.000*** (0.00) (0.00)	0.692*** (0.00) (0.00)	0.001** (0.02) (0.00)	0.000 (0.60) (0.00)
$\sum_{l=1}^5 PQSPR_{i,t-l}$	0.953*** (0.00) (0.00)	-0.018*** (0.00) (0.00)	-0.031*** (0.00) (0.00)	-0.003*** (0.00) (0.00)
<i>Stock</i> × <i>Day</i> FE	Yes	Yes	Yes	Yes
<i>Exchange</i> × <i>Hour</i> FE	Yes	Yes	Yes	Yes
Adj. R^2 [%]	98.01%	89.51%	59.62%	34.50%
Observations	110,386,929	110,386,929	110,386,929	110,386,929

Table OA24
Intraday Vector autoregressions and Granger causality, 2011 - 2016

Compared to Panel A of Table 5 I only use data starting from 2011.

This table reports results from panel vector autoregressions using two dimensional stock-day and exchange-hour fixed effects. The vector autoregression is estimated using absolute order imbalance from single trades and from pair trades, price deviations, and proportional quoted spreads as endogenous variables with five lags.

$$LHS_{i,t} = FE + \sum_{l=1}^5 \beta_{1,l} \ln|OIB|_{single,i,t-l} + \sum_{l=1}^5 \beta_{2,l} |\Delta Price|_{i,t-l} + \sum_{l=1}^5 \beta_{3,l} \ln|OIB|_{pair,i,t-l} + \sum_{l=1}^5 \beta_{4,l} PQSPR_{i,t-l} + \epsilon_{i,t}$$

where $\ln|OIB|_{single}$ and $\ln|OIB|_{pair}$ is estimated as the logarithm of one plus the absolute difference in buyer and seller initiated single and pair trades during 1-minute interval t , respectively; $|\Delta Price|$ is the maximum price deviation within interval t (from Eq. 1); and $PQSPR$ is average proportional quoted spread within interval t . All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. For a detailed description of these variables I refer to Table 1. In all cases the table reports the sum of all lagged variables for each endogenous variable. p -values are in parentheses below the coefficients, based on a χ^2 test for testing the null hypothesis that the sum of the lagged coefficients is equal to zero when explaining the column variable. The next row reports the p -value of testing joint significance of the lagged values. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. All data underlying the computations are from TRTH.

	$PQSPR_{i,t}$	$ \Delta Price _{i,t}$	$ OIB _{single,i,t}$	$ OIB _{pair,i,t}$
$\sum_{l=1}^5 OIB _{pair,i,t-l}$	-0.001*** (0.00) (0.00)	-0.025*** (0.00) (0.02)	0.056*** (0.00) (0.00)	0.263*** (0.00) (0.00)
$\sum_{l=1}^5 OIB _{single,i,t-l}$	0.002*** (0.00) (0.00)	0.049*** (0.00) (0.02)	0.319*** (0.00) (0.00)	0.062*** (0.00) (0.00)
$\sum_{l=1}^5 \Delta Price _{i,t-l}$	0.000*** (0.00) (0.00)	0.690*** (0.00) (0.00)	0.001* (0.06) (0.00)	0.000 (0.12) (0.00)
$\sum_{l=1}^5 PQSPR_{i,t-l}$	0.964*** (0.00) (0.00)	-0.027*** (0.01) (0.00)	-0.038*** (0.00) (0.00)	-0.005*** (0.00) (0.00)
<i>Stock</i> × <i>Day</i> FE	Yes	Yes	Yes	Yes
<i>Exchange</i> × <i>Hour</i> FE	Yes	Yes	Yes	Yes
Adj. R^2 [%]	98.26%	88.94%	58.69%	33.64%
Observations	66,006,144	66,006,144	66,006,144	66,006,144

Table OA25
Intraday Vector autoregressions and Granger causality, 2001 - 2016

Compared to Panel A of Table 5 I only use stock-days with at least 100 trades.

This table reports results from panel vector autoregressions using two dimensional stock-day and exchange-hour fixed effects. The vector autoregression is estimated using absolute order imbalance from single trades and from pair trades, price deviations, and proportional quoted spreads as endogeneous variables with five lags.

$$LHS_{i,t} = FE + \sum_{l=1}^5 \beta_{1,l} \ln|OIB|_{single,i,t-l} + \sum_{l=1}^5 \beta_{2,l} |\Delta Price|_{i,t-l} + \sum_{l=1}^5 \beta_{3,l} \ln|OIB|_{pair,i,t-l} + \sum_{l=1}^5 \beta_{4,l} PQSPR_{i,t-l} + \epsilon_{i,t}$$

where $\ln|OIB|_{single}$ and $\ln|OIB|_{pair}$ is estimated as the logarithm of one plus the absolute difference in buyer and seller initiated single and pair trades during 1-minute interval t , respectively; $|\Delta Price|$ is the maximum price deviation within interval t (from Eq. 1); and $PQSPR$ is average proportional quoted spread within interval t . All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. For a detailed description of these variables I refer to Table 1. In all cases the table reports the sum of all lagged variables for each endogenous variable. p -values are in parentheses below the coefficients, based on a χ^2 test for testing the null hypothesis that the sum of the lagged coefficients is equal to zero when explaining the column variable. The next row reports the p -value of testing joined significance of the lagged values. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. All data underlying the computations are from TRTH.

	$PQSPR_{i,t}$	$ \Delta Price _{i,t}$	$ OIB _{single,i,t}$	$ OIB _{pair,i,t}$
$\sum_{l=1}^5 OIB _{pair,i,t-l}$	-0.0003 (0.25) (0.00)	-0.010*** (0.01) (0.00)	0.074*** (0.00) (0.00)	0.251*** (0.00) (0.00)
$\sum_{l=1}^5 OIB _{single,i,t-l}$	0.002*** (0.00) (0.00)	0.024*** (0.00) (0.00)	0.291*** (0.00) (0.00)	0.055*** (0.00) (0.00)
$\sum_{l=1}^5 \Delta Price _{i,t-l}$	0.000*** (0.00) (0.00)	0.713*** (0.00) (0.00)	0.003*** (0.00) (0.00)	0.000 (0.32) (0.00)
$\sum_{l=1}^5 PQSPR_{i,t-l}$	0.906*** (0.00) (0.00)	-0.022*** (0.01) (0.00)	-0.082*** (0.00) (0.00)	-0.008*** (0.00) (0.00)
<i>Stock</i> × <i>Day</i> FE	Yes	Yes	Yes	Yes
<i>Exchange</i> × <i>Hour</i> FE	Yes	Yes	Yes	Yes
Adj. R^2 [%]	87.46%	92.89%	48.33%	31.73%
Observations	103,436,271	103,436,271	103,436,271	103,436,271

Table OA26
Intraday Vector autoregressions and Granger causality, 2001 - 2016

Compared to Panel A of Table 5 I use 5-minute intervals.

This table reports results from panel vector autoregressions using two dimensional stock-day and exchange-hour fixed effects. The vector autoregression is estimated using absolute order imbalance from single trades and from pair trades, price deviations, and proportional quoted spreads as endogenous variables with five lags.

$$LHS_{i,t} = FE + \sum_{l=1}^5 \beta_{1,l} \ln|OIB|_{single,i,t-l} + \sum_{l=1}^5 \beta_{2,l} |\Delta Price|_{i,t-l} + \sum_{l=1}^5 \beta_{3,l} \ln|OIB|_{pair,i,t-l} + \sum_{l=1}^5 \beta_{4,l} PQSPR_{i,t-l} + \epsilon_{i,t}$$

where $\ln|OIB|_{single}$ and $\ln|OIB|_{pair}$ is estimated as the logarithm of one plus the absolute difference in buyer and seller initiated single and pair trades during 5-minute interval t , respectively; $|\Delta Price|$ is the maximum price deviation within interval t (from Eq. 1); and $PQSPR$ is average proportional quoted spread within interval t . All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. For a detailed description of these variables I refer to Table 1. In all cases the table reports the sum of all lagged variables for each endogenous variable. p -values are in parentheses below the coefficients, based on a χ^2 test for testing the null hypothesis that the sum of the lagged coefficients is equal to zero when explaining the column variable. The next row reports the p -value of testing joint significance of the lagged values. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. All data underlying the computations are from TRTH.

	$PQSPR_{i,t}$	$ \Delta Price _{i,t}$	$ OIB _{single,i,t}$	$ OIB _{pair,i,t}$
$\sum_{l=1}^5 OIB _{pair,i,t-l}$	0.001 (0.17) (0.00)	-0.006 (0.14) (0.00)	0.093*** (0.00) (0.00)	0.163*** (0.00) (0.00)
$\sum_{l=1}^5 OIB _{single,i,t-l}$	0.003*** (0.00) (0.00)	0.044*** (0.00) (0.00)	0.169*** (0.00) (0.00)	0.068*** (0.00) (0.00)
$\sum_{l=1}^5 \Delta Price _{i,t-l}$	0.000*** (0.00) (0.00)	0.420*** (0.00) (0.00)	0.003*** (0.00) (0.00)	0.000 (0.35) (0.62)
$\sum_{l=1}^5 PQSPR_{i,t-l}$	0.875*** (0.00) (0.00)	-0.031*** (0.00) (0.00)	-0.049*** (0.00) (0.00)	-0.005*** (0.00) (0.00)
<i>Stock</i> × <i>Day</i> FE	Yes	Yes	Yes	Yes
<i>Exchange</i> × <i>Hour</i> FE	Yes	Yes	Yes	Yes
Adj. R^2 [%]	96.02%	85.31%	69.94%	56.42%
Observations	28,743,738	28,743,738	28,743,738	28,743,738

Table OA27
Intraday Vector autoregressions and Granger causality, 2001 - 2016

Compared to Panel A of Table 5 I only use home-market stocks.

This table reports results from panel vector autoregressions using two dimensional stock-day and exchange-hour fixed effects. The vector autoregression is estimated using absolute order imbalance from single trades and from pair trades, price deviations, and proportional quoted spreads as endogeneous variables with five lags.

$$LHS_{i,t} = FE + \sum_{l=1}^5 \beta_{1,l} \ln|OIB|_{single,i,t-l} + \sum_{l=1}^5 \beta_{2,l} |\Delta Price|_{i,t-l} + \sum_{l=1}^5 \beta_{3,l} \ln|OIB|_{pair,i,t-l} + \sum_{l=1}^5 \beta_{4,l} PQSPR_{i,t-l} + \epsilon_{i,t}$$

where $\ln|OIB|_{single}$ and $\ln|OIB|_{pair}$ is estimated as the logarithm of one plus the absolute difference in buyer and seller initiated single and pair trades during 1-minute interval t , respectively; $|\Delta Price|$ is the maximum price deviation within interval t (from Eq. 1); and $PQSPR$ is average proportional quoted spread within interval t . All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. For a detailed description of these variables I refer to Table 1. In all cases the table reports the sum of all lagged variables for each endogenous variable. p -values are in parentheses below the coefficients, based on a χ^2 test for testing the null hypothesis that the sum of the lagged coefficients is equal to zero when explaining the column variable. The next row reports the p -value of testing joined significance of the lagged values. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. All data underlying the computations are from TRTH.

	$PQSPR_{i,t}$	$ \Delta Price _{i,t}$	$ OIB _{single,i,t}$	$ OIB _{pair,i,t}$
$\sum_{l=1}^5 OIB _{pair,i,t-l}$	-0.0021*** (0.00) (0.00)	-0.023*** (0.00) (0.00)	0.067*** (0.00) (0.00)	0.256*** (0.00) (0.00)
$\sum_{l=1}^5 OIB _{single,i,t-l}$	0.005*** (0.00) (0.00)	0.048*** (0.00) (0.00)	0.314*** (0.00) (0.00)	0.061*** (0.00) (0.00)
$\sum_{l=1}^5 \Delta Price _{i,t-l}$	0.000*** (0.00) (0.00)	0.699*** (0.00) (0.00)	0.001** (0.03) (0.00)	0.000*** (0.00) (0.00)
$\sum_{l=1}^5 PQSPR_{i,t-l}$	0.939*** (0.00) (0.00)	-0.005** (0.02) (0.00)	-0.019*** (0.00) (0.00)	-0.001*** (0.00) (0.00)
<i>Stock</i> × <i>Day</i> FE	Yes	Yes	Yes	Yes
<i>Exchange</i> × <i>Hour</i> FE	Yes	Yes	Yes	Yes
Adj. R^2 [%]	97.62%	89.23%	56.36%	34.07%
Observations	77,879,518	77,879,518	77,879,518	77,879,518

Table OA28
Intraday Vector autoregressions and Granger causality, 2001 - 2016

Compared to Panel A of Table 5 I only use host-market stocks.

This table reports results from panel vector autoregressions using two dimensional stock-day and exchange-hour fixed effects. The vector autoregression is estimated using absolute order imbalance from single trades and from pair trades, price deviations, and proportional quoted spreads as endogeneous variables with five lags.

$$LHS_{i,t} = FE + \sum_{l=1}^5 \beta_{1,l} \ln|OIB|_{single,i,t-l} + \sum_{l=1}^5 \beta_{2,l} |\Delta Price|_{i,t-l} + \sum_{l=1}^5 \beta_{3,l} \ln|OIB|_{pair,i,t-l} + \sum_{l=1}^5 \beta_{4,l} PQSPR_{i,t-l} + \epsilon_{i,t}$$

where $\ln|OIB|_{single}$ and $\ln|OIB|_{pair}$ is estimated as the logarithm of one plus the absolute difference in buyer and seller initiated single and pair trades during 1-minute interval t , respectively; $|\Delta Price|$ is the maximum price deviation within interval t (from Eq. 1); and $PQSPR$ is average proportional quoted spread within interval t . All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. For a detailed description of these variables I refer to Table 1. In all cases the table reports the sum of all lagged variables for each endogenous variable. p -values are in parentheses below the coefficients, based on a χ^2 test for testing the null hypothesis that the sum of the lagged coefficients is equal to zero when explaining the column variable. The next row reports the p -value of testing joined significance of the lagged values. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. All data underlying the computations are from TRTH.

	$PQSPR_{i,t}$	$ \Delta Price _{i,t}$	$\ln OIB _{single,i,t}$	$\ln OIB _{pair,i,t}$
$\sum_{l=1}^5 \ln OIB_{pair,i,t-l}$	-0.0003** (0.04) (0.00)	-0.005* (0.08) (0.00)	0.102*** (0.00) (0.00)	0.243*** (0.00) (0.00)
$\sum_{l=1}^5 \ln OIB_{single,i,t-l}$	0.003*** (0.00) (0.00)	0.017*** (0.00) (0.02)	0.241*** (0.00) (0.00)	0.047*** (0.00) (0.00)
$\sum_{l=1}^5 \Delta Price _{i,t-l}$	0.000* (0.05) (0.00)	0.699*** (0.00) (0.00)	0.001*** (0.00) (0.00)	0.000* (0.06) (0.00)
$\sum_{l=1}^5 PQSPR_{i,t-l}$	0.964*** (0.00) (0.00)	-0.022*** (0.00) (0.00)	-0.029*** (0.00) (0.00)	-0.003*** (0.00) (0.00)
<i>Stock</i> × <i>Day</i> FE	Yes	Yes	Yes	Yes
<i>Exchange</i> × <i>Hour</i> FE	Yes	Yes	Yes	Yes
Adj. R^2 [%]	98.64%	89.23%	64.06%	36.29%
Observations	77,879,717	77,879,717	77,879,717	77,879,717

Table OA29
Intraday Vector autoregressions and Granger causality, 2001 - 2016

Compared to Panel A of Table 5 I estimate panel regressions with variables from both home- and host-market stocks entering the regression simultaneously.

This table reports results from panel vector autoregressions using two dimensional stock-day and exchange-hour fixed effects. The vector autoregression is estimated using absolute order imbalance from single trades and from pair trades, price deviations, and proportional quoted spreads as endogenous variables with five lags.

$$LHS_{i,t} = FE + \sum_{l=1}^5 \beta_{1,l} \ln|OIB|_{single,i,t-l} + \sum_{l=1}^5 \beta_{2,l} |\Delta Price|_{i,t-l} + \sum_{l=1}^5 \beta_{3,l} \ln|OIB|_{pair,i,t-l} + \sum_{l=1}^5 \beta_{4,l} PQSPR_{i,t-l} + \epsilon_{i,t}$$

where $\ln|OIB|_{single}$ and $\ln|OIB|_{pair}$ is estimated as the logarithm of one plus the absolute difference in buyer and seller initiated single and pair trades during 1-minute interval t , respectively; $|\Delta Price|$ is the maximum price deviation within interval t (from Eq. 1); and $PQSPR$ is average proportional quoted spread within interval t . All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. For a detailed description of these variables I refer to Table 1. In all cases the table reports the sum of all lagged variables for each endogenous variable. p -values are in parentheses below the coefficients, based on a χ^2 test for testing the null hypothesis that the sum of the lagged coefficients is equal to zero when explaining the column variable. The next row reports the p -value of testing joint significance of the lagged values. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. All data underlying the computations are from TRTH.

	$PQSPR_{home,i,t}$	$PQSPR_{host,i,t}$	$ \Delta Price _{i,t}$	$\ln OIB _{home,single,i,t}$	$\ln OIB _{host,single,i,t}$	$\ln OIB _{pair,i,t}$
$\sum_{l=1}^5 \ln OIB_{pair,i,t-l}$	-0.0025*** (0.00) (0.00)	-0.0006*** (0.00) (0.00)	-0.030*** (0.00) (0.00)	0.062*** (0.00) (0.00)	0.092*** (0.00) (0.00)	0.218*** (0.00) (0.00)
$\sum_{l=1}^5 \ln OIB_{home,single,i,t-l}$	-0.005*** (0.00) (0.00)	0.001*** (0.00) (0.00)	0.046*** (0.00) (0.00)	0.319*** (0.00) (0.00)	0.017*** (0.00) (0.00)	0.045*** (0.00) (0.00)
$\sum_{l=1}^5 \ln OIB_{host,single,i,t-l}$	-0.001*** (0.00) (0.00)	0.003*** (0.00) (0.00)	0.012*** (0.00) (0.11)	-0.001 (0.87) (0.00)	0.239*** (0.00) (0.00)	0.042*** (0.00) (0.00)
$\sum_{l=1}^5 \Delta Price _{i,t-l}$	-0.0003*** (0.00) (0.00)	-0.0001** (0.05) (0.00)	0.699*** (0.00) (0.00)	0.001** (0.04) (0.00)	0.000*** (0.00) (0.00)	0.000*** (0.00) (0.00)
$\sum_{l=1}^5 PQSPR_{home,i,t-l}$	0.939*** (0.00) (0.00)	0.000 (0.95) (0.04)	-0.005* (0.06) (0.00)	-0.019*** (0.00) (0.00)	0.001*** (0.00) (0.00)	-0.001*** (0.00) (0.00)
$\sum_{l=1}^5 PQSPR_{host,i,t-l}$	0.001*** (0.01) (0.01)	0.964*** (0.00) (0.00)	-0.022*** (0.00) (0.00)	-0.011*** (0.00) (0.00)	-0.029*** (0.00) (0.00)	-0.003*** (0.00) (0.00)
<i>Stock</i> × <i>Day</i> FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>Exchange</i> × <i>Hour</i> FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2 [%]	97.62%	98.64%	89.23%	56.35%	64.07%	36.40%

Table OA29 continued

	$PQSPR_{home,i,t}$	$PQSPR_{host,i,t}$	$ \Delta Price _{i,t}$	$\ln OIB _{home,single,i,t}$	$\ln OIB _{host,single,i,t}$	$\ln OIB _{pair,i,t}$
Observations	77,879,350	77,879,350	77,879,350	77,879,350	77,879,350	77,879,350

Table OA30
Intraday Vector autoregressions and Granger causality, 2001 - 2016

Compared to Panel A of Table 5 I use the number of trades, instead of *OIB*.

This table reports results from panel vector autoregressions using two dimensional stock-day and exchange-hour fixed effects. The vector autoregression is estimated using absolute order imbalance from single trades and from pair trades, price deviations, and proportional quoted spreads as endogenous variables with five lags.

$$LHS_{i,t} = FE + \sum_{l=1}^5 \beta_{1,l} \ln Trades_{single,i,t-l} + \sum_{l=1}^5 \beta_{2,l} |\Delta Price|_{i,t-l} + \sum_{l=1}^5 \beta_{3,l} \ln Trades_{pair,i,t-l} + \sum_{l=1}^5 \beta_{4,l} PQSPR_{i,t-l} + \epsilon_{i,t}$$

where $\ln Trades_{single}$ and $\ln Trades_{pair}$ is estimated as the logarithm of one plus the the number of single and pair trades during 1-minute interval t , respectively; $|\Delta Price|$ is the maximum price deviation within interval t (from Eq. 1); and $PQSPR$ is average proportional quoted spread within interval t . All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. For a detailed description of these variables I refer to Table 1. In all cases the table reports the sum of all lagged variables for each endogenous variable. p -values are in parentheses below the coefficients, based on a χ^2 test for testing the null hypothesis that the sum of the lagged coefficients is equal to zero when explaining the column variable. The next row reports the p -value of testing joined significance of the lagged values. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. All data underlying the computations are from TRTH.

	$PQSPR_{i,t}$	$ \Delta Price _{i,t}$	$\ln Trades_{single,i,t}$	$\ln Trades_{pair,i,t}$
$\sum_{l=1}^5 \ln Trades_{pair,i,t-l}$	-0.0017*** (0.00) (0.00)	-0.021*** (0.00) (0.00)	0.002 (0.76) (0.00)	0.301*** (0.00) (0.00)
$\sum_{l=1}^5 \ln Trades_{single,i,t-l}$	0.004*** (0.00) (0.00)	0.036*** (0.00) (0.00)	0.482*** (0.00) (0.00)	0.062*** (0.00) (0.00)
$\sum_{l=1}^5 \Delta Price _{i,t-l}$	0.000*** (0.00) (0.00)	0.699*** (0.00) (0.00)	0.000 (0.27) (0.00)	0.000* (0.05) (0.00)
$\sum_{l=1}^5 PQSPR_{i,t-l}$	0.949*** (0.00) (0.00)	-0.012*** (0.00) (0.00)	-0.025*** (0.00) (0.00)	-0.002*** (0.00) (0.00)
<i>Stock</i> × <i>Day</i> FE	Yes	Yes	Yes	Yes
<i>Exchange</i> × <i>Hour</i> FE	Yes	Yes	Yes	Yes
Adj. R^2 [%]	98.02%	89.23%	76.40%	41.12%
Observations	155,759,235	155,759,235	155,759,235	155,759,235

Table OA31
Intraday Vector autoregressions and Granger causality, 2001 - 2016

Compared to Panel A of Table 5 I estimate OIB as the difference in buyer and seller initiated dollars trading within each 1-minute interval, instead of as the difference in the number of buyer and seller initiated trades.

This table reports results from panel vector autoregressions using two dimensional stock-day and exchange-hour fixed effects. The vector autoregression is estimated using absolute order imbalance from single trades and from pair trades, price deviations, and proportional quoted spreads as endogenous variables with five lags.

$$LHS_{i,t} = FE + \sum_{l=1}^5 \beta_{1,l} \ln|OIB|_{single,i,t-l} + \sum_{l=1}^5 \beta_{2,l} |\Delta Price|_{i,t-l} + \sum_{l=1}^5 \beta_{3,l} \ln|OIB|_{pair,i,t-l} + \sum_{l=1}^5 \beta_{4,l} PQSPR_{i,t-l} + \epsilon_{i,t}$$

where $\ln|OIB|_{single}$ and $\ln|OIB|_{pair}$ is estimated as the logarithm of one plus the absolute difference in buyer and seller initiated single and pair trades during 1-minute interval t , respectively; $|\Delta Price|$ is the maximum price deviation within interval t (from Eq. 1); and $PQSPR$ is average proportional quoted spread within interval t . All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. For a detailed description of these variables I refer to Table 1. In all cases the table reports the sum of all lagged variables for each endogenous variable. p -values are in parentheses below the coefficients, based on a χ^2 test for testing the null hypothesis that the sum of the lagged coefficients is equal to zero when explaining the column variable. The next row reports the p -value of testing joined significance of the lagged values. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. All data underlying the computations are from TRTH.

	$PQSPR_{i,t}$	$ \Delta Price _{i,t}$	$ OIB _{single,i,t}$	$ OIB _{pair,i,t}$
$\sum_{l=1}^5 OIB _{pair,i,t-l}$	-0.00003 (0.39) (0.00)	-0.001*** (0.01) (0.00)	0.058*** (0.00) (0.00)	0.227*** (0.00) (0.00)
$\sum_{l=1}^5 OIB _{single,i,t-l}$	0.001*** (0.00) (0.00)	0.012*** (0.00) (0.00)	0.334*** (0.00) (0.00)	0.064*** (0.00) (0.00)
$\sum_{l=1}^5 \Delta Price _{i,t-l}$	0.000*** (0.00) (0.00)	0.699*** (0.00) (0.00)	0.004*** (0.00) (0.00)	-0.001* (0.08) (0.00)
$\sum_{l=1}^5 PQSPR_{i,t-l}$	0.949*** (0.00) (0.00)	-0.011*** (0.00) (0.00)	-0.146*** (0.00) (0.00)	-0.012*** (0.00) (0.00)
<i>Stock</i> $\times$ <i>Day</i> FE	Yes	Yes	Yes	Yes
<i>Exchange</i> $\times$ <i>Hour</i> FE	Yes	Yes	Yes	Yes
Adj. R^2 [%]	98.02%	89.23%	67.05%	39.06%
Observations	155,759,235	155,759,235	155,759,235	155,759,235

Table OA32
Intraday Vector autoregressions and Granger causality, 2001 - 2016

Compared to Panel A of Table 5 I use price improvements instead of *PQSPR*.

This table reports results from panel vector autoregressions using two dimensional stock-day and exchange-hour fixed effects. The vector autoregression is estimated using absolute order imbalance from single trades and from pair trades, price deviations, and price improvements as endogeneous variables with five lags.

$$LHS_{i,t} = FE + \sum_{l=1}^5 \beta_{1,l} \ln|OIB|_{single,i,t-l} + \sum_{l=1}^5 \beta_{2,l} |\Delta Price|_{i,t-l} + \sum_{l=1}^5 \beta_{3,l} \ln|OIB|_{pair,i,t-l} + \sum_{l=1}^5 \beta_{4,l} PriceImpr_{i,t-l} + \epsilon_{i,t}$$

where $\ln|OIB|_{single}$ and $\ln|OIB|_{pair}$ is estimated as the logarithm of one plus the absolute difference in buyer and seller initiated single and pair trades during 1-minute interval t , respectively; $|\Delta Price|$ is the maximum price deviation within interval t (from Eq. 1); and $PriceImpr$ is the average difference in proportional quoted spreads prevailing trades and proportional effective spreads ($PQSPR[t] - PESPR[t]$) within interval t . All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. For a detailed description of these variables I refer to Table 1. In all cases the table reports the sum of all lagged variables for each endogenous variable. p -values are in parentheses below the coefficients, based on a χ^2 test for testing the null hypothesis that the sum of the lagged coefficients is equal to zero when explaining the column variable. The next row reports the p -value of testing joined significance of the lagged values. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. All data underlying the computations are from TRTH.

	$PriceImpr_{i,t}$	$ \Delta Price _{i,t}$	$ OIB _{single,i,t}$	$ OIB _{pair,i,t}$
$\sum_{l=1}^5 OIB _{pair,i,t-l}$	-0.00037 (0.27) (0.00)	-0.013*** (0.00) (0.00)	0.08*** (0.00) (0.00)	0.26*** (0.00) (0.00)
$\sum_{l=1}^5 OIB _{single,i,t-l}$	-0.0062*** (0.00) (0.00)	0.024*** (0.00) (0.00)	0.3*** (0.00) (0.00)	0.053*** (0.00) (0.00)
$\sum_{l=1}^5 \Delta Price _{i,t-l}$	-0.00032*** (0.00) (0.00)	0.7*** (0.00) (0.00)	0.0014*** (0.00) (0.00)	0.000054 (0.39) (0.00)
$\sum_{l=1}^5 PriceImpr_{i,t-l}$	0.29*** (0.00) (0.00)	-0.29*** (0.00) (0.00)	-0.035*** (0.01) (0.00)	0.035*** (0.00) (0.00)
<i>Stock</i> $\times$ <i>Day</i> FE	Yes	Yes	Yes	Yes
<i>Exchange</i> $\times$ <i>Hour</i> FE	Yes	Yes	Yes	Yes
Adj. R^2 [%]	17.39%	89.24%	57.62%	34.02%
Observations	155,702,353	155,702,353	155,702,353	155,702,353

Table OA33

How days between corporate actions affect trading, 2001 - 2016

This table reports event fixed-effect panel regression results of explaining trading twenty days around days between corporate actions (the *Event*), i.e., when either the host or the home-market is cum-dividend but the other is ex-dividend. The table reports the ratio of pair trades to single trades (Panel A), the number of all trades (Panel B), and the number of single trades (Panel C). Single trades for the host (home) market are trades without a simultaneous trade in the home (host) market. Pair trades are simultaneous trades of both the host- and the home-market share with opposite signs. All trades and single trades are reported separately for the home and the host market.

$$T_{i,d} = FE_e + \sum_{t=-5}^5 \alpha_t D_t + \epsilon_{i,d}$$

$T_{i,d}$ is the ratio of pair to single trades (Panel A) or the number of trades (Panels B and C) for stock i on day d so that d is within 20 days of an Event for stock i . D_t are dummy variables set equal to one if the Event is t days away and zero otherwise. FE_e are event fixed effects. Standard errors are clustered by stock and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. p -values are in parentheses below the coefficients. All data underlying the computations are from TRTH.

	Panel A: <i>Pair/Single</i>	Panel B: All Trades Home	Host	Panel C: Single Trades Home	Host
α_{-5}	-0.083 (0.74)	31.032 (0.47)	25.414 (0.51)	8.687 (0.67)	3.067 (0.86)
α_{-4}	0.206 (0.45)	24.877 (0.63)	84.327* (0.06)	5.206 (0.88)	64.651** (0.02)
α_{-3}	-0.181 (0.47)	13.793 (0.86)	44.866 (0.59)	-2.321 (0.94)	28.756 (0.24)
α_{-2}	-0.914*** (0.00)	-132.139*** (0.00)	-113.142*** (0.00)	0.631 (0.98)	19.637 (0.23)
α_{-1}	-1.259*** (0.00)	-136.057*** (0.01)	-134.078*** (0.00)	19.003 (0.50)	20.988 (0.32)
Event (α_0)	-0.850*** (0.01)	-8.457 (0.88)	-4.152 (0.93)	86.894*** (0.01)	91.201*** (0.00)
α_{+1}	-0.231 (0.25)	68.403 (0.25)	45.442 (0.24)	78.811** (0.04)	55.849** (0.01)
α_{+2}	-0.177 (0.44)	-24.831 (0.74)	-51.846 (0.20)	32.970 (0.51)	5.960 (0.75)
α_{+3}	-0.226 (0.48)	-81.209 (0.24)	-51.610 (0.41)	-20.710 (0.54)	8.894 (0.71)
α_{+4}	-0.087 (0.74)	-33.419 (0.57)	57.636 (0.39)	-2.239 (0.93)	88.818** (0.01)
α_{+5}	0.030 (0.88)	45.586 (0.49)	37.102 (0.39)	40.929 (0.35)	32.425* (0.09)
Event FE	Yes	Yes	Yes	Yes	Yes
Adj. R^2 [%]	67.85	75.59	72.11	83.24	80.55
Observations	43,296	43,296	45 43,296	43,296	43,296

Table OA34

Instrumental variable regressions to estimate arbitrage on illiquidity, 2001 - 2016

Compared to Table 8 of the paper I use stock-days with at least five pair trades.

This table presents the second stage regressions of instrumenting the ratio of pair to single trades when explaining illiquidity by a dummy variable which is one on days between corporate actions, i.e., when either the host or the home-market is cum-dividend but the other is ex-dividend. Each column presents the results of the following panel regression with two dimensional stock-month fixed effects:

$$Illi_{i,d} = FE + \beta_1 \times \overline{Pair/Single}_{i,d} + \zeta_1 \times \mathbf{Controls}_{i,d} + \epsilon_{i,d}$$

where illiquidity ($Illi_{i,d}$) of home- or host-market stock i on day d is measured by the average proportional quoted spread ($PQSPR$), proportional effective spread ($PESPR$), or the logarithm of the average absolute pricing error ($\ln|Has|$) immediately after *pair* trades. Averages are taken only after trades that occur when the stock is mispriced (i.e., when the stock is *Cheap*, *Costly*, or in between corporate actions) and days with less than five *pair* trades are dropped from the analysis. $\overline{Pair/Single}_{i,d}$ is the fitted value from first stage regressions. And **Controls** is a vector of control variables. For a description of these variables I refer to Tables 1 and 7. All stock specific variables (except *price*) are measured during the overlapping trading time, i.e. when both the home- and host-market are in their continuous trading session. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. R^2 statistics are not reported, because they cannot be properly interpreted in two stage regressions. p -values are in parentheses below the coefficients. Data to compute *price* are from Datastream, *Ted* is from the Federal Reserve Bank of St. Louis, and all other data underlying the computations are from TRTH.

	$PQSPR_{i,d}$		$PESPR_{i,d}$		$\ln Has _{i,d}$	
$\overline{Pair/Single}_{i,d}$	-0.811** (0.04)	-0.735** (0.04)	-0.138** (0.01)	-0.128** (0.01)	-1.612*** (0.00)	-1.562*** (0.00)
$CorpAct_{i,d}$		-0.008 (0.33)		-0.001 (0.52)		-0.017 (0.19)
$\log(price)_{i,d}$		-0.064* (0.07)		-0.056*** (0.00)		-0.377*** (0.00)
$Vola_{i,d}$		0.608** (0.01)		0.170*** (0.00)		2.311*** (0.00)
$OIB_{i,d}$		0.000 (0.13)		0.000 (0.14)		0.000 (0.37)
$Trades_{i,d}$		0.067* (0.09)		0.012** (0.05)		-0.060* (0.06)
$Quotes_{i,d}$		-0.013 (0.12)		-0.003* (0.08)		-0.003 (0.65)
$Trsize_{i,d}$		0.087* (0.05)		0.003 (0.67)		0.029 (0.48)
Ted_d		0.002 (0.63)		0.004*** (0.00)		0.068*** (0.00)
$Stock \times Month$ FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	455,116	445,234	455,116	445,234	453,470	443,603

Table OA35

Instrumental variable regressions to estimate arbitrage on illiquidity, 2001 - 2016

Compared to Table 8 of the paper I use stock-days with at least twenty pair trades.

This table presents the second stage regressions of instrumenting the ratio of pair to single trades when explaining illiquidity by a dummy variable which is one on days between corporate actions, i.e., when either the host or the home-market is cum-dividend but the other is ex-dividend. Each column presents the results of the following panel regression with two dimensional stock-month fixed effects:

$$Illi_{i,d} = FE + \beta_1 \times \overline{Pair/Single}_{i,d} + \zeta_1 \times \mathbf{Controls}_{i,d} + \epsilon_{i,d}$$

where illiquidity ($Illi_{i,d}$) of home- or host-market stock i on day d is measured by the average proportional quoted spread ($PQSPR$), proportional effective spread ($PESPR$), or the logarithm of the average absolute pricing error ($\ln|Has|$) immediately after *pair* trades. Averages are taken only after trades that occur when the stock is mispriced (i.e., when the stock is *Cheap*, *Costly*, or in between corporate actions) and days with less than twenty *pair* trades are dropped from the analysis. $\overline{Pair/Single}_{i,d}$ is the fitted value from first stage regressions. And **Controls** is a vector of control variables. For a description of these variables I refer to Tables 1 and 7. All stock specific variables (except *price*) are measured during the overlapping trading time, i.e. when both the home- and host-market are in their continuous trading session. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. R^2 statistics are not reported, because they cannot be properly interpreted in two stage regressions. p -values are in parentheses below the coefficients. Data to compute *price* are from Datastream, *Ted* is from the Federal Reserve Bank of St. Louis, and all other data underlying the computations are from TRTH.

	$PQSPR_{i,d}$		$PESPR_{i,d}$		$\ln Has _{i,d}$	
$\overline{Pair/Single}_{i,d}$	-0.193*	-0.171*	-0.073*	-0.069*	-1.437***	-1.386***
	(0.06)	(0.05)	(0.08)	(0.07)	(0.00)	(0.00)
$CorpAct_{i,d}$		-0.004		-0.000		-0.006
		(0.54)		(0.97)		(0.51)
$\log(price)_{i,d}$		-0.084***		-0.038***		-0.339***
		(0.00)		(0.00)		(0.00)
$Vola_{i,d}$		0.274***		0.140***		2.381***
		(0.00)		(0.00)		(0.00)
$OIB_{i,d}$		0.000		0.000		0.000
		(0.12)		(0.19)		(0.45)
$Trades_{i,d}$		0.013		0.006		-0.078**
		(0.13)		(0.13)		(0.01)
$Quotes_{i,d}$		-0.005*		-0.002*		-0.005
		(0.08)		(0.10)		(0.43)
$Trsize_{i,d}$		0.022		0.009		0.082
		(0.13)		(0.13)		(0.22)
Ted_d		0.009***		0.004***		0.061***
		(0.00)		(0.00)		(0.00)
$Stock \times Month$ FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	362,663	354,840	362,663	354,840	361,337	353,521

Table OA36

Instrumental variable regressions to estimate arbitrage on illiquidity, 2001 - 2016

Compared to Table 8 of the paper I use the daily maximum price deviations as an inverse measure of arbitrage activity.

This table presents the second stage regressions of instrumenting the daily maximum price deviations ($\max(\Delta Price)$) when explaining illiquidity by a dummy variable which is one on days between corporate actions, i.e., when either the host or the home-market is cum-dividend but the other is ex-dividend. Each column presents the results of the following panel regression with two dimensional stock-month fixed effects:

$$Illiq_{i,d} = FE + \beta_1 \times \overline{\max(\Delta Price)} + \zeta_1 \times \mathbf{Controls}_{i,d} + \epsilon_{i,d}$$

where illiquidity ($Illiq_{i,d}$) of home- or host-market stock i on day d is measured by the average proportional quoted spread ($PQSPR$), proportional effective spread ($PESPR$), or the logarithm of the average absolute pricing error ($\ln|Has|$) immediately after *pair* trades. Averages are taken only after trades that occur when the stock is mispriced (i.e., when the stock is *Cheap*, *Costly*, or in between corporate actions) and days with less than twenty *pair* trades are dropped from the analysis. $\overline{\max(\Delta Price)}$ is the fitted value from first stage regressions. And **Controls** is a vector of control variables. For a description of these variables I refer to Tables 1 and 7. All stock specific variables (except *price*) are measured during the overlapping trading time, i.e. when both the home- and host-market are in their continuous trading session. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. R^2 statistics are not reported, because they cannot be properly interpreted in two stage regressions. p -values are in parentheses below the coefficients. Data to compute *price* are from Datastream, *Ted* is from the Federal Reserve Bank of St. Louis, and all other data underlying the computations are from TRTH.

	$PQSPR_{i,d}$		$PESPR_{i,d}$		$\ln Has _{i,d}$	
$\overline{\max(\Delta Price)}$	0.005** (0.04)	0.005** (0.05)	0.004* (0.05)	0.004* (0.05)	0.026*** (0.01)	0.028*** (0.00)
$CorpAct_{i,d}$		-0.005 (0.42)		-0.000 (0.81)		-0.009 (0.23)
$\log(price)_{i,d}$		-0.111*** (0.00)		-0.103*** (0.00)		-0.468*** (0.00)
$Vola_{i,d}$		0.109*** (0.00)		0.160*** (0.00)		1.171*** (0.00)
$OIB_{i,d}$		0.000 (0.47)		-0.000 (0.85)		-0.000** (0.04)
$Trades_{i,d}$		-0.002** (0.02)		-0.001 (0.15)		-0.205*** (0.00)
$Quotes_{i,d}$		-0.001 (0.39)		-0.001 (0.12)		0.023* (0.09)
$Trsize_{i,d}$		0.000 (0.98)		-0.003 (0.66)		-0.080** (0.02)
Ted_d		0.010*** (0.00)		0.009*** (0.00)		0.077*** (0.00)
$Stock \times Month$ FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	407,684	398,822	407,684	398,822	406,274	397,418

Table OA36 continued

Table OA37

Instrumental variable regressions to estimate arbitrage on illiquidity, 2001 - 2016

Compared to Table 8 of the paper I use the speed of mean-reversion as an alternative measure of arbitrage activity.

This table presents the second stage regressions of instrumenting the speed of mean-reversion (θ) when explaining illiquidity by a dummy variable which is one on days between corporate actions, i.e., when either the host or the home-market is cum-dividend but the other is ex-dividend. Each column presents the results of the following panel regression with two dimensional stock-month fixed effects:

$$Illiq_{i,d} = FE + \beta_1 \times \hat{\theta} + \zeta_1 \times \mathbf{Controls}_{i,d} + \epsilon_{i,d}$$

where illiquidity ($Illiq_{i,d}$) of home- or host-market stock i on day d is measured by the average proportional quoted spread ($PQSPR$), proportional effective spread ($PESPR$), or the logarithm of the average absolute pricing error ($\ln|Has|$) immediately after *pair* trades. Averages are taken only after trades that occur when the stock is mispriced (i.e., when the stock is *Cheap*, *Costly*, or in between corporate actions) and days with less than twenty *pair* trades are dropped from the analysis. $\hat{\theta}$ is the fitted value from first stage regressions. And **Controls** is a vector of control variables. For a description of these variables I refer to Tables 1 and 7. All stock specific variables (except *price*) are measured during the overlapping trading time, i.e. when both the home- and host-market are in their continuous trading session. Standard errors are clustered by stock-pair and statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. R^2 statistics are not reported, because they cannot be properly interpreted in two stage regressions. p -values are in parentheses below the coefficients. Data to compute *price* are from Datastream, *Ted* is from the Federal Reserve Bank of St. Louis, and all other data underlying the computations are from TRTH.

	$PQSPR_{i,d}$		$PESPR_{i,d}$		$\ln Has _{i,d}$	
$\hat{\theta}$	-0.523** (0.02)	-0.529** (0.02)	-0.420** (0.03)	-0.439** (0.03)	-2.739*** (0.00)	-2.991*** (0.00)
$CorpAct_{i,d}$		-0.005 (0.36)		-0.001 (0.61)		-0.013 (0.11)
$\log(price)_{i,d}$		-0.113*** (0.00)		-0.104*** (0.00)		-0.474*** (0.00)
$Vola_{i,d}$		0.170*** (0.00)		0.211*** (0.00)		1.515*** (0.00)
$OIB_{i,d}$		0.000 (0.32)		0.000 (0.78)		-0.000** (0.04)
$Trades_{i,d}$		-0.000 (0.78)		0.001 (0.51)		-0.196*** (0.00)
$Quotes_{i,d}$		0.001 (0.43)		0.001 (0.59)		0.034* (0.07)
$Trsize_{i,d}$		0.001 (0.96)		-0.003 (0.69)		-0.078** (0.03)
Ted_d		0.010*** (0.00)		0.009*** (0.00)		0.077*** (0.00)
$Stock \times Month$ FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	407,655	398,795	407,655	398,795	406,247	397,393

Table OA37 continued

Fig. OA1. Intraday Impulse response functions from single and pair trades and quoted spreads, 2001- 2016

Compared to Figure 1 of the paper I report IRFs from all variables.

This figure shows impulse response functions (IRF) from panel vector autoregression (VAR) estimated similar as in Panel A of Table 5 of the paper, as described in more details in the main text. For a description of the VAR and these variables I refer to Tables 1 and 5 of the paper. All IRF in the first row show responses in OIB_{single} , $|\Delta Price|$, OIB_{pair} , and $PQSPR$ to a Cholesky one standard-deviation shock to OIB_{single} in minute 0 (the contemporaneous effect) to minute 5. The following rows show responses to shocks to the other variables in the system. All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. Each figure shows bootstrapped 95% confidence bands based on 1000 runs (lower, upper), which in most cases completely overlap with the IRF. All data underlying the computations are from TRTH.

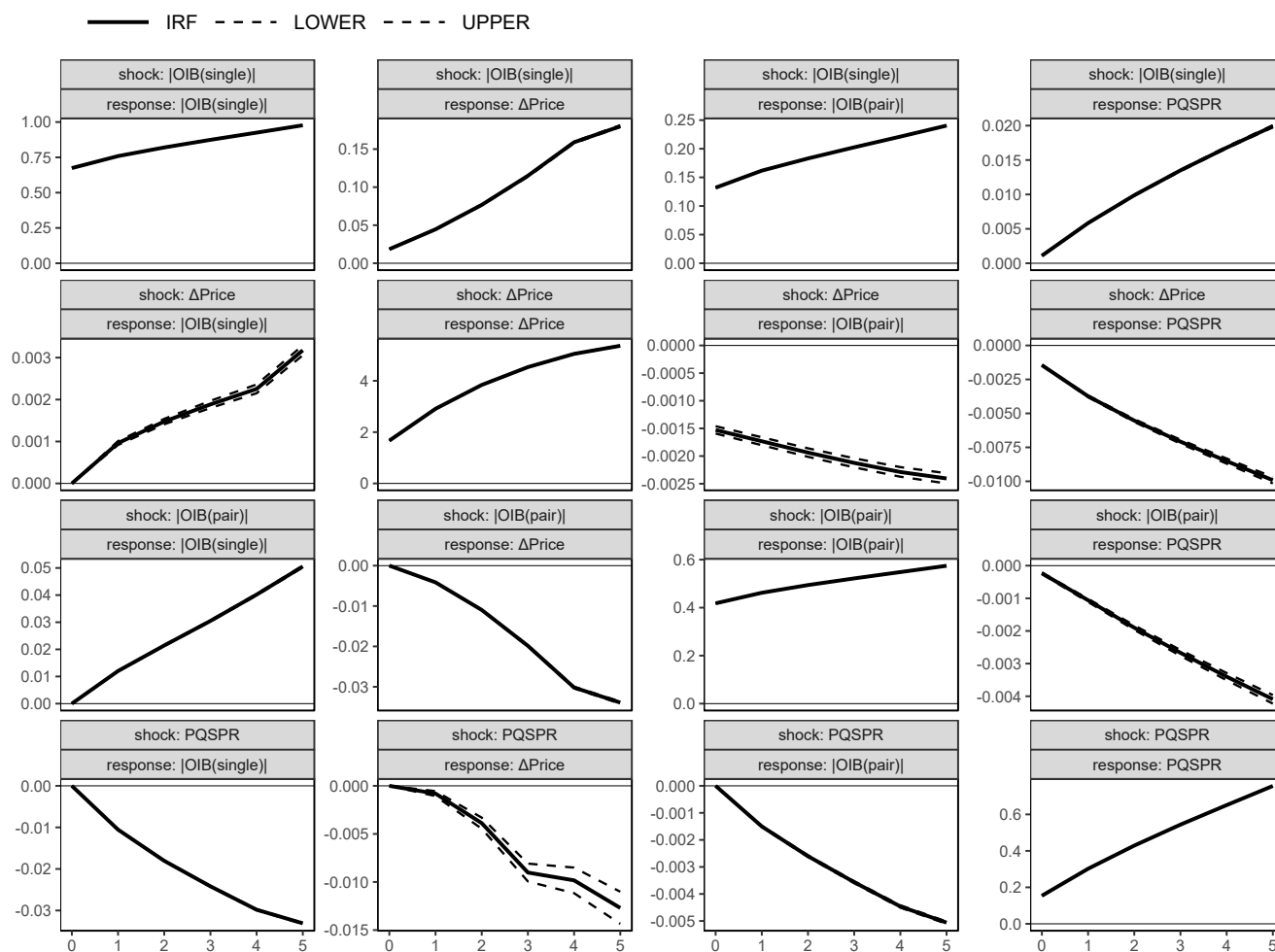


Fig. OA2. Intraday Impulse response functions from single and pair trades and quoted spreads, 2001- 2016

Compared to Figure 1 of the paper I only use OIB from single and pair trades and price deviations as endogeneous variables.

This figure shows impulse response functions (IRF) from panel vector autoregression (VAR) estimated similar as in Panel A of Table 5 of the paper, as described in more details in the main text. For a description of the VAR and these variables I refer to Tables 1 and 5 of the paper. All IRF in the first row show responses in OIB_{single} , $|\Delta Price|$, and OIB_{pair} to a Cholesky one standard-deviation shock to OIB_{single} in minute 0 (the contemporaneous effect) to minute 5. The following rows show responses to shocks to the other variables in the system. All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. Each figure shows bootstrapped 95% confidence bands based on 1000 runs (lower, upper). All data underlying the computations are from TRTH.

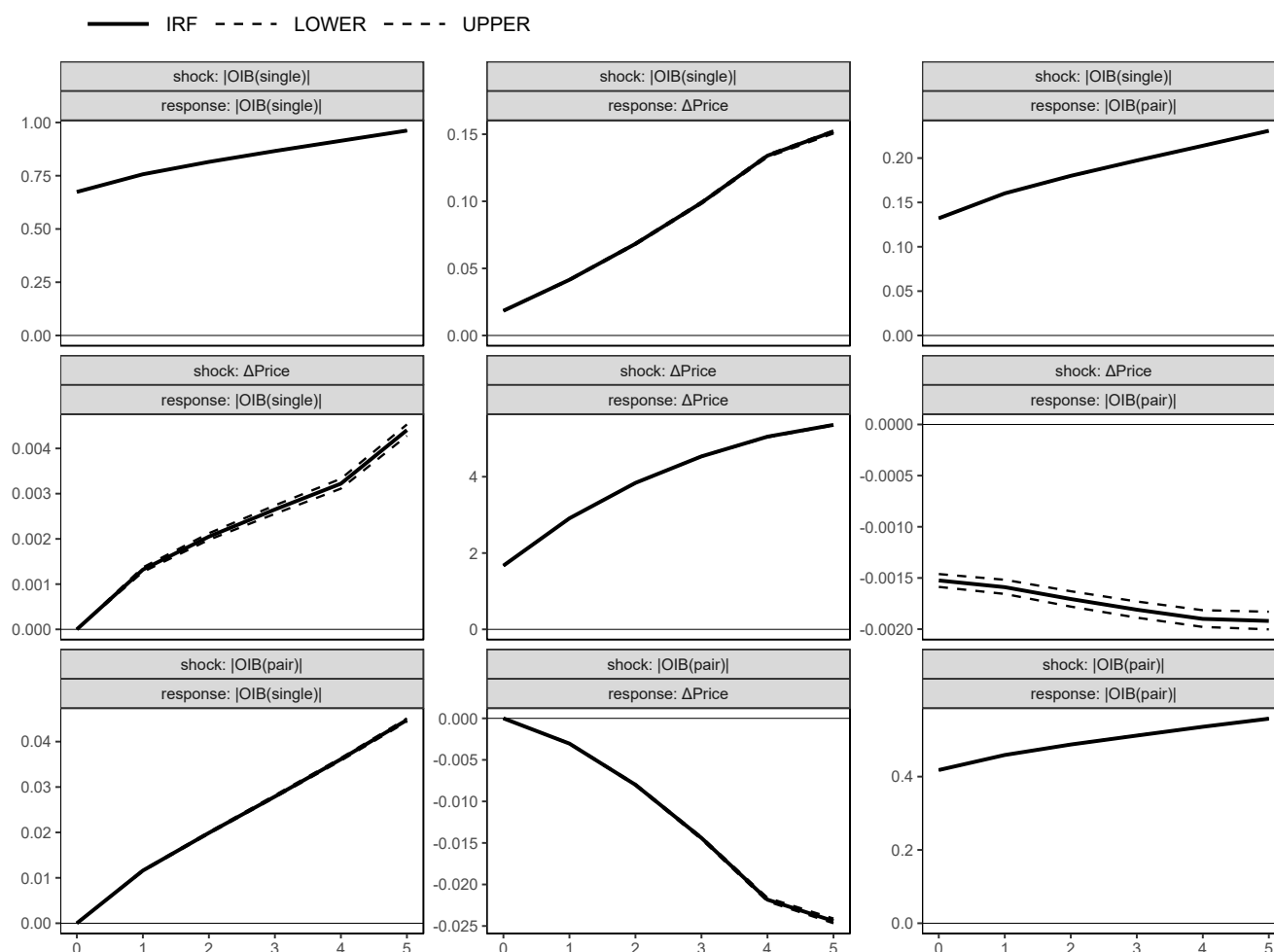


Fig. OA3. Intraday Impulse response functions from single and pair trades and quoted spreads, 2001- 2016

Compared to Figure 1 of the paper I only use OIB from single and pair trades and quoted spreads as endogeneous variables.

This figure shows impulse response functions (IRF) from panel vector autoregression (VAR) estimated similar as in Panel A of Table 5 of the paper, as described in more details in the main text. For a description of the VAR and these variables I refer to Tables 1 and 5 of the paper. All IRF in the first row show responses in OIB_{single} , $PQSPR$, and OIB_{pair} to a Cholesky one standard-deviation shock to OIB_{single} in minute 0 (the contemporaneous effect) to minute 5. The following rows show responses to shocks to the other variables in the system. All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. Each figure shows bootstrapped 95% confidence bands based on 1000 runs (lower, upper). All data underlying the computations are from TRTH.

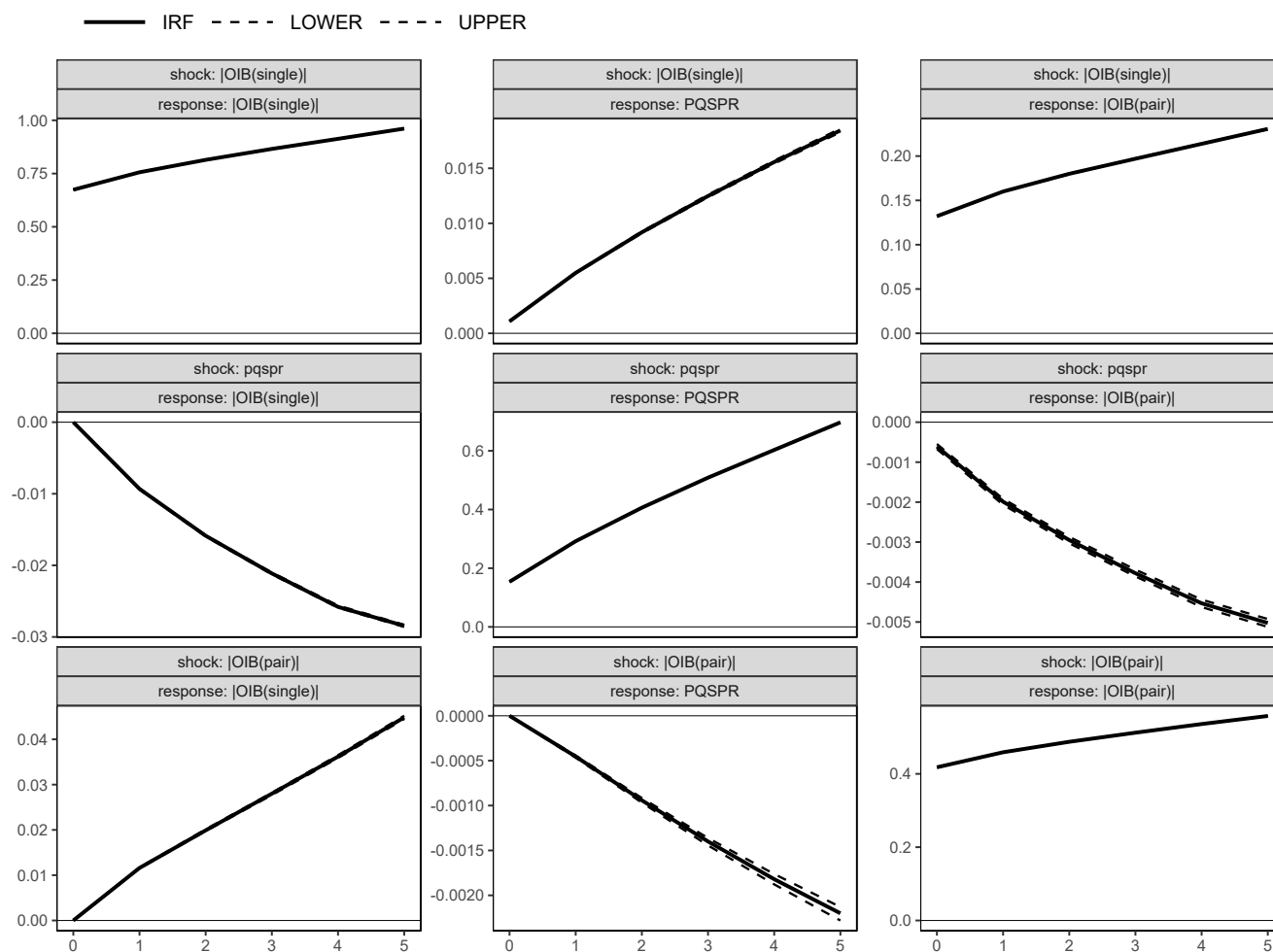


Fig. OA4. Intraday Impulse response functions from single and pair trades and quoted spreads, 2001- 2016

Compared to Figure 1 of the paper I estimate IRFs based on 5-minute intervals.

This figure shows impulse response functions (IRF) from panel vector autoregression (VAR) estimated similar as in Panel A of Table 5 of the paper, as described in more details in the main text. For a description of the VAR and these variables I refer to Tables 1 and 5 of the paper. All IRF in the first row show responses in OIB_{single} , $|\Delta Price|$, and OIB_{pair} to a Cholesky one standard-deviation shock to OIB_{single} in minute 0 (the contemporaneous effect) to minute 25. The following rows show responses to shocks to the other variables in the system. All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. Each figure shows bootstrapped 95% confidence bands based on 1000 runs (lower, upper). All data underlying the computations are from TRTH.

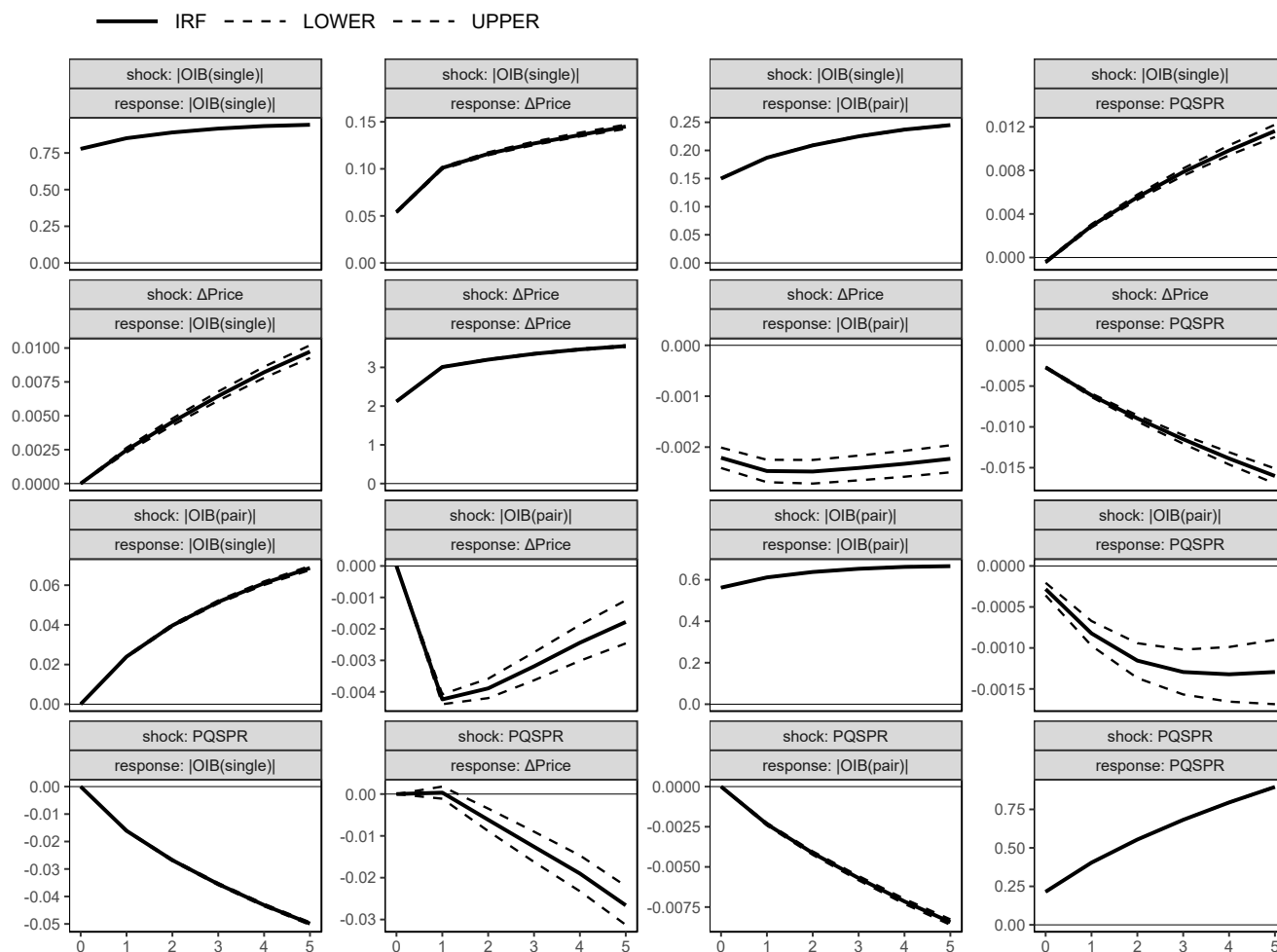


Fig. OA5. Intraday Impulse response functions from single and pair trades and quoted spreads, 2001- 2016

Compared to Figure 1 of the paper I estimate OIB as the absolute difference in buyer and seller initiated trading in USD.

This figure shows impulse response functions (IRF) from panel vector autoregression (VAR) estimated similar as in Panel A of Table 5 of the paper, as described in more details in the main text. For a description of the VAR and these variables I refer to Tables 1 and 5 of the paper. All IRF in the first row show responses in OIB_{single} , $|\Delta Price|$, and OIB_{pair} to a Cholesky one standard-deviation shock to OIB_{single} in minute 0 (the contemporaneous effect) to minute 5. The following rows show responses to shocks to the other variables in the system. All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. Each figure shows bootstrapped 95% confidence bands based on 1000 runs (lower, upper). All data underlying the computations are from TRTH.

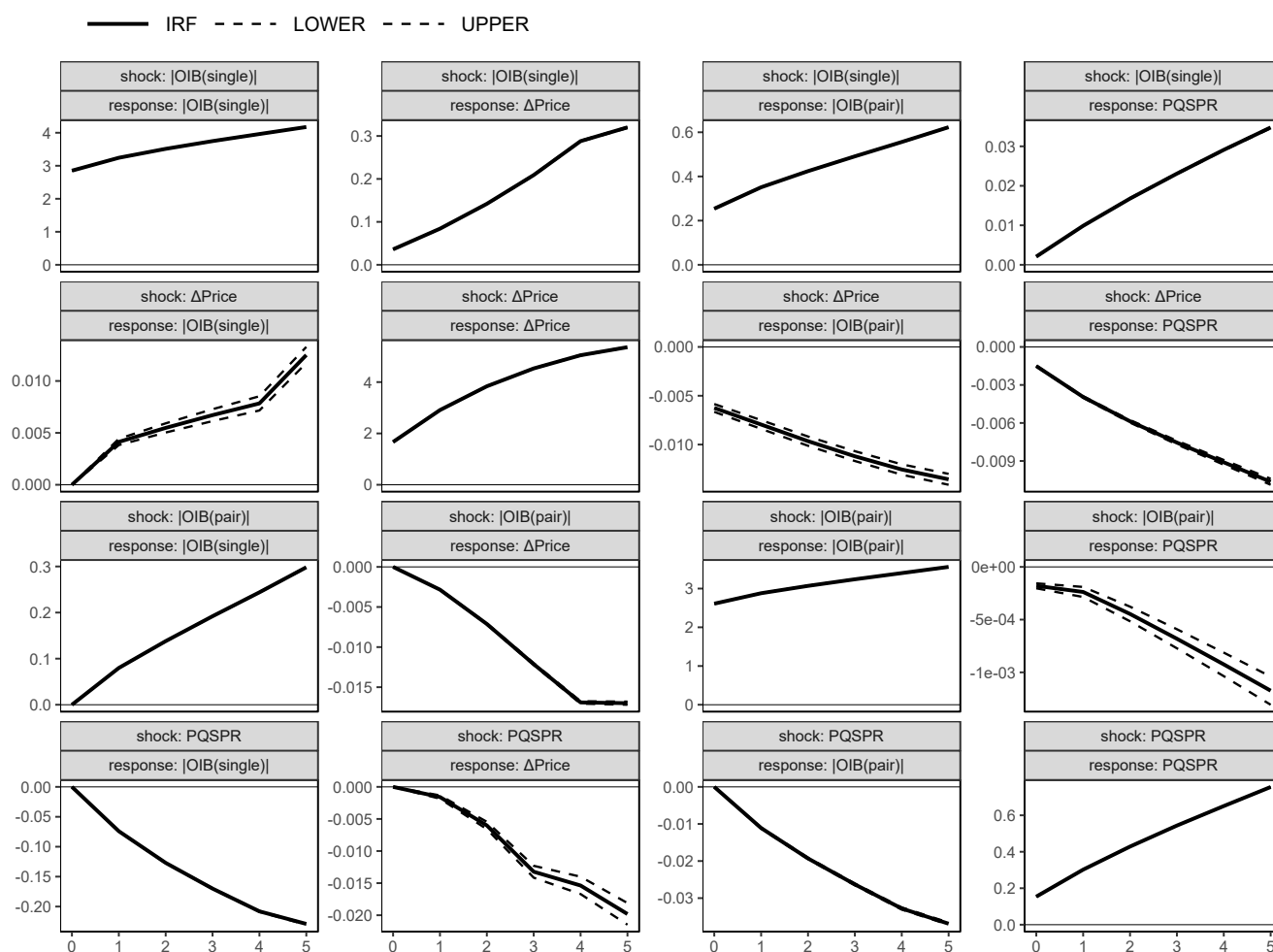


Fig. OA6. Intraday Impulse response functions from single and pair trades and quoted spreads, 2001- 2016

Compared to Figure 1 of the paper instead of $PQSPR$ I estimate the IRF using price improvements.

This figure shows impulse response functions (IRF) from panel vector autoregression (VAR) estimated similar as in Panel A of Table 5 of the paper, as described in more details in the main text. For a description of the VAR and these variables I refer to Tables 1 and 5 of the paper. *Priceimprovement* is the average difference in proportional quoted spreads prevailing trades and proportional effective spreads ($PQSPR[t] - PESPR[t]$) in each 1-minute interval. All IRF in the first row show responses in OIB_{single} , $|\Delta Price|$, and OIB_{pair} to a Cholesky one standard-deviation shock to OIB_{single} in minute 0 (the contemporaneous effect) to minute 5. The following rows show responses to shocks to the other variables in the system. All variables are measured during the overlapping trading time, i.e. when both the home market and the cross-listed market are in their continuous trading session. Each figure shows bootstrapped 95% confidence bands based on 1000 runs (lower, upper). All data underlying the computations are from TRTH.

